
Universal Approximation with Softmax Attention

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Abstract

We prove that with linear transformations, both (i) two-layer self-attention and (ii) one-layer self-attention followed by a softmax function are universal approximators for continuous sequence-to-sequence functions on compact domains. Our main technique is a new interpolation-based method for analyzing attention’s internal mechanism. This leads to our key insight: self-attention is able to approximate a generalized version of ReLU to arbitrary precision, and hence subsumes many known universal approximators. Building on these, we show that two-layer multi-head attention or *even* one-layer multi-head attention followed by a softmax function suffices as a sequence-to-sequence universal approximator. In contrast, prior works rely on feed-forward networks to establish universal approximation in Transformers. Furthermore, we extend our techniques to show that, (softmax-)attention-only layers are capable of approximating gradient descent in-context. We believe these techniques hold independent interest. Code is available at https://github.com/MAGICS-LAB/UAP_Attention.

1 Introduction

We study the universal approximation ability of the attention mechanism (Vaswani et al., 2017). We prove that either *two-layer self-attention* or *one-layer self-attention followed by a softmax* (each equipped only with linear transformations) is capable of approximating any sequence-to-sequence contin-

uous function on a compact domain. Different from previous studies (Yun et al., 2020a; Jiang & Li, 2024; Takakura & Suzuki, 2023; Kajitsuka & Sato, 2024; Hu et al., 2025), our results highlight the expressive power of Transformers derived *only* from the attention module. By focusing exclusively on attention, our analysis demonstrates that the softmax operation itself suffices as a piecewise linear approximator. Furthermore, we extend this framework to broader applications, such as in-context learning (Brown et al., 2020; Bai et al., 2024), using the same attention-only architecture.

Prior studies of Transformer-based universality lean on deep attention stacks (Yun et al., 2020a) or feed-forward (FFN) sub-layers (Kajitsuka & Sato, 2024; Hu et al., 2025) or strong assumptions on data or architecture (Takakura & Suzuki, 2023; Petrov et al., 2024). These results make it unclear whether attention alone is essential or auxiliary.

To combat this, we develop a new *interpolation-based* technique for analyzing attention. We discretize the target function’s output range into a uniform set of “anchors,” embed them into the key-query-value transformations of softmax attention, and leverage softmax for a near-argmax selection. Effectively and surprisingly, this procedure turns attention into a one- or two-layer piecewise linear approximator (i.e., a generalized notation of ReLU). Consequently, attention alone suffices for universal approximation — no large FFN blocks or complex positional encodings are needed. This leads to our main results — even a single- or two-layer attention configuration suffices to approximate continuous functions for sequence-to-sequence tasks.

Beyond pure universal approximation, we also extend the same technique to *in-context learning* scenarios (Brown et al., 2020; Bai et al., 2024), showing that attention alone is capable of mimicking gradient-descent-like updates and approximate statistical models, akin to (Bai et al., 2024).

Altogether, our results reveal a minimalistic yet powerful principle: *attention itself* captures the core expressive power needed for sequence-to-sequence universality. By isolating attention from other Transformer components, we affirm that the softmax-based mechanism has a direct route to approximate continuous mappings across a compact domain.

Contributions. Our contributions are four-fold:

- **Attention Approximation via Interpolation Selection.**

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We present a new *interpolation-based* method to analyze attention’s internal mechanism. First, we partition the target function’s range into uniformly spaced “anchors” and embed these anchors in the key-query-value transformations. Then, by approximating an argmax-style choice over these anchors, the softmax operation replicates piecewise linear behavior. Consequently, attention simulates an interpolation scheme for approximating known universal approximators. This insight eliminates reliance on auxiliary feed-forward layers to facilitate universal approximation of transformer architectures and highlights attention’s inherent ability to approximate target functions with minimal overhead. See [Figure 1](#) for a visualization.

- **One-Layer Single-Head (Softmax-)Attention Approximates Generalized ReLUs.** With our interpolation technique, we show that, for length- n input, single-head and H -head (softmax-)attention approximate n generalized ReLUs with $O(1/n)$, and $O(1/(nH))$ precision a token-wise manner ([Theorems 3.1](#) and [3.2](#)), respectively.
- **Two-Layer Multi-Head Attention Suffices to Be Sequence-to-Sequence Universal Approximator.** We show that (i) stacking two *attention-only* layers or (ii) one *attention layer* followed by a *softmax function* suffice for universal approximation of continuous sequence-to-sequence functions ([Theorem 3.3](#) and [Corollary 3.3.1](#) or a more Transformer-native extension in [Section H](#)). Compared to existing Transformer-based universal approximation results ([Yun et al., 2020a](#); [Kajitsuka & Sato, 2024](#); [Hu et al., 2025](#)), our result demonstrates that attention provides sufficient expressiveness. These findings highlight the core expressive power of attention and depart from prior works that rely on deep attention or feed-forward sub-layers for universality guarantees.
- **In-Context Approximation and Gradient Descent.** We extend our techniques and results to in-context learning settings. We prove that attention approximates generalized ReLUs in-context ([Theorem C.1](#)). Furthermore, we show that multi-head softmax attention is capable of In-Context Gradient Descent (ICGD) ([Theorem 4.1](#)), and hence simulates various statistical models, such as ridge regression and generalized linear models. These results improve upon ([Bai et al., 2024](#)), which is limited to ReLU attention and sometimes requires FFNs to facilitate ICGD.

We highlight that our results are general and require minimal assumptions. Our theory assumes only the target function is continuous on the compact domain. No assumptions are made about the data or model, making our results and techniques widely applicable.

This generality departs from prior studies ([Yun et al., 2020a](#); [Jiang & Li, 2024](#); [Takakura & Suzuki, 2023](#); [Kajitsuka & Sato, 2024](#); [Hu et al., 2025](#)). In particular, [Yun et al. \(2020a\)](#);

[Kajitsuka & Sato \(2024\)](#); [Hu et al. \(2025\)](#) rely on the concept of *contextual mapping* and assume a minimal separation condition on the data. [Jiang & Li \(2024\)](#) achieve Jackson-type universal approximation and require target space to have finite complexity measure, which acts like smoothness conditions in classical approximation theory. [Takakura & Suzuki \(2023\)](#) assume infinite-dimensional data. Moreover, while most existing works require many attention or FFN layers to achieve universal approximation of transformer blocks, our theory requires only one or two *attention-only* layers. This is, to the best of our knowledge, the first work on universal approximation of the attention mechanism.

Organization. Our main theorems progress in three steps. First, [Theorem 3.1](#) provides a single-head warm-up result to demonstrate our interpolation selection technique, showing that attention with linear transform approximate truncated linear functions. Next, [Theorem 3.2](#) extends this construction to multi-head attention setting aligning with practical transformer. Finally, [Theorem 3.3](#) upgrades the multi-head formulation to sequence-to-sequence universal approximation, and [Theorem 4.1](#) further applies the same ideas to in-context approximation of gradient descent.

Related Work. [Section D](#) offers additional details on the necessary related work discussed above.

2 Preliminaries

Notation. We use lower-case letters (e.g., v) for vectors and upper-case letters (e.g., M) for matrices. The vector e_k denotes the one-hot vector with a 1 in position k and 0 elsewhere. Let $X \in \mathbb{R}^{d \times n}$ denote the input sequence, where d is the token dimension and n is the sequence length; intermediate inputs/outputs are denoted by $Z \in \mathbb{R}^{d \times n}$. For a matrix A , $A_{:,j}$ is its j -th column, $A_{i,:}$ is its i -th row, and A_{ij} is the entry in row i and column j . We write $\|\cdot\|_\infty$ (or $\|\cdot\|_2$) for the vector ∞ -norm (resp., 2-norm). For a matrix $Z \in \mathbb{R}^{d \times n}$, we define the (p, q) -norm as

$$\|Z\|_{p,q} := \left(\sum_{j=1}^n \left(\sum_{i=1}^d |Z_{ij}|^p \right)^{\frac{q}{p}} \right)^{\frac{1}{q}}, \quad \|Z\|_{\infty,\infty} := \max_{i,j} |Z_{ij}|.$$

For a function f , $\|f\|_{L_\infty} := \sup_{x \in \Omega} |f(x)|$ denotes its supremum norm on the given domain Ω . More generally, we define the L_p -norm of function f as

$$\|f\|_{L_p} := \left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}}. \quad (2.1)$$

The full summary of table of notion is in [Section A](#).

Attention Layer. Let $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$ be input sequence of length n .

Definition 2.1 (Attention Layer). Let H denote the number of heads of self-attention block. For any input sequence $X \in \mathbb{R}^{d \times n}$, we define the multi-head self-attention layer as

$$\text{Attn}_m(X) = \sum_{h=1}^H W_V^{(h)} X \text{Softmax}((W_K^{(h)} X)^\top W_Q^{(h)} X) W_O^{(h)},$$

where $W_K^{(h)}, W_Q^{(h)} \in \mathbb{R}^{d_h \times d}$, $W_V^{(h)} \in \mathbb{R}^{d_o \times d}$, $W_O^{(h)} \in \mathbb{R}^{n \times n_o}$ for $h \in [H]$. We use Attn_s to denote *single-head* self-attention.

Here we pick non-identical dimensions for W_K, W_Q, W_V, W_O for generality of our analysis.

3 Main Theory

In this section, we introduce an interpolation-based method to characterize the internal mechanism of a single-head attention block. Building on this technique, we establish the universal approximation capability of attention from single-head to multi-head, then to in-context learning, and to the general sequence-to-sequence setting. Specifically, in [Section 3.1](#) we use a single-head self-attention with a sequence-wise linear transformation¹ to illustrate our interpolation techniques. It approximates n generalized ReLUs with $O(1/n)$ precision ([Theorem 3.1](#)). Build on top of this, in [Section 3.2](#) we construct the multi-head version with token-wise linear map aligns with standard linear map in transformer. We demonstrate that increasing the number of heads reduces the required computational complexity per head for the same approximation error ϵ . Explicitly, H -head attention yields $O(1/(nH))$ precision for approximating generalized ReLUs. In [Section 4](#), we extend the method to in-context learning, showing that a single-head self-attention with a linear layer approximates n generalized ReLUs in-context. Lastly, in [Section 3.3](#), we prove that such a minimalist attention layer suffices as a sequence-to-sequence universal approximator.

3.1 Attention Approximation as Interpolation Selection: Approximating Generalized ReLUs with $O(1/n)$ Precision

A key insight of our work is that single-head self-attention approximates a generalized ReLU function. Since ReLU neural network is a well-known universal approximator, this result implies that even a *minimalist* attention configuration subsumes many established universal approximators.

Truncated Linear Functions as Generalized ReLUs. We first formalize the generalized ReLU function using the

¹We remark that, this sequence-wise linear transformation is not essential to our analysis and can be removed without loss of generality. We adopt it in [Section 3.1](#) only for proof simplicity. Please also see [Remark F.1](#).

concept of a truncated linear function $\text{Range}_{[a,b]}(\cdot)$:

Definition 3.1 (Truncated Linear Function). We define the truncated linear function as follows:

$$\text{Range}_{[a,b]}(x) = \begin{cases} a & x \leq a, \\ x & a \leq x \leq b, \\ b & b \leq x. \end{cases}$$

Intuitively, the truncated linear function is a segment of a linear function, with output value ranging from a to b .

Definition 3.2 (Truncated Linear Model). Define a *truncated linear model* as $\text{Range}_{[a,b]}(w^\top x + t)$, where $w \in \mathbb{R}^d$ is a learnable weight and $t \in \mathbb{R}$ is a bias.

We remark that the truncated linear model is a generalized ReLU and subsumes many known universal approximators, including ReLU ([Example 1](#)), Hard Tanh ([Example 2](#)) and Clipped ReLU ([Example 3](#)). Please see [Section G](#) for explicit expressions. These bounded activations appear in many practical scenarios where output constraints or gradient stability are desired.

Our goal here is to show attention is able to approximate $\text{Range}_{[a,b]}(w^\top x + t)$ with arbitrary precision. ReLU networks are classic universal function approximators ([Lu et al., 2017](#); [Sonoda & Murata, 2017](#); [Hanin, 2019](#); [Park et al., 2021](#)). By demonstrating that single-head attention approximates $\text{Range}_{[a,b]}(x)$ to arbitrary precision, we show that *attention alone* replicates the essential behavior of ReLUs (and even more general piecewise linear transformations). This provides a foundation for proving broader universal approximation results using *only* attention mechanisms.

Interpolation Scheme. To approximate $\text{Range}_{[a,b]}(\cdot)$ with attention, we partition $[a, b]$ into p uniform segments:

Definition 3.3 (Interpolation). Let $[a, b] \subset \mathbb{R}$ be an interval with $a \leq b$ and let $p \geq 2$. Define p interpolation anchors

$$\begin{aligned} \tilde{L}_1^{[a,b]} &:= a, & \tilde{L}_p^{[a,b]} &:= b, \\ \tilde{L}_k^{[a,b]} &:= a + \frac{k-1}{p-1}(b-a), & k &\in [p]. \end{aligned}$$

Hence, $\tilde{L}_1 < \dots < \tilde{L}_p$ forms a uniform partition of $[a, b]$. We also write

$$\Delta L := \tilde{L}_k^{[a,b]} - \tilde{L}_{k-1}^{[a,b]} = \frac{b-a}{p-1}, \quad k \in \{1, \dots, p-1\}.$$

We often omit the superscript $[a, b]$ when the context is clear.

Importantly, these segments $\{\tilde{L}_k\}_{k=1}^p$ serve as “targets” for the attention mechanism in later parts.

Interpolation Method for Attention Approximation.

Now we present our fundamental result — a single-head self-attention with a linear transformation is capable of approximating truncated linear models in a token-wise manner. Let $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$ be the input sequence. We define δ as follows. For i -th column of attention score matrix $K^\top Q$, let $x_{1,i}$ and $x_{2,i}$ be its largest and second-largest entries and define $\delta_i := x_{1,i} - x_{2,i}$, and denote $\delta = \min_{i \in [n]} \delta_i$ to be the smallest such gap over all columns.

Theorem 3.1 (Single-Head Attention Approximates a Truncated Linear Model, Input-Adaptive β , Arbitrary $G(\cdot)$). Fix real numbers $a < b$, and let $\text{Range}_{[a,b]}(\cdot)$ be the truncation operator from Definition 3.1. Let $p \geq \max\{n+1, 3\}$. Fix $w \in \mathbb{R}^d$, $t \in \mathbb{R}$. For every $\epsilon_0 \in (0, 1)$, suppose $\beta \geq \frac{1}{\delta} \log \frac{p-1}{\epsilon_0}$. Then, there exists a single-layer, single-head self-attention layer Attn and a sequence-wise affine map $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$ such that, for every $X \in \mathbb{R}^{d \times n}$ and every $i \in [n]$,

$$\begin{aligned} & \|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty \\ & \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error by Lemma E.1}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}}. \end{aligned}$$

Here $e_{\tilde{k}_i}$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \underset{k \in [p]}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k|, \quad \tilde{k}_i := G(k_i) \in [d_o]. \quad (3.1)$$

Here $k_i \in [p]$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : [p] \rightarrow [d_o]$ denotes any set-to-set function sending the interpolation index $k \in [p]$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output.

Corollary 3.1.1 (Extension to Many Truncated Linear Models). For n truncated linear models $\{w_i, t_i\}_{i \in [n]}$, the same conditions give

$$\begin{aligned} & \|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty \\ & \leq \max\{|a|, |b|\} \cdot \epsilon_0 + \frac{b-a}{p}. \end{aligned}$$

Corollary 3.1.2 (Single-Head Attention Approximates Many Truncated Linear Models, Input-Agnostic β , Constant $G(\cdot)$). Under the same conditions as Theorem 3.1 but choose $\beta \geq \frac{2}{(\Delta L)^2} \log \frac{2(p-2)}{\epsilon_0}$ we have

$$\|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty$$

$$\leq \max\{|a|, |b|\} \cdot \epsilon_0 + \frac{b-a}{p},$$

where k_i and $\tilde{k}_i$ are defined as (3.1), but now $G : [p] \rightarrow [d_o]$ denotes any set-to-constant function sending the interpolation index $k \in [p]$ into a constant $\tilde{k} \in [d_o]$ specifying in the desired row of the output.

Intuitively, Theorem 3.1 and the corollaries ensure that a single-head self-attention layer with a suitable linear layer is capable of approximating n “truncated” linear models with token-level granularity. We accomplish this via an interpolation method. To elaborate, a few remarks are in order.

Remark 3.1 (Interpolation Selection with Softmax Attention). Here, we provide a high-level overview of our proof techniques: we approximate the target function (truncated linear models of interest) using interpolation points and leverage softmax attention for interpolation point selection. We also provide conceptual visualization in Figure 1.

For the i -th column (token) of $\text{Attn} \circ A(X) \in \mathbb{R}^{d_o \times n}$, our goal is to approximate the one-hot vector $\text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}$, where $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ is a scalar (the truncated linear output), and $e_{\tilde{k}_i}$ is a one-hot vector of dimension d_o . To achieve this, we require *at least* n column vectors in $\text{Softmax}(K^\top Q)$ to represent potential outputs of the n truncated linear models.

Since the output of $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ lies within $[a, b]$ (Definition 3.1), we apply the interpolation scheme (Definition 3.3) to partition $[a, b]$ into p points. For each $i \in [n]$, there exists an interpolation point $\tilde{L}_{k_i}$ closest to truncated linear model $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$, where $k_i := \underset{k \in [p]}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k|$ is the selected interpolation index.

Our key idea is to

1. **Select Interpolation Index.** Express $\{k_i\}_{i \in [n]}$ as one-hot column vectors $\{e_{k_i}\}_{i \in [n]} \in \mathbb{R}^p$.
2. **Approximate Anchors Design.** Design $K^\top Q$ such that $\text{Softmax}(K^\top Q)_{:,i}$ approximates e_{k_i} for all $i \in [n]$.
3. **Recover the Selected Interpolation Point in Value Space.** Encode interpolation point $\{\tilde{L}_k\}_{k=1}^p$ into V such that, for each $i \in [n]$, the largest entry of $V \text{Softmax}(K^\top Q)_{:,i}$ to be the interpolation point $\tilde{L}_{k_i}$, ($k_i \in [p]$). Recall that, $\tilde{L}_{k_i}$ is selected as the closest interpolation point to $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$

We visualize in Figure 1 and summarize as follows:

$$\max_{j \in [d_o]} \left[\underbrace{\overbrace{p \text{ column vectors containing } p \text{ interp. points } \{\tilde{L}_k\}_{k=1}^p}^V}_{\text{approximate one-hot representation of selected interpolation index } k_i} \text{Softmax}(K^\top Q)_{:,i} \right]_{j,i}$$

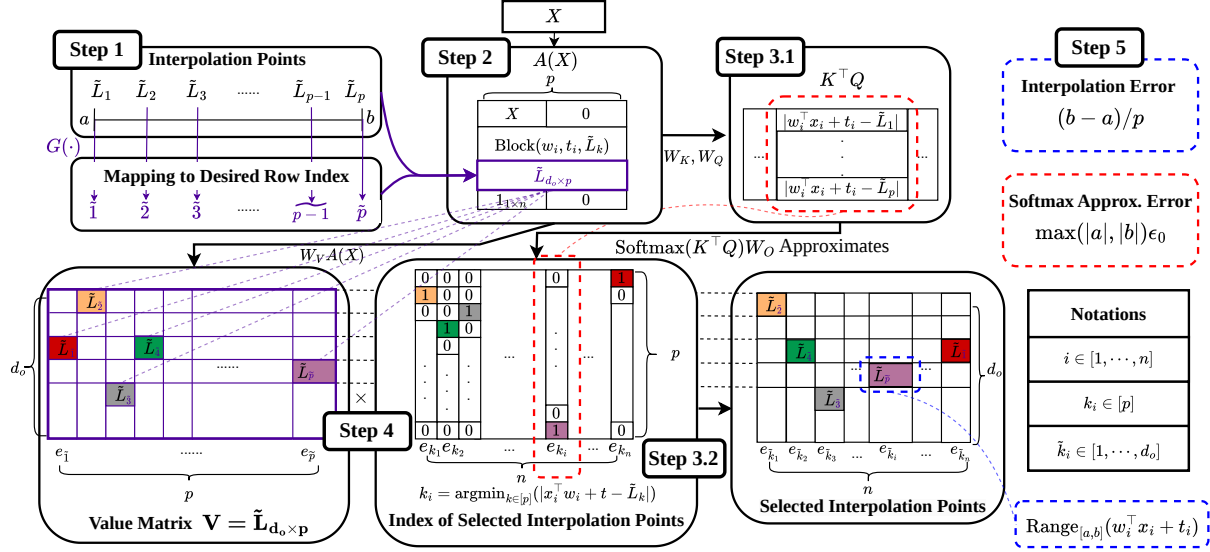


Figure 1. Visualization of Proposed Interpolation Technique (Theorem 3.1). Every step in the figure corresponds to a step in the proof sketch in Section 3.1. Our goal is to use softmax attention mechanism to approximate n truncated linear models $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ for $i \in [n]$, and hence establish universality. To achieve this, we first divide the output range $[a, b]$ into p interpolation points, and encode them into the value matrix V . Then, we treat the attention score $\text{Softmax}(K^\top Q)$ as a *selector* to select an interpolation point closest to the desired output from V . Specifically, each column of $\text{Softmax}(K^\top Q)_{:,i}$ (for $i \in [n]$) approximates a one-hot vector e_{k_i} , where k_i is the index of closest interpolation point to $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$. Hence, when multiplying with V , $V \cdot \text{Softmax}(K^\top Q)$ selects out the closest interpolation points for every truncated linear model from V . The **same color** across matrices indicates the same interpolation point chosen by the softmax function. The color purple indicates how G maps each interpolation point index k into the desired row index $\tilde{k}$. The **grey dashed lines** highlight that the position of $\tilde{L}_k$ in the value matrix and the output matrix are the same, indicating each closest interpolation point of i -th token is placed correctly at the output. The **blue dashed line** illustrates the interpolation error, while the **red dashed line** shows the softmax approximation error. For simplicity, we highlight the error only for a token x_i .

$$\begin{aligned}
 &= \underbrace{\tilde{L}_{k_i}}_{\underset{\{ \tilde{L}_1, \dots, \tilde{L}_p \}}{\operatorname{argmin}} | \text{Range}_{[a,b]}(w_i^\top x_i + t_i) - \tilde{L}_k |} \\
 &+ \underbrace{\text{error}}_{\text{(By finite-}\beta \text{ softmax approximation Lemma E.1)}}. \quad (3.2)
 \end{aligned}$$

This way we use attention mechanism to perform interpolation approximation to each truncated linear model output.

Remark 3.2 (Why $A(\cdot)$ and Its Connection to Practice). To accomplish (3.2), we embed the p interpolation points into $A(X)$ such that $K^\top Q = A(X)^\top W_K^\top W_Q A(X)$ contains these points among its entries. The linear map $A(\cdot)$ here includes a sequence-wise operation to only simplify the proof. It is not a standard component of Transformer architecture. Importantly, Theorem 3.1 serve as the simplest illustrative example for our interpolation selection techniques. In Theorem 3.2, we extend this technique to the multi-head setting and replace the *sequence-wise* $A(\cdot)$ with a *token-wise linear* transformation that aligns with practical Transformer architectures. See also Section 5.

Remark 3.3 (Meaning of k_i , $\tilde{k}_i$, and $G(\cdot)$). Here, we clarify the distinction between k_i and $\tilde{k}_i$. The difference lies in

their roles within the interpolation and output spaces. Given the i -th token x_i , $k_i \in [p]$ identifies the closest interpolation point $\tilde{L}_{k_i}$ to the target value $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$. In contrast, $\tilde{k}_i \in [d_o]$ is an output coordinate index: it specifies in which coordinate of the d_o -dimensional output vector we place the selected point $\tilde{L}_{k_i}$ (grey dashed lines in Figure 1). The mapping $G : [p] \rightarrow [d_o]$ connects these two roles by assigning each interpolation index k a coordinate $\tilde{k} := G(k)$, and for each token i we then have $\tilde{k}_i = G(k_i)$ (purple font in Figure 1). In the simplest case one take $G(k) \equiv 1$ for all $k \in [p]$, so that every $\tilde{L}_{k_i}$ is placed in the first row of the output matrix. This flexibility allows G to be tailored to the scenarios considered. See Section F.2.1 for detailed discussion.

Remark 3.4 (Universal Approximation Implications). Since $\text{Range}_{[a,b]}(\cdot)$ acts as a bounded ReLU, demonstrating that a single-head attention layer approximates it arbitrarily well implies that attention alone is capable of replicating and generalizing known piecewise linear networks. We leverage this result to establish the universal approximation properties of attention-based architectures in broader settings (e.g., multi-head, seq-to-seq) in subsequent sections.

Proof Sketch. We design the key-query matrices such that, for each token x_i , the column $\text{Softmax}(K^\top Q)_{:,i}$ selects the closest interpolation point $\tilde{L}_{k_i}$ to $w_i^\top x_i + t_i$. This yields a single-head attention output approximating the truncated linear model at each token.

Our proof consists of five conceptual steps:

Step 1: Partitioning. Partition the range $[a, b]$ into $p - 1$ segments, defining interpolation points $\{\tilde{L}_k\}_{k=1}^p$, so that for any $\text{Range}_{[a,b]}(w_i^\top x_i + t_i) \in [a, b]$, there exists a nearest interpolation point $\tilde{L}_{k_i}$ satisfying

$$|x_i^\top w_i + t_i - \tilde{L}_{k_i}| \leq \frac{b-a}{p}, \quad \text{for all } i \in [n].$$

Step 2: Linear Encoding. Apply a linear transformation

$$A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p},$$

augmenting the input X with additional rows and columns to include: (i) the input tokens x_i , (ii) the weights $\{w_i\}_{i=1}^n$ and biases $\{t_i\}_{i=1}^n$ (to construct truncated linear models), (iii) the interpolation points $\{\tilde{L}_k\}_{k=1}^p$, and (iv) auxiliary entries for constructing the desired key-query scores.

Step 3: Key-Query Construction. Design W_K, W_Q such that each column of $K^\top Q \in \mathbb{R}^{p \times p}$ has entries of the form

$$[K^\top Q]_{k,i} = (-2x_i^\top w_i - 2t_i + \tilde{L}_1 + \tilde{L}_k) \cdot (k-1).$$

The rationale behind this design is the equivalence between the following two objectives (see (F.8) for a proof):

$$\begin{aligned} & \underset{k \in [p]}{\text{argmin}} (-2x_i^\top w_i - 2t_i + \tilde{L}_1 + \tilde{L}_k) \cdot (k-1) \\ &= \underset{k \in [p]}{\text{argmin}} |x_i^\top w_i + t_i - \tilde{L}_k|, \end{aligned}$$

where the second objective selects the interpolation point $\tilde{L}_{k_i}$ (see (3.1)) closest to $x_i^\top w_i + t_i$ among p interpolation points. Thus, $[K^\top Q]_{k,i}$ indicates the interpolation point $\tilde{L}_k$ closest to $w_i^\top x_i + t_i$. Using Lemma E.1, the softmax function approximates the argmax, ensuring that the column vector $\text{Softmax}(K^\top Q)_{:,i}$ approximates a one-hot selection of $\tilde{L}_{k_i}$, the closest interpolation point. Specifically, $\text{Softmax}(K^\top Q)_{:,i}$ approximates $e_{k_i} \in \mathbb{R}^p$.

Step 4: Value Mapping. Design W_V such that $V = W_V A(X)$ encodes the interpolation points $\{\tilde{L}_k\}_{k=1}^p$ from $A(X)$ into the column vectors of $V \in \mathbb{R}^{d_o \times p}$. Specifically, for $k \in [p]$, the k -th column of V is $\tilde{L}_k e_{\tilde{k}}$. Then, multiplying V with $\text{Softmax}(K^\top Q) \in \mathbb{R}^{p \times p}$, where the i -th column approximates $e_{k_i} \in \mathbb{R}^p$ (from Step 3), gives

$$\underbrace{V}_{d_o \times p} \underbrace{\text{Softmax}(K^\top Q)}_{p \times 1}_{:,i} \in \mathbb{R}^{d_o}.$$

The largest entry of this product approximates the closest interpolation point $\tilde{L}_{k_i}$. Post-multiplication by the projection matrix W_O discards the extra $(p-n)$ columns beyond the original sequence length n .

Step 5: Error Control. We must bound two types of errors. (i) Interpolation Error: Partitioning $[a, b]$ into p segments ensures each $w_i^\top x_i + t_i \in [a, b]$ lies within $(b-a)/p$ of some interpolation point $\tilde{L}_{k_i}$. (ii) Softmax Approximation Error: Using Softmax_β instead of a hard $\arg \max$ introduces ϵ_0 (Lemma E.1). Moreover, because $\max_k |\tilde{L}_k| \leq \max\{|a|, |b|\}$, the softmax spread contributes at most $\max\{|a|, |b|\} \cdot \epsilon_0$. Consequently, for each token i ,

$$\begin{aligned} & \underbrace{|\text{Range}_{[a,b]}(w_i^\top x_i + t_i) - \tilde{L}_{k_i}|}_{\text{Interpolation error} \leq \frac{b-a}{p}} \\ & + \underbrace{\|\text{Softmax}_\beta(\cdot) - e_{k_i}\| \cdot \max\{|a|, |b|\}}_{\text{Softmax approx. error} \leq \max\{|a|, |b|\} \epsilon_0} \\ & \leq \frac{b-a}{p} + \max\{|a|, |b|\} \epsilon_0. \end{aligned}$$

By tuning p and the softmax β , we make these errors arbitrarily small, proving that single-head attention approximates $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ for each token with arbitrary precision. Please see Section F.2 for a detailed proof. $\square$

In summary, increasing the partition size p (reducing ϵ) improves the approximation to arbitrary precision $O(1/n)$. As $p > n$, a longer input sequence (with larger n and hence larger p) yields a larger attention score matrix (i.e., $\text{Softmax}(K^\top Q)$), enabling higher-resolution interpolation. This highlights the expressive power of the minimalist attention layer. In contrast, typical Transformers rely on multi-head structures and feed-forward layers.

3.2 H -Head Attention Approximates Generalized ReLU with $O(1/(nH))$ Precision

In Section 3.1, we show how a single-head self-attention layer approximates n truncated linear models by embedding p interpolation points into its key-query-value matrices. Here, we extend this construction to *multi-head* attention. We show that H -head attention improves the approximation precision from $O(1/n)$ (Theorem 3.1) to $O(1/(nH))$ for approximating generalized ReLU. This establishes a tradeoff between the number of heads and the per-head complexity, determined by the size of the linear layer A . Intuitively, more heads allow each head to focus on a smaller subset of interpolation points, reducing the partition size p needed per head to achieve the same overall error.

Theorem 3.2 (Multi-Head Attention Approximate Truncated Linear Models). Fix real numbers $a < b$, and let the truncation operator $\text{Range}_{[a,b]}(\cdot)$ follow Definition 3.1. For a precision parameter $p > n$ with $\epsilon = O(1/p)$, number of

head $H = p/(n-2)$ there exists a single-layer, H -head self-attention Attn^H with a linear transformation $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(d+n) \times n}$, such that $\text{Attn}^H \circ A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d_o \times n}$ satisfies, for any $i \in [n]$,

$$\|\text{Attn}^H \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{(n-2)H}}_{\text{interpolation error}}.$$

Here $e_{\tilde{k}_i}$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \underset{k \in [p]}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k|,$$

where $\tilde{k}_i = G(k_i) \in [d_o]$. Here $k_i \in [p]$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : [p] \rightarrow [d_o]$ denotes any set-to-constant function sending the interpolation index $k \in [p]$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output.

Corollary 3.2.1 (Approximation Error). The approximation error scales as $O(1/(nH))$.

Tradeoff: Multiple Heads H vs. Partition Size p . Whereas the single-head construction in [Theorem 3.1](#) places all p interpolation points into one attention head (possibly requiring $\ell = p - n$ extra columns in $A(X)$), multi-head attention splits these p points across different heads.

Consequently, each head only needs to handle a fraction of the total interpolation range, allowing for fewer effective points per head. In practice, this reduces per-head computation (both in forming K, Q, V and in performing the softmax) while preserving the same global partition resolution (i.e., the same overall approximation error ϵ).

Proof Sketch. Our proof strategy follows [Theorem 3.1](#), but distributes the interpolation workload:

1. **Partition the Points Across Heads.** Suppose we have H attention heads and want to approximate $\text{Range}_{[a,b]}(\cdot)$ with total precision $O(1/p)$. We split the p interpolation points into H groups, each group containing $p/H = n-2$ points.
2. **Local Encoding.** In each head, we store (in V) only the portion of the $(n-2)$ interpolation points assigned to that head. We also add two sentinel columns representing “no contribution” outside the local interpolation range. This ensures that if a token’s value is not covered by head h , the head h remains inactive (outputs zero).
3. **Head Selection.** We design the key-query matrices such that each token x_i “selects” the head whose local inter-

polarization range covers $w_i^\top x_i + t_i$. Softmax in that head’s output then acts as an approximate arg max among the assigned interpolation points. We also discuss the case where the value $w_i^\top x_i + t_i$ happen at the shared endpoint of two adjacent heads.

4. **Combine Heads.** [Lemma F.2](#) tells us every token is either (1) strictly inside one head’s interval or (2) exactly on a shared endpoint of two consecutive intervals. In the interior case only that head contributes; at a boundary the two neighbouring heads output a convex sum of the same two grid points. Either way the total error is the interpolation error $(b-a)/p$ plus the softmax error ϵ_0 added at most $(O(H) + |b|)\epsilon_0$, ϵ_0 can be arbitrarily small by setting a large enough β .

By splitting p points across H heads, each head handles only p/H points. Thus, the *per-head* complexity decreases while achieving the same global approximation $\epsilon = O(1/p)$. Moreover, $\epsilon = (1/(nH))$ by $H = p/(n-2)$. Please see [Section F.3](#) for a detailed proof. $\square$

3.3 Sequence-to-Sequence Universal Approximation by Self-Attention

Building on the results so far, we now show that a two-layer multi-head attention — augmented with simple linear transformations — achieves *sequence-to-sequence* universal approximation.

Overview of Our Proof Strategy. [Theorem 3.1](#) establishes that a single-head or multi-head attention layer is capable of approximating generalized ReLUs (truncated linear models) on a token-by-token basis. To extend this capability to more general sequence-to-sequence settings, we:

- **Step 1: Construct a Two-Layer ReLU Network as a Vector-to-Scalar Universal Approximator.** We construct a two-layer ReLU neural network in [Lemma 3.1](#) that serves as a universal approximator for any continuous function $f : \mathbb{R}^N \rightarrow \mathbb{R}$ on a compact domain, with a p -norm error.
- **Step 2: Approximate the Constructed ReLU Neural Network with Attentions.** In [Lemma 3.2](#), we prove that one layer multi-head attention plus one layer single head attention approximate the constructed ReLU neural network from [Lemma 3.1](#). This proves that two-layer attention approximates any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ on compact domain with a p -norm error.
- **Step 3: Extend to Sequence-to-Sequence Approximation.** We generalize [Lemma 3.2](#) to sequence-to-sequence approximation in [Theorem 3.3](#). This involves decomposing an arbitrary continuous map $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ into $d \cdot n$ scalar-valued functions $f_{ij} : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$. We approximate each f_{ij} with different attention layers construct in [Lemma 3.2](#), and then aggregate these scalar outputs into a

matrix form with an additional multi-head attention layer. This shows that a two-layer attention mechanism suffices as a sequence-to-sequence universal approximator. We also extend to ∞ -norm error in [Theorem E.1](#).

Below, we elaborate on the conceptual steps in detail and defer the proofs to appendices.

Step 1: Universal Approximation via Two-Layer ReLU Networks. We start with the universal approximation theorem of a two-layer feed-forward network with ReLU activation. Let $\mathcal{X} \subset \mathbb{R}^N$ be a compact domain, and $\|f\|_{L_p}$ be the L_p -norm following (2.1), for function f on its given domain.

Lemma 3.1 (Explicit Construction of ReLU Neural Network as Universal Approximator). Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a continuous function defined on $\mathcal{X}$. For any $\epsilon > 0$, there exists a two-layer feed-forward neural network $\text{FFN} : \mathbb{R}^N \rightarrow \mathbb{R}$ with ReLU activation functions such that for all $x \in \mathcal{X}$

$$\|\text{FFN}(x) - f(x)\|_{L_p} \leq \epsilon. \quad (3.3)$$

Proof. Please see [Section F.4](#) for a detailed proof. $\square$

With the constructed ReLU NN, we proceed to step 2, approximating it using a two-layer attention mechanism. We achieve this by utilizing [Theorem 3.1](#) that attention approximate $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ in a tokenwise manner.

Step 2: Approximate the Constructed ReLU Neural Network with Attentions. Now we prove the universal sequence-to-scalar approximation of multi-head attention.

Lemma 3.2 (Sequence-to-Scalar Universal Approximation of Two Layer Attention). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when composed with linear transformations, there exists a one layer multi-head attention Attn_m stacked with one layer single-head attention Attn_s composed with linear connections A_1 and A_2 , such that

$$\|f - \text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1\|_{L_p} \leq \epsilon.$$

Proof. Please see [Section F.5](#) for a detailed proof. $\square$

Note that in [Lemma 3.2](#) the function of the second single-head attention is to utilize the softmax function to pick out the closest grid point v to the input X . Hence we derive a one layer multi-head attention version of [Lemma 3.2](#) below.

Lemma 3.3 (Single-Layer Multi-Head Attention Version of [Lemma 3.2](#)). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when composed with linear transformations, there exists a one layer multi-head attention Attn_m followed by a Softmax function and attached with linear connections A_1 and A_2 ,

such that

$$\|f - A_2 \circ \text{Softmax} \circ \text{Attn}_m \circ A_1\|_{L_p} \leq \epsilon.$$

Proof. Please see [Section F.6](#) for a detailed proof. $\square$
We now state our final result of sequence-to-sequence universal approximation of two-layer attention.

Step 3. By combining dn two-layer attention blocks $\text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1$ from [Lemma 3.2](#), we approximate each output entry of $f(X)$ individually.

Theorem 3.3 (Two-Layer-Sequence-to-Sequence Approximation). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when composed with linear transformations, there exists a two layer multi-head attention Attn_m stacked with one layer multi-head attention Attn_m , attached with linear connection A_1 and A_2 , such that

$$\|f - \text{Attn}_m^{(2)} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1\|_{L_p} \leq \epsilon.$$

Corollary 3.3.1 (Single-Layer Attention Sequence-to-Sequence Approximation). There exists a single-layer multi-head attention Attn_m followed by a Softmax function and attached with linear connections A_1 and A_2^{ij} for $i \in [d], j \in [n]$, such that

$$\|f - \sum_{i \in [d], j \in [n]} A_2^{ij} \circ \text{Softmax} \circ \text{Attn}_m^{(1)} \circ A_1\|_{L_p} \leq \epsilon.$$

Proof. Please see [Section F.7](#) for a detailed proof. $\square$

4 In-Context Learning

We extend the interpolation selection technique and [Theorem 3.1](#) to the in-context learning setting ([Brown et al., 2020](#); [Bai et al., 2024](#)). In [Theorem 4.1](#), we show that standard softmax attention perform in-context gradient descent, broadening the results established for ReLU attention in ([Bai et al., 2024](#)). Specifically, we demonstrate that softmax attention is capable of doing in-context gradient descent on convex loss functions. We first define the problem setting.

Definition 4.1 (In-Context Learning Problem Formulation). The sequential input X in the in-context learning scenario is defined as

$$X := \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n \\ w & w & \cdots & w \\ 1 & 1 & \cdots & 1 \end{bmatrix},$$

where $w^\top x_i$ ($i \in [n]$) denote the input-output pairs. w parametrize the model connecting x_i and y_i , and is altered (trained) between layers. The task of in-context learning is to using the given input-output pairs (x_i, y_i) to predict the

output of a newcomer input x_u .

Then, we prove a multi-head Softmax attention is capable of doing in-context gradient descent on loss functions parametrized by $w^\top x_i$ ($i \in [n]$) and t (as linear coefficient and bias), as well as giving an according prediction on x_u .

Theorem 4.1 (In-Context Gradient Descent). Let $l : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be any C^1 loss function defined on $(w^\top x_i, y_i)$. With input X in the form of Definition 4.1, when X is bounded, there exists a multi-head self-attention Attn_m with skip connections and each attached with a linear layer, such that for any $\epsilon > 0$, irrelevant of X , we have

$$\left\| \text{Attn}_m \circ A(X) - \begin{bmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \\ w - \eta \nabla L(w) & \cdots & w - \eta \nabla L(w) \\ 1 & \cdots & 1 \end{bmatrix} \right\|_\infty \leq \epsilon,$$

where η denotes the learning rate and $L(w) := \sum_{i=1}^n l(w^\top x_i, y_i)$ is an empirical loss upon the given input-output pairs.

Proof. Please see Section C for a proof sketch and Section F.10 for a detailed proof. $\square$

We note that in the original proof of (Bai et al., 2024), they rely on the approximation ability of ReLU neural networks to approximate the derivative of the loss function. Therefore, they use ReLU-based attention to approximate a sum of ReLU functions. In contrast, by leveraging Theorem 3.1, we show softmax attention approximates generalized ReLU function by approximating truncated linear models, and hence approximates in-context gradient descent. Since softmax attention is the dominant mechanism used in practice, our results provide a more realistic foundation for understanding in-context learning tasks. In Section C, we provide two research directions inspired by Theorem 4.1.

5 Discussion and Conclusion

We establish the universal approximation theory of simple softmax attention layer for any continuous sequence-to-sequence function on a compact domain. Our key technique is to cast attention as softmax-based selection mechanism of the interpolation points in the output domain (Remark 3.1). This enables softmax attentions with simple linear transform to approximate the generalized ReLUs (and hence many known universal approximators) in a token-wise manner (Theorems 3.1 and 3.2). Based on this, we derive the universal approximation theory for sequence-to-sequence functions using (i) two softmax-attention layers (Theorem 3.3) or (ii) one softmax-attention layer followed by a softmax function (Corollary 3.3.1). We also extend our results to in-context learning (Section 4).

Connecting to Practical Attention/Transformer. Beyond such isolated analysis (i.e., attention-only), our results also extend to practical transformers in Section H. We remark

that our sequence-to-sequence universal approximation in Section 3.3 uses the single-head result of Theorem 3.1 for simplicity of presentation. The same proofs hold if we replace it with the multi-head result of Theorem 3.2. That is, the two theorems are interchangeable for establishing universal sequence-to-sequence approximation. The main differences lie in that the A mapping is a sequence-wise linear operation in Theorem 3.1, while A is an ordinary token-wise linear transform in Theorem 3.2. The construction of A in Theorem 3.2 aligns with a practical transformer/attention. It is a token-wise linear layer augmented with positional encoding like an ordinary embedding layer. We further derive the sequence-to-sequence universal approximation result based on Theorem 3.2 explicitly in Section H.

Limitation and Future Work. Our in-context learning results only simulate gradient descent and do not establish in-context universal approximation. Also we only study the architectural capabilities of softmax attention, i.e., expressiveness. We do not comment on their optimization and generalization. We leave them for future work.

[Note Added after Acceptance]: After the original arXiv posting of this paper, several followups by the authors pursued directions suggested by this paper: Transformer in-context algorithm emulation and prompt-programmable computations beyond gradient descent (Hu et al., 2026a); in-context gradient descent via chain-of-thought (Chen et al., 2026a); in-context universal approximation and compositional generalization (Chen et al., 2026b); and translations of ReLU target-specific approximation results to softmax-attention Transformers (Hu et al., 2026b). The results of this paper do not depend on these later developments.

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Impact Statement

By the theoretical nature of this work, we do not anticipate any negative social impact.

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A Table of Notation

Table 1. Notations and Symbols

Symbol	Description
d	Input dimension of each sequence element
n	Input sequence length
H	Number of attention heads
p	Number of interpolation anchors ($p > n$)
d_o	Output (value) dimension per head
$X = [x_1, \dots, x_n]$	Input sequence matrix in $\mathbb{R}^{d \times n}$
x_i	i -th token (column) of X
w_i, t_i	Weight and bias for the i -th truncated linear model
f	Target continuous function being approximated
$\mathcal{X}$	Compact domain of inputs considered
$\text{Range}[a, b](\cdot)$	Truncated linear (generalized ReLU) with range $[a, b]$
a, b	Lower and upper bounds of truncation (with $a < b$)
$\tilde{L}_k$	k -th uniformly spaced interpolation anchor in $[a, b]$
ΔL	Anchor spacing: $\tilde{L}_k - \tilde{L}_{k-1}$
k	Interpolation index, $k \in [p]$
$\tilde{k}$	Row index of interpolation point $\tilde{L}_k$ in the output space, $\tilde{k} \in [d_o]$
$G(\cdot)$	Mapping $G : [p] \rightarrow [d_o]$ with $\tilde{k} = G(k)$
k_i	Index of anchor closest to $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$
$\tilde{k}_i$	i -th token's row index of the chosen anchor in the output space
e_k	One-hot vector with 1 in position k
β	Inverse temperature in softmax
δ	Gap between the two largest input entries of softmax
γ	Gap between the first largest and third largest input entries of softmax
ϵ	Desired overall approximation error
ϵ_0	Softmax approximation error

B Experimental Studies

In this section, we provide proof-of-concept numerical experiments to back up our theoretical results. We divide our experiments into the following two objectives.

- **Objective 1: Validating the Proposed Interpolation Selection Scheme (Theorem 3.1 and Theorem 3.2).** We aim to verify the theoretical approximation rates (Figure 2): $O(1/p)$ with respect to the number of interpolation points p , linear scaling in the interval length $|b - a|$, and $O(1/H)$ in terms of the number of heads H for multi-head attention. Furthermore, we print out the attention weights to determine that each column of $\text{Softmax}(K^\top Q)$ becomes close to one-hot indicators selecting interpolation points (Figure 3).
- **Objective 2: Sequence-to-Sequence Approximation (Theorem 3.3).** We create synthetic data for a sequence-to-sequence task to verify that the approximation rate is again $O(1/p)$ and $O(1/H)$. We use two-layer ReLU network with flatten input $X_{\text{flatten}} \in \mathbb{R}^{dn}$ to mix information from all dn input dimensions. In this formulation, the output token at each position depends on the entire input sequence. The result in Figure 4 shows that it aligns with the theoretical result.

B.1 Validating the $O(1/p)$ and $O(1/H)$ Approximation Rates

Model Architecture. We train single/multi-head single-layer softmax attention to model truncated linear model, with an extra linear layer A applied on the input and create $p - n$ extra empty tokens. For the attention weight experiments, we guide the model by encoding the interpolation points in the last row of the key matrix K as in the proof of Theorem 3.1. We also encode the interpolation point onto random column indices in the value matrix V as $\tilde{k}_i$ in the theorem.

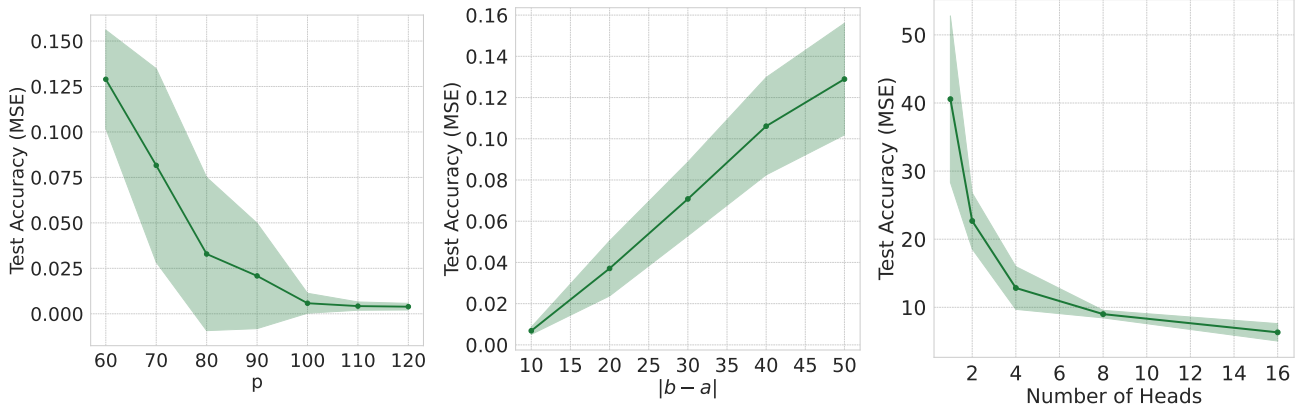


Figure 2. Ablation Study for Three Key Parameters in Our One-Layer Attention (Theorem 3.1 and Theorem 3.2). All the results align with the theoretical analysis that the approximation error scales as $O(1/p)$, $O(1/H)$, and grows linearly in $|b-a|$. We report test accuracy (MSE) as the mean and one standard deviation (shaded region) over 10 random seed runs. The synthetic dataset consists of 1000 samples with a 80/20 train-test split. All other hyperparameters remain fixed for three experiments ($d = 10$, $n = 50$, hidden dimension = 32, learning rate = 0.001, epoch = 50 and batch size = 32). The experiments are run on an NVIDIA A100 GPU.

Data Generation. For the experiments of the truncated linear model, we represent each sample as $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$ and with ground truth have each label y_i encoded on the first rows of the matrix $Y \in \mathbb{R}^{d \times n} := [y_1 e_1, y_2 e_1, \dots, y_n e_1]$. For $i \in [n]$, we first fix a truncated linear model by sampling a weight vector $w_i \sim N(0, I_d)$ and bias $t_i \sim N(0, 1)$. These token-specific parameters are fixed for all samples (in one run). Then for each sample X , we draw every token $x_i \sim \text{Uniform}(-5, 5)$, and compute the label y_i as

$$y_i = \text{Range}_{[a,b]}(w_i^\top x_i + t_i).$$

We generate N samples using this process, we set $N = 1000$ and train for 50 epochs. For attention weight experiments, the difference is that we encode the ground truth y_i onto the same random column indices as described in the model architecture paragraph.

Metrics. We train the model with the following Mean Squared Error (MSE) loss

$$\mathcal{L}_{\text{MSE}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where $\hat{y}_i \in \mathbb{R}^d$ is the prediction of the attention layer.

Results. We present our findings in Figure 2 and Figure 3.

- **Approximation Performance.** Figure 2 shows that the MSE follows the theoretical approximation rates. It decreases as $O(1/p)$ when we increase the number of interpolation points p . It also scales linearly with the truncation interval $|b-a|$ and behaves as $O(1/H)$ with H heads in multi-head attention. Increasing p not only reduces the approximation error but also stabilizes training, as indicated by the smaller standard deviations across 10 runs.
- **Attention Heatmaps.** Figure 3 confirms the “one-hot” interpolation selection phenomenon. For a small truncated range $a = -0.5, b = 0.5$, most ground truths of y lie at $\tilde{L}_0$ and $\tilde{L}p - 1$. As the figure shows, each token x_i (query index) puts most of its attention on the 0-th and 29-th keys (the interpolation points). When we increase $|b-a|$, the attention weight spreads across more key indices.

B.2 Sequence-to-Sequence Approximation Rates

Model Architecture. We train a small model with 2-layer multi-head attention with linear mapping A_1 and A_2 as in Theorem 3.3. For the experiment of number of interpolation points, we set $H = 2$ to speed up the experiment. All the parameters are randomly initialized instead of hard-set to the form of weight in the proof.

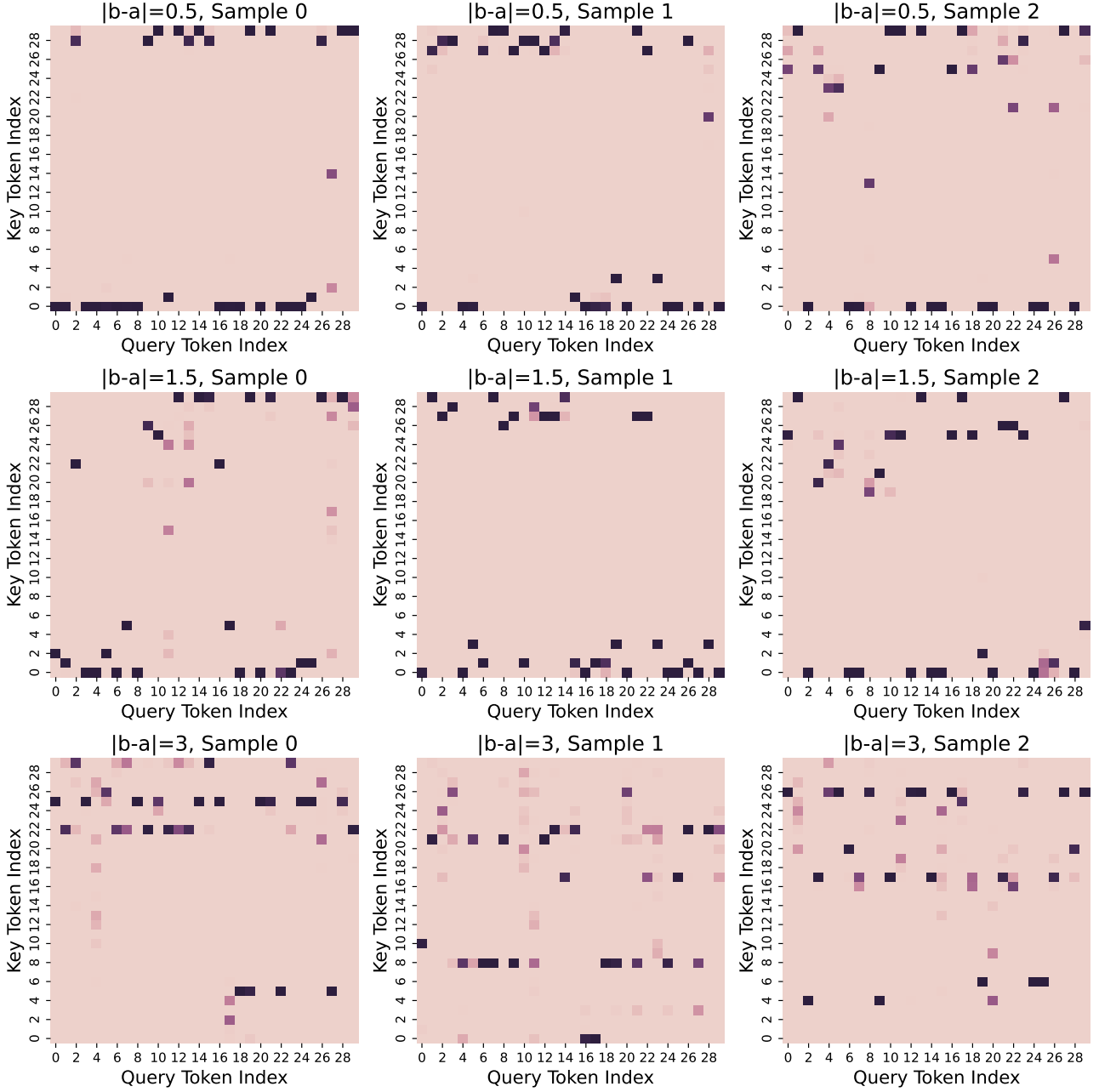


Figure 3. Attention Heatmap for $|b-a| = 0.5, 1.5, 3$. The figure shows the attention heatmap $\text{Softmax}(K^\top Q)$ ((F.5)) for 3 random test samples with parameters $p, n = 30$. In particular, for smaller truncation intervals ($|b-a| = 0.5$), the attention distribution concentrates on boundary interpolation points, as our theoretical analysis anticipates. When expanding the truncation interval width, the attention weights transition to selecting intermediate interpolation points. We set the hyperparameters to 100 epochs, learning rate = 0.001, batch size 32, hidden dimension = 10, $\beta = 30$, and random seed = 1234.

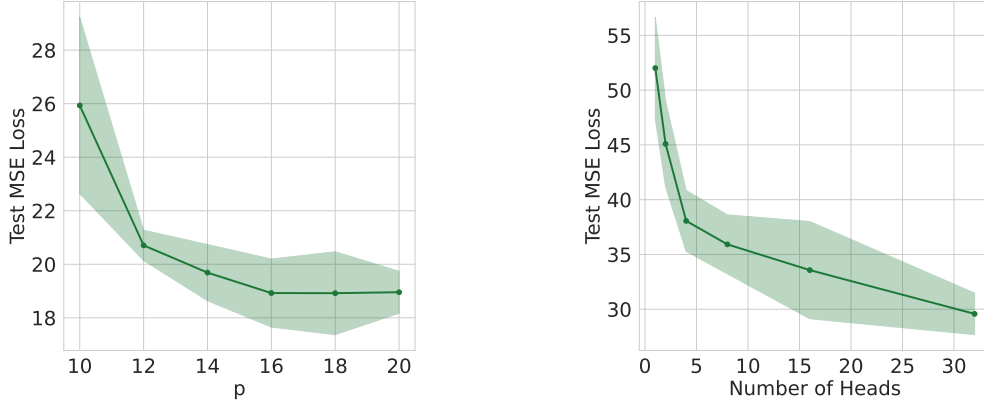


Figure 4. Ablation Study for Two Parameters p and H in Theorem 3.3. The results align with our theoretical approximation rate as $O(1/p)$ and $O(1/H)$. We report test accuracy (MSE) as the mean and one standard deviation (shaded region) over 10 random seed runs. The synthetic dataset consists of 50000 data points with $d = 5$ and a 80/20 train-test split. For both experiments, we set the learning rate = 0.001, epoch = 3, and batch size = 32. For the number of interpolation points p experiment (left figure), to speed up the training process, we set $n = 10$ and the hidden dimension of the model to be 16. For number of heads experiment (right figure), we increase the sequence length to $n = 20$ to make the task harder so we can see the trend when increasing the number of heads, and also increase the hidden dimension to 32 so it can be divided by $H = 32$. The experiments are run on an NVIDIA A100 GPU.

Data Generation. Same as Section B.1, each sample is in the form of $X = [x_1, \dots, x_n] \in \mathbb{R}^{d \times n}$, and we generate each x_i from a uniform distribution $\text{Uniform}(0, 1)$. We generate targets via a global sequence-to-sequence mapping. Concretely, for each sample X , we first flatten it into a vector $X_{\text{flatten}} \in \mathbb{R}^{dn}$ and then pass it through a two-layer ReLU network:

$$Y_{\text{flatten}} = W_2 \text{ReLU}(W_1 X_{\text{flatten}} + b_1) + b_2,$$

where $W_1 \in \mathbb{R}^{m \times dn}$, $b_1 \in \mathbb{R}^m$ are the weights and bias of the hidden layer, and $W_2 \in \mathbb{R}^{dn \times m}$, $b_2 \in \mathbb{R}^{dn}$ are those of the output layer. Finally, Y_{flatten} is reshaped into a sequence in $Y = [y_1, \dots, y_n] \in \mathbb{R}^{d \times n}$. The above data generation ensures that each output token $y_i \in \mathbb{R}^d$ is a function of the entire input sequence. In our experiments, we generate $N = 50000$ samples and again use a 80/20 train-test split. The input dimension and hidden dimension of the ReLU network to generate synthetic data are $d = 5$ and $m = 10$.

Metrics. We use Mean Squared Error (MSE) loss

$$\mathcal{L}_{\text{MSE}} = \sum_{i=1}^n (y_i - \hat{y}_i)^2,$$

where $\hat{y}_i \in \mathbb{R}^d$ is the prediction of the attention layer.

Results. As shown in Figure 4, the approximation rate is again in the trend of $O(1/p)$ and $O(1/H)$. This result validates the theoretical analysis for sequence-to-sequence approximation of the attention-only layer considered in Theorem 3.3. The decrease in MSE error indicates this small model (2-layer and hidden dimension = 16) captures the global dependencies in the sequence-to-sequence task.

In summary, our empirical observations confirm that (i) attention emulates truncated linear model via “selection” of key-value pairs, and (ii) a straightforward attention-only network learns nontrivial sequence-to-sequence dependence. Both align with the approximation rate provided by our theoretical results.

C Detailed Results and Discussion of In-Context Learning

Here we provide an application to showcase the generality of our theory and techniques.

C.1 Attention Approximates Truncated Linear Models In-Context

We extend the interpolation selection technique and [Theorem 3.1](#) to the in-context learning setting ([Brown et al., 2020](#); [Bai et al., 2024](#)). The next theorem shows that when the length- n input includes weights $\{w_i\}_{i \in [n]}$ and biases $\{t_i\}_{i \in [n]}$, attention is able to approximate n truncated linear models $\{\text{Range}_{[a,b]}(w_i^\top x_i + t_i)\}_{i \in [n]}$ in-context.

Theorem C.1 (Attention Approximates Truncated Linear Models In-Context). Fix real numbers $a < b$, and let the truncation operator $\text{Range}_{[a,b]}(\cdot)$ follow [Definition 3.1](#). Let the input be

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ w & w & \cdots & w \\ t_1 & t_2 & \cdots & t_n \end{bmatrix} \in \mathbb{R}^{(2d+1) \times n},$$

where $\{w, x_i\}_{i \in [n]}$ are bounded. Let $\epsilon_0 \geq 0$. For a precision parameter $p > n$, there exists a single-layer, single-head self-attention Attn with a linear transformation $A : \mathbb{R}^{(2d+1) \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$, such that $\text{Attn} \circ A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d_o \times n}$ satisfies, for any $i \in [n]$,

$$\|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}^\top\|_\infty \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}},$$

each w_i is a elementwise multiplication of w by a vector v_i . Here $e_{\tilde{k}_i}^\top$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \underset{k \in [p]}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k| \quad \text{where} \quad \tilde{k}_i := G(k_i) \in [d_o].$$

Here $k_i \in [p]$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : [p] \rightarrow [d_o]$ denotes any set-to-set function sending the interpolation index $k \in [p]$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output. By setting $\beta \geq \delta^{-1} \ln \frac{n-1}{\epsilon_0}$, we make ϵ_0 arbitrarily small, though the theorem fails on a arbitrarily small volume in $\mathbb{R}^{d \times n}$. When $\tilde{k}_i$ a constant for all $i \in [n]$, by setting $\beta \geq \frac{2}{(\Delta L)^2} \ln \frac{p-2}{\epsilon_0}$, we achieve arbitrary small ϵ_0 without any failure region.

Proof Sketch. Applying our interpolation method in the in-context learning setting, we compute the interpolation point closest to the linear transformation result within attention. This involves comparing $|w^\top x - \tilde{L}_k|_2^2$ for $k \in [p]$. Consequently, attention must compute a *quartic* polynomial of the input, while standard transformers only produce *quadratic* expressions. To combat this, we propose a technique enabling attention to perform equivalent computations for higher-order polynomials in our setting. Please see [Remark F.8](#) and [Section F.9](#) for a detailed proof. $\square$

C.2 In-Context Gradient Descent

We extend [Theorem C.1](#) to show that standard softmax attention perform in-context gradient descent, broadening the results established for ReLU attention in ([Bai et al., 2024](#)). Specifically, we demonstrate that softmax attention is capable of doing in-context gradient descent on convex loss functions.

We first define the problem setting similar to theirs.

Definition C.1 (In-Context Learning Problem Formulation). The sequential input X in the in-context learning scenario is defined as

$$X := \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n \\ w & w & \cdots & w \\ 1 & 1 & \cdots & 1 \end{bmatrix},$$

where (x_i, y_i) , $i \in [n]$ denote the input-output pairs. w parametrize the model connecting x_i and y_i , and is altered(trained) between layers.

Remark C.1. The task of in-context learning is simplified to using the given input-output pairs (x_i, y_i) to predict the output of a new input x_u .

In this setting, we prove a multi-head Softmax attention is capable of doing in-context gradient descent on loss functions parametrized by $w^\top x_i$ ($i \in [n]$) and t (as linear coefficient and bias), as well as giving an according prediction to the output on x_u .

Theorem C.2 (In-Context Gradient Descent). Let $l : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be any C^1 loss function defined on $(w^\top x_i, y_i)$. With input X in the form of [Definition 4.1](#), when X is bounded, there exists a multi-head self-attention Attn_m with skip connections and each attached with a linear layer, such that for any $\epsilon > 0$, irrelevant of X , we have

$$\left\| \text{Attn}_m \circ A(X) - \begin{bmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \\ w - \eta \nabla L(w) & \cdots & w - \eta \nabla L(w) \\ 1 & \cdots & 1 \end{bmatrix} \right\|_\infty \leq \epsilon,$$

where η denotes the learning rate and $L(w) := \sum_{i=1}^n l(w^\top x_i, y_i)$ is an empirical loss upon the given input-output pairs.

Proof Sketch. We know the universal approximation theorem for ReLU neural networks ([Pinkus, 1999](#)) ensures that there exists ReLU network $\sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)})$ approximate r -th coordinate of the derivative of loss function on token x_i as $\frac{\partial}{\partial w} l(w^\top x_i, y_i) \in \mathbb{R}^d$ for $r \in [d]$. By [Theorem C.1](#), we construct multi-head attention with linear mapping $\sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r} \circ A_{h,r}(\cdot)$ to approximate the above ReLU neural network on every coordinate r , hence also approximate $\frac{\partial}{\partial w} l(w^\top x_i, y_i) \in \mathbb{R}^d$ for $r \in [d]$. By designing W_O^* we sum up the derivative of loss function $\frac{\partial}{\partial w} l(w^\top x_i, y_i)$ on different in-context example (x_i, y_i) , hence approximate $\nabla L(w)$. Please see [Section F.10](#) for a detailed proof. $\square$

We note that in the original proof of ([Bai et al., 2024](#)), they also rely on the approximation ability of ReLU neural networks to approximate the derivative of the loss function. Therefore, they use ReLU-based attention in their proof to approximate a sum of ReLU functions. In contrast, by leveraging [Theorem 3.1](#), we show that softmax attention approximates generalized ReLU function by approximating truncated linear models, and hence approximates in-context gradient descent. Since softmax attention is the dominant mechanism used in practice, our results provide a more realistic foundation for understanding in-context learning tasks.

Beyond this, [Theorem C.2](#) also suggest two advanced future works.

- **Task Composition.** Our construction naturally extends to task composition from subtasks. Suppose we have N subtasks (e.g., gradient descent, lasso, linear regression, etc.). [Theorem C.2](#) imply there exists an attention layer to approximate this task in-context (such as Attn_{GD} for the gradient-descent subtask). Our universal approximation results allow a *frozen* attention module to approximate these task-specific attention maps in-context, so that a single attention module realize all N subtasks from input-output examples. We also remark that this step is not trivial. On top of this frozen layer, we introduce additional “routing” attention layer to select and compose these subtasks into the task of our interest in-context. This way extends our techniques to meta-learning or task composition naturally.
- **Simulation of Learning Algorithms In-Context.** By stacking [Theorem C.2](#), softmax attention simulate multi-step in-context gradient descent and thereby recover a wide range of learning algorithms. Specifically, our [Theorem C.2](#) shows that a single softmax-attention layer implement one step of gradient descent for any C^1 loss of the form $\ell(w^\top x_i, y_i)$. This contains a wide range of loss functions, including ridge, GLM and lasso loss functions. By stacking copies of this layer, standard convergence results for gradient descent (as in Lemma 14, Lemma C.1, Proposition A.2, and Proposition A.3 of ([Bai et al., 2024](#))) imply that a depth- T transformer approximate the T -step in-context learning dynamics for these algorithms.

We view these two directions as promising next steps for developing a more systematic theory of task composition and algorithm learning in transformers.

D Related Work

Universal Approximation of Transformer. We first introduce the most relevant previous works about the universal approximation ability of transformer, and move on to other works investigating the expressive power of transformer with different target function classes.

[Yun et al. \(2020a\)](#); [Kajitsuka & Sato \(2024\)](#) treat attention layer as the contextual mappings and derive that attention layer attached with FFNs is a universal approximator on continuous sequence-to-sequence permutation equivariant function. Specifically, [Yun et al. \(2020a\)](#) prove that multi-head attention with two-layer FFN approximate continuous permutation equivariance sequence-to-sequence functions on a compact domain. Their construction of transformer block maintains constant width but requires $O(n(1/\delta)^{dn}/n!)$ layers, where δ is the fixed grid width of the input domain and n is the sequence length. For any continuous sequence-to-sequence function, removing the factorial term $n!$ in the denominator leaves the remaining term growing exponentially with n . [Kajitsuka & Sato \(2024\)](#) further show that one-layer and single-head attention, with low-rank weight matrices, is able to carry out contextual mapping, simplifying the construction in terms of the number of layers. [Takakura & Suzuki \(2023\)](#) prove that one-layer transformer (attention + token-wise FFN) with one embedding layer (with positional encoding) approximates shift-equivariant α -smoothness function. They show that the approximation error of the above function class is independent of input and output dimension hence achieving an infinite-dimension approximation result. Their result require $O(\log(1/\epsilon)^{1/\alpha})$ number of heads. [Jiang & Li \(2024\)](#) use Kolmogorov representation theorem to get Jackson-type approximation rate, which hinges on explicit smoothness assumptions that yield quantitative convergence rates for single-layer single-head transformer. Despite these advances, prior works proving the universality of Transformers often depend on the FFNs attached after attention to perform token-wise transformations. Our work is different from these papers by removing the need for FFN from transformer to demonstrate the first universal approximation result of attention mechanism. Our construction requires 2-layer multi-head attention with linear transformation, with head complexity $H = O(d(1/\delta)^{dn})$. See [Remark D.1](#) for detailed discussion.

Several other works investigate the universal approximation theorem of transformer with different variants. [Yun et al. \(2020b\)](#) prove that universal approximation of sparse transformer. [Kratsios et al. \(2022\)](#) prove the constrained universal approximation theorem of probabilistic transformer. [Likhoshesterov et al. \(2023\)](#) and [Edelman et al. \(2022\)](#) demonstrate that a single-layer self-attention mechanism is able to learn sparse functions of the input sequence, with sample complexity and hidden size are logarithmic relative to the sequence length. For the representation power on matrix, [Bhojanapalli et al. \(2020\)](#) show that when the hidden dimension of attention is smaller than sequence, multi-head attention cannot output certain positive column-stochastic matrices, and [Likhoshesterov et al. \(2023\)](#) show that self-attention approximates any sparse matrices. Other works investigate the universal approximation of in-context learning setting. [Furuya et al. \(2025\)](#) shows a deep transformer block approximating any continuous mapping from an arbitrary-length prompt to its next token. [Li et al. \(2025\)](#) characterize a broad family of functions $f : \mathbb{R}^d \rightarrow \mathbb{R}$ that admit a sparse expansion in a fixed finite feature basis, and show that a transformer with sigmoid activation attention simulates a Lasso objective to recover those coefficients from the in-context examples. In contrast, our universal approximation result targets standard sequence-to-sequence functions outside the ICL framework and removes FFN, establishing universality for the softmax attention mechanism alone.

Interpolation Methods for Universal Approximations. We also summarize and discuss prior works that utilize interpolation-based approaches to establish universal approximation theory:

- [Kratsios \(2023\)](#) build a “Probabilistic Transformer” that approximates regular conditional distributions by combining a feed-forward network with an attention-based final layer. They use softmax weights to form convex combinations of “anchor” distributions. Our work also relies on anchor selection via softmax, but we focus on deterministic sequence-to-sequence tasks rather than mapping to probability measures.
- [Shen et al. \(2022\)](#) prove that ReLU MLPs can achieve optimal approximation rates by partitioning the domain into dyadic grids. This is a classic piecewise-linear interpolation strategy. Our approach shares the same interpolation philosophy but implements it through attention (without requiring deep ReLU layers).
- [Galimberti \(2025\)](#) address inputs from non-metric or infinite-dimensional spaces by projecting them onto finite “anchors” and then applying MLP-like operators. This resembles our anchor-based selection, though we rely on standard self-attention rather than specialized infinite-dimensional layers.
- [Kratsios & Furuya \(2025\)](#) show that MLPs with trainable activations are universal in-context learners. They construct Voronoi partitions of the context space. Our work also employs anchor-driven interpolation for in-context tasks, but via attention-based selection rather than partitioning MLPs.

- [Furuya et al. \(2025\)](#) demonstrate that standard Transformers, with multi-head softmax attention and feed-forward layers, are universal in-context learners for arbitrary-size contexts. We focus on attention-alone universality. Our proofs show that even a single or two-layer softmax-attention mechanism can learn continuous sequence-to-sequence or in-context functions.

All these works use interpolation as a unifying theme, either via MLPs, attention, or trainable activations. Some rely on argmax (hard selection), others on softmax (continuous weighting). We extend this line by showing that a minimal attention-only setup suffices for universal approximation. Our method needs no deep stacks or feed-forward blocks, and it extends naturally to in-context learning.

Remark D.1 (Head Complexity and Parameters Complexity). In [Section F.6](#), we use $2dg^{dn} + 1$ heads to achieve sequence-to-scalar universal approximation. The first term is because we need $2d$ heads per grid point and there are $|G_D| = g^{dn}$ grid points, see [\(F.44\)](#). Also note that $p = |G_D|$ in those attentions. The extra one head in second term select $f(v)$, see [\(F.48\)](#). For sequence-to-sequence universal approximation, the first layer is the same as multi-head attention that quantizes the input as in [Section F.6](#), but increase the selection heads to dn . The total number of heads is therefore $2dg^{dn} + dn$. If we represent the head complexity with grid width $\delta_{\text{grid}} := 2B/g$ as in [Yun et al. \(2020a\)](#), the head complexity satisfies $H = O(d(1/\delta_{\text{grid}})^{dn})$.

Next we derive the parameters complexity. In the multi-head construction [\(F.45\)](#) for grid-point approximation and in the single-head selection attention [\(F.48\)](#), each head has constant dimension, so the parameters of each head are $O(1)$. The linear maps A_0 and A^* in the proof of [Lemma F.3](#) (and A_1 in [Lemma 3.2](#)) are shared by all $2dg^{dn}$ heads and do not scale with the number of heads. The only component scale with the $2dg^{dn}$ head is W_0 . That one contribute to $O(n g^{dn})$ parameters per head, hence $O(dn g^{2dn})$ parameters in total. For dn single-head attention Attn_s^{ij} in sequence-to-sequence approximation [\(F.55\)](#), each head has dimension $O(dn)$ and contributes $O((dn)^2)$ parameters. The map A_2 is shared by all dn single-head Attn_s^{ij} , so its parameter count does not scale with dn , but its dimension is $O(dn g^{dn})$. Hence the total parameters complexity is $O(dn g^{2dn} + (dn)^2) = O(dn(1/\delta_{\text{grid}}^2)^{dn})$. For [Theorem H.1](#), the sequence-to-sequence approximation result based on [Theorem 3.2](#), the head complexity is $H = O(N)$, where N is the number of neuron of ReLU feed-forward network it aim to approximate. By classical FFN universal approximation results ([Pinkus, 1999](#)) we have $N = O((1/\delta_{\text{grid}})^{dn})$, hence $H = O((1/\delta_{\text{grid}})^{dn})$. The parameters complexity is dominated by the first $O(N(d^2 + dn))$ and the third layer $O(n^2 N^2)$ of attention, and it's clear $O(n^2 N^2) = O(n^2(1/\delta_{\text{grid}}^2)^{dn})$ dominate since $N \gg d$. Finally, as noted also in [Yun et al. \(2020a\)](#), this exponential dependence cannot be avoided when approximating arbitrary continuous functions, since in the worst case the model must memorize an independent output for each of the g^{dn} grid.

Remark D.2. One may reduce the parameter complexity above by:

- Replacing or improving the sequence-wise linear transformation, for example by using an additional attention layer to replace the sequence-wise layer.
- Only calculating the learnable parameters (e.g., w, t) attribute to universal approximation ability. There is large amount of entries in the constructed matrices are fixed as 0 and 1. One can also optimize the constructive proof to optimize the use of those entries.

We leave this direction to future work.

E Additional Theoretical Results

E.1 Approximating Hardmax with Finite Temperature Softmax

A central step in our proofs is to replace a hard arg max operation with a continuous softmax using a sufficiently large inverse temperature β . Intuitively, as $\beta \rightarrow \infty$, the softmax output approaches a one-hot vector that selects the largest entry of x . The following lemma provides a precise bound on how large β must be to achieve a desired approximation error.

Lemma E.1 (Approximating Hardmax with Finite-Temperature Softmax). Let $x = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$, $\epsilon > 0$. Define $\text{Softmax}_\beta(\cdot)$ as

$$\text{Softmax}_\beta(x) := \left[\frac{\exp(\beta x_1)}{\sum_{j=1}^n \exp(\beta x_j)}, \dots, \frac{\exp(\beta x_n)}{\sum_{j=1}^n \exp(\beta x_j)} \right].$$

The following statements hold:

- **Case of a Unique Largest Entry.** Assume $x_1 = \max_{i \in [n]} x_i$ is unique, and $x_2 = \max_{i \in [n] \setminus \{1\}} x_i$. Then, if $\beta \geq (\ln(n-1) - \ln(\epsilon))/(x_1 - x_2)$, we have

$$\left\| \text{Softmax}_\beta(x) - e_1 \right\|_\infty \leq \epsilon,$$

where $e_1 \in \mathbb{R}^n$ is the one-hot vector corresponding to the maximal entry of x (i.e., x_1 .)

- **Case of Two Largest Entries (Tied or Separated by δ).** Assume x_1 and x_2 are the first and second largest entries, respectively, with $\delta = x_1 - x_2 \geq 0$. Let x_3 be the third largest entry and is smaller than x_1 by a constant $\gamma > 0$ irrelevant to the input. Then, if $\beta \geq (\ln(n-2) - \ln \epsilon)/\gamma$, we have

$$\left\| \text{Softmax}_\beta(x) - \frac{1}{1 + e^{-\beta\delta}} e_1 - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} e_2 \right\|_\infty \leq \epsilon.$$

Proof. Please see [Section F.1](#) for a detailed proof. □

E.2 Sequence-to-Sequence Universal Approximation with ℓ_∞ -Error

Here, we present the result that a two-layer multi-head attention mechanism achieves sequence-to-sequence universal approximation with respect to the ∞ -norm error.

We refine our sequence-to-sequence approximation result from [Theorem 3.3](#) to an ∞ -norm guarantee in [Theorem E.1](#). We achieve this by combining the existing ReLU neural networks approximation result in ∞ -norm with our attention-approximate-generalized-ReLU result from [Theorem 3.1](#).

Theorem E.1 (Sequence-to-Sequence Approximation in Infinity Norm). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when attached with linear transformations, there exists a one layer multi-head attention Attn_m stacked with one layer multi-head attention Attn_m , such that when the precision parameter in [Theorem F.5](#) is $p = \Omega(n^{5/2})$, for any $X \in \mathcal{X}$

$$\|f(X) - \text{Attn}_m^{(2)} \circ A \circ \text{Attn}_m^{(1)} \circ A(X)\|_\infty \leq \epsilon.$$

Proof. Please see [Section F.8](#) for a detailed proof. □

F Proofs of Main Text

Throughout this paper, we use p interpolation anchors indexed by $k \in [p]$ to interpolate interval $[a, b]$:

$$L_k := a + \frac{k-1}{p-1}(b-a), \quad k \in [p], \quad \Delta L := \frac{b-a}{p-1}.$$

We also write

$$m_k := k - 1,$$

so that

$$L_k - L_1 = m_k \Delta L.$$

For token $i \in [n]$, write

$$z_i := w_i^\top x_i + t_i.$$

F.1 Proof of Lemma E.1

Lemma F.1 (Lemma E.1 Restated: Approximating Hardmax with Finite-Temperature Softmax). Let $x = [x_1, x_2, \dots, x_n] \in \mathbb{R}^n$, $\epsilon > 0$. Define $\text{Softmax}_\beta(\cdot)$ as

$$\text{Softmax}_\beta(x) := \left[\frac{\exp(\beta x_1)}{\sum_{j=1}^n \exp(\beta x_j)}, \dots, \frac{\exp(\beta x_n)}{\sum_{j=1}^n \exp(\beta x_j)} \right].$$

The following statements hold:

- **Case of a Unique Largest Entry.** Assume $x_1 = \max_{i \in [n]} x_i$ is unique, and $x_2 = \max_{i \in [n] \setminus \{1\}} x_i$. Then, if $\beta \geq (\ln(n-1) - \ln(\epsilon)) / (x_1 - x_2)$, we have

$$\left\| \text{Softmax}_\beta(x) - e_1 \right\|_\infty \leq \epsilon,$$

where $e_1 \in \mathbb{R}^n$ is the one-hot vector corresponding to the maximal entry of x (i.e., x_1).

- **Case of Two Largest Entries (Tied or Separated by δ).** Assume x_1 and x_2 are the first and second largest entries, respectively, with $\delta = x_1 - x_2 \geq 0$. Let x_3 be the third largest entry and is smaller than x_1 by a constant $\gamma > 0$ irrelevant to the input. Then, if $\beta \geq (\ln(n-2) - \ln(\epsilon)) / \gamma$, we have

$$\left\| \text{Softmax}_\beta(x) - \frac{1}{1 + e^{-\beta\delta}} e_1 - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} e_2 \right\|_\infty \leq \epsilon.$$

Proof. In the following proof, we denote $\text{Softmax}(\cdot)$ function as $\sigma(\cdot)$ for simplicity.

For the first condition that x with unique maximal entry x_1 , denote $\exp(\beta x_i) / \sum_{j=1}^n \exp(\beta x_j)$ as $\sigma_\beta(x)_i$. we have:

$$\begin{aligned} & \left\| \sigma_\beta(x_1, x_2, x_3, \dots, x_n)_1 - e_1 \right\|_\infty \\ &= \max\{1 - \sigma(x)_1, \sigma(x)_2, \dots, \sigma(x)_n\} \\ &= \max\{1 - \sigma(x)_1, 1 - \sigma(x)_1 - \sum_{i \neq 1,2} \sigma(x)_i, \dots, 1 - \sigma(x)_1 - \sum_{i \neq 1,n} \sigma(x)_i\} \quad (\text{By } \sum_{i=1}^n \sigma(x)_i = 1) \\ &\leq 1 - \sigma(x)_1 \\ &= 1 - \frac{1}{1 + \sum_{j=2}^n e^{\beta(x_j - x_1)}} \quad (\text{By dividing } \sigma(x)_1 \text{ by } e^{\beta x_1}) \end{aligned}$$

$$\begin{aligned}
 &= \frac{\sum_{j=2}^n e^{\beta(x_j - x_1)}}{1 + \sum_{j=2}^n e^{\beta(x_j - x_1)}} \\
 &\leq \sum_{j=2}^n e^{\beta(x_j - x_1)} \\
 &\leq (n-1)e^{\beta(x_2 - x_1)} \quad (\text{Since } x_2 \text{ is the second largest entry}) \\
 &\leq \epsilon.
 \end{aligned}$$

For the second occasion, we have:

$$\begin{aligned}
 &\left\| \sigma_\beta(x) - \frac{1}{1 + e^{-\beta\delta}} e_1 - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} e_2 \right\|_\infty \\
 &\leq \max\left\{ \frac{1}{1 + e^{-\beta\delta}} - \sigma_\beta(x)_1, \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} - \sigma_\beta(x)_2, \sigma_\beta(x)_3, \dots, \sigma_\beta(x)_n \right\}, \quad (\text{F.1})
 \end{aligned}$$

where the last inequality comes from the definition of infinity norm. Plug in $\delta = x_1 - x_2$ we calculate the first two term to be:

$$\begin{aligned}
 \frac{1}{1 + e^{-\beta\delta}} - \sigma_\beta(x)_1 &= \frac{1}{1 + e^{\beta(x_2 - x_1)}} - \frac{e^{\beta x_1}}{\sum_{i=1}^n e^{\beta x_i}} \\
 &= e^{\beta x_1} \left(\frac{1}{e^{\beta x_1} + e^{\beta x_2}} - \frac{1}{\sum_{i=1}^n e^{\beta x_i}} \right) \quad \left(\frac{1}{1 + e^{\beta(x_2 - x_1)}} = \frac{e^{\beta x_1}}{e^{\beta x_1} + e^{\beta x_2}} \right) \\
 &= e^{\beta x_1} \left(\frac{\sum_{i=1}^n e^{\beta x_i} - (e^{\beta x_1} + e^{\beta x_2})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})} \right) = \frac{e^{\beta x_1} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})}.
 \end{aligned}$$

Follows the same calculation we get

$$\frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} - \sigma_\beta(x)_2 = \frac{e^{\beta x_2} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})}.$$

Hence we have

$$\begin{aligned}
 &\max\left\{ \frac{e^{\beta x_1} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})}, \frac{e^{\beta x_2} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})} \right\} \\
 &\leq \frac{e^{\beta x_1} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2})(\sum_{i=1}^n e^{\beta x_i})} \quad (x_1 \geq x_2 \text{ by assumption}) \\
 &\leq \sum_{i=3}^n e^{\beta(x_i - x_1)} \\
 &\leq (n-2) \cdot e^{\beta(x_3 - x_1)} \\
 &\leq \epsilon. \quad (\text{F.2})
 \end{aligned}$$

Furthermore we have

$$\begin{aligned}
 |\sigma_\beta(x)_i| &\leq |\sigma_\beta(x)_3| \quad (\text{By assumption } x_3 \text{ is the third largest elements}) \\
 &= \frac{e^{\beta x_3}}{\sum_{j=1}^n e^{\beta x_j}} \\
 &\leq \frac{e^{\beta x_3}}{e^{\beta x_1}} \\
 &= e^{\beta(x_3 - x_1)}
 \end{aligned}$$

$$\begin{aligned}
 &\leq e^{\frac{\ln(n-2) - \ln(\epsilon)}{x_1 - x_3}} (x_3 - x_1) && \text{(By the assumption of } \beta \text{ in the main text)} \\
 &\leq \frac{\epsilon}{n-2}. && \text{(F.3)}
 \end{aligned}$$

Combining (F.2) and (F.3) yields that (F.1) is

$$\begin{aligned}
 &\left\| \sigma_\beta(x) - \frac{1}{1 + \exp^{-\beta\delta}} e_1 - \frac{\exp^{-\beta\delta}}{1 + \exp^{-\beta\delta}} e_2 \right\|_\infty \\
 &= \max\left\{ \sigma_\beta(x)_1 - \frac{1}{1 + \exp^{-\beta\delta}}, \frac{\exp^{-\beta\delta}}{1 + \exp^{-\beta\delta}} - \sigma_\beta(x)_2, \sigma_\beta(x)_3 \right\} \\
 &\leq \max\left\{ \frac{e^{\beta x_1} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2}) (\sum_{i=1}^n e^{\beta x_i})}, \frac{e^{\beta x_2} (\sum_{i=3}^n e^{\beta x_i})}{(e^{\beta x_1} + e^{\beta x_2}) (\sum_{i=1}^n e^{\beta x_i})}, \frac{e^{\beta x_3}}{\sum_{j=1}^n e^{\beta x_j}} \right\} \\
 &\leq \max\left\{ \epsilon, \frac{\epsilon}{n-2} \right\} && \text{(By (F.2) and (F.3))} \\
 &\leq \epsilon.
 \end{aligned}$$

This completes the proof. $\square$

F.2 Proof of Theorem 3.1, Corollary 3.1.1, and Corollary 3.1.2

We first define δ used in this theorem. For i -th column of attention score matrix $K^\top Q$, let $x_{1,i}$ and $x_{2,i}$ be its largest and second-largest entries and define $\delta_i := x_{1,i} - x_{2,i}$, and denote $\delta = \min_{i \in [n]} \delta_i$ to be the smallest such gap over all columns.

These sections provide three versions of attention approximating truncated linear models. We first state the assumptions for each case below.

- **Same Truncated Linear Models for Every Token, Input-Adaptive β , Arbitrary $G(\cdot)$ (Section F.2.1).** First, the attention approximate the same linear model (w, t) for all tokens: $\text{Range}_{[a,b]}(w^\top x_i + t)$ for all $i \in [n]$. Second, the largest entry in $K^\top Q$ is unique. In this case, β scale with $O(1/\delta)$. These have two drawbacks: (i) β depends on the input instead of the model architecture and (ii) to make the error ϵ_0 arbitrarily small and when δ is close to zero, one needs very large β , and even then the guarantee excludes an arbitrarily small volume in $\mathbb{R}^{d \times n}$.
- **Extension to Different Truncated Linear Models for Every Token (Section F.2.2).** We extend the above setting to construct attention approximating different linear model (w_i, t_i) for each token: $\text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ for all $i \in [n]$.
- **Input-Agnostic β , Constant $G(\cdot)$ (Section F.2.3).** The top two entries are either tied or separated by a small gap $\delta \geq 0$. By contrast, later in the proof we show that when applying case (ii) of Lemma E.1, β scale with $O(1/\gamma)$ and $\gamma = O((\Delta L)^2)$, a constant for fixed model and irrelevant to the input. This better align with practices, so the theory statement emphasizes case (ii) in the main text.

F.2.1 PROOF OF THEOREM 3.1

Theorem F.1 (Theorem 3.1 Restated: Single-Head Attention Approximates Truncated Linear Model). Fix real numbers $a < b$, and let $\text{Range}_{[a,b]}(\cdot)$ be the truncation operator from Definition 3.1. Let $p \geq \max\{n+1, 3\}$. Fix $w \in \mathbb{R}^d, t \in \mathbb{R}$. For every $\epsilon_0 \in (0, 1)$, suppose $\beta \geq \frac{1}{\delta} \log \frac{p-1}{\epsilon_0}$. Then, there exists a single-layer, single-head self-attention layer Attn and a sequence-wise affine map $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$ such that, for every $X \in \mathbb{R}^{d \times n}$ and every $i \in [n]$,

$$\left\| \text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w^\top x_i + t) e_{\tilde{k}_i} \right\|_\infty \leq \underbrace{\max\{|a|, |b|\}}_{\text{finite-}\beta \text{ softmax error}} \cdot \epsilon_0 + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}}.$$

Here $e_{\tilde{k}_i}$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \underset{k \in [p]}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k| \quad \text{where} \quad \tilde{k}_i := G(k_i) \in [d_o].$$

Here $k_i \in [p]$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : [p] \rightarrow [d_o]$ denotes any set-to-set function sending the interpolation index $k \in [p]$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output.

Proof. We begin with the common setup used by both cases.

First we denote

$$\ell_k := (k - 1)(\tilde{L}_k + \tilde{L}_1 - 2t), \quad k \in [p],$$

following [Definition 3.3](#).

Then, we specify the linear transformation A prepended to attention layer Attn

$$\begin{aligned} A(X) &= \underbrace{\begin{bmatrix} I_d & & \\ & 0_{(d+d_o+2) \times d} & \end{bmatrix}}_{(2d+d_o+2) \times d} X \underbrace{\begin{bmatrix} I_n, 0_{n \times (p-n)} \end{bmatrix}}_{n \times p} + \underbrace{\begin{bmatrix} 0_d & 0_d & \cdots & 0_d & 0_d & \cdots & 0_d \\ 0_d & w & \cdots & (n-1)w & nw & \cdots & (p-1)w \\ 0 & \ell_2 & \cdots & \ell_n & \ell_{n+1} & \cdots & \ell_p \\ & & & \tilde{L}_{d_o \times p} & & & \\ 1 & 1 & \cdots & 1 & 0 & \cdots & 0 \end{bmatrix}}_{(2d+d_o+2) \times p} \\ &= \begin{bmatrix} x_1 & x_2 & \cdots & x_n & 0 & \cdots & 0 \\ 0_d & w & \cdots & (n-1)w & nw & \cdots & (p-1)w \\ 0 & \ell_2 & \cdots & \ell_n & \ell_{n+1} & \cdots & \ell_p \\ & & & \tilde{L}_{d_o \times p} & & & \\ 1 & 1 & \cdots & 1 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{(2d+d_o+2) \times p}, \end{aligned} \quad (\text{F.4})$$

where

$$\tilde{L} = [\tilde{L}_1 e_{\tilde{1}}, \dots, \tilde{L}_k e_{\tilde{k}}, \dots, \tilde{L}_p e_{\tilde{p}}] \in \mathbb{R}^{d_o \times p}.$$

Here, $e_{\tilde{k}} \in \mathbb{R}^{d_o}$ denotes a one-hot vector where only the $\tilde{k}$ -th index has a value of 1.

Namely, before feeding the input token into the self-attention mechanism Attn, we preprocess it with linear transformations $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$. Note that the precision parameter $p \in \mathbb{N}$, defined in [Definition 3.3](#), is required to be larger than the input sequence length n .

Essentially, A extends the input sequence with extra rows/columns for the latter use of interpolation approximation.

Remark F.1. The A here is a sequence-wise linear transformation for the simplicity of demonstrating our method. For a practical, token-wise implementation, see [Theorem 3.2](#). As noted at [Section 5](#), one can interchange [Theorem 3.1](#) and [Theorem 3.2](#) in all subsequent proofs since both yield the same approximation result. We also note that eliminating the sequence-wise operator $\begin{bmatrix} I_n, 0_{n \times (p-n)} \end{bmatrix}_{n \times p}$ in linear transformation A (F.4) is doable. One can achieve this by simply padding input sequence $X \in \mathbb{R}^{d \times n}$ to have sequence length p and remove A .

For the attention matrices of the self-attention layer, we construct their parameters to be

$$\begin{aligned} W_Q &= -\beta \begin{bmatrix} I_d & 0_{d \times d} & 0_{d \times 1} & 0_{d \times d_o} & 0_{d \times 1} \\ 0_{1 \times d} & 0_{1 \times d} & 0 & 0_{1 \times d_o} & 1 \end{bmatrix} \in \mathbb{R}^{(d+1) \times (2d+d_o+2)}, \\ W_K &= \beta \begin{bmatrix} 0_{d \times d} & -2I_d & 0_d & 0_{d \times d_o} & 0_{d \times 1} \\ 0_{1 \times d} & 0_{1 \times d} & 1 & 0_{1 \times d_o} & 0 \end{bmatrix} \in \mathbb{R}^{(d+1) \times (2d+d_o+2)}. \end{aligned}$$

In this setting, we construct the query and key matrix Q, K as

$$Q = W_Q A(X) = -\beta \begin{bmatrix} x_1 & x_2 & \cdots & x_n & 0 & \cdots & 0 \\ 1 & 1 & \cdots & 1 & 0 & \cdots & 0 \end{bmatrix} \in \mathbb{R}^{(d+1) \times p},$$

and

$$K = \beta \begin{bmatrix} 0 & -2w & \cdots & -2(p-1)w \\ 0 & \tilde{L}_1 + \tilde{L}_2 - 2t & \cdots & (p-1)(\tilde{L}_1 + \tilde{L}_p - 2t) \end{bmatrix}.$$

Thus for $K^\top Q$, we have

$$K^\top Q = -\beta^2 \begin{bmatrix} (-2x_1^\top w - 2t + \tilde{L}_1 + \tilde{L}_1) \cdot 0 & \cdots & (-2x_n^\top w - 2t + \tilde{L}_1 + \tilde{L}_1) \cdot 0 & 0 & \cdots \\ (-2x_1^\top w - 2t + \tilde{L}_1 + \tilde{L}_2) \cdot 1 & \cdots & (-2x_n^\top w - 2t + \tilde{L}_1 + \tilde{L}_2) \cdot 1 & 0 & \cdots \\ \vdots & \ddots & \vdots & \vdots & \ddots \\ (-2x_1^\top w - 2t + \tilde{L}_1 + \tilde{L}_p) \cdot (p-1) & \cdots & (-2x_n^\top w - 2t + \tilde{L}_1 + \tilde{L}_p) \cdot (p-1) & 0 & \cdots \end{bmatrix}. \quad (\text{F.5})$$

Next, we use [Lemma E.1](#) that softmax approximate hardmax to find the smallest entry in each column of $K^\top Q$. Then we find the closest interpolation point $\tilde{L}_i$ to $w^\top x_i + t$.

Until now we finish the shared proof for [Theorem 3.1](#), Corollary 3.1.1 and Corollary 3.1.2.

Starting below, we begin the proof specific to [Theorem 3.1](#). Consider the case when the largest entry in every column of $K^\top Q$ is unique and larger than the second largest entry by at least δ . This case has no constraint on $\tilde{k}_i$ but with the tradeoff that β scales with $O(1/\delta)$ depending on the input, see [Section F.1](#) for the detailed discussion.

By [Lemma E.1](#) case (i), for arbitrary $\epsilon_0 > 0$, when every column has a unique minimum entry u_1 that's larger than the second largest u_2 for a constant at least δ , and β to be sufficiently large such that

$$\beta \geq \frac{\ln(n-1) - \ln \epsilon_0}{u_1 - u_2},$$

the following holds

$$\|\text{Softmax}_\beta((K^\top Q)_{:,i}) - e_{k_i}\|_\infty \leq \epsilon_0, \quad (\text{F.6})$$

where k_i is defined as

$$k_i := \underset{k \in [p]}{\operatorname{argmin}} (-2x_i^\top w - 2t + \tilde{L}_0 + \tilde{L}_k) \cdot (k-1).$$

The meaning of k_i correspond to the interpolation point index k that minimizes $|x_i^\top w + t - \tilde{L}_k|$ for $k \in [p]$.

We further deduce this result as follows

$$\begin{aligned} k_i &= \underset{k \in [p]}{\operatorname{argmin}} (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (k-1) \\ &= \underset{k \in [p]}{\operatorname{argmin}} (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (k-1) \Delta L && (\text{Multiply by a positive constant}) \\ &= \underset{k \in [p]}{\operatorname{argmin}} (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (\tilde{L}_k - \tilde{L}_1) && (\text{By } (k-1)\Delta L = \tilde{L}_k - \tilde{L}_1) \\ &= \underset{k \in [p]}{\operatorname{argmin}} (-2x_i^\top w - 2t) \tilde{L}_k + \tilde{L}_k^2 + (x_i^\top w + t)^2 && (\text{Drop constants independent of } k) \\ &= \underset{k \in [p]}{\operatorname{argmin}} (x_i^\top w + t - \tilde{L}_k)^2 && (\text{F.7}) \end{aligned}$$

$$= \underset{k \in [p]}{\operatorname{argmin}} |x_i^\top w + t - \tilde{L}_k|. \quad (\text{F.8})$$

Until now we find the right interpolation point index k that minimizes $|x_i^\top w + t - \tilde{L}_k|$ for $k \in [p]$.

Next, we construct value matrix V to map out the desired interpolation point $\tilde{L}_{k_i}$ according to $\tilde{k}_i$.

Define W_V to pick up the matrix $\tilde{L} = [\tilde{L}_1 e_{\tilde{1}}, \dots, \tilde{L}_p e_{\tilde{p}}] \in \mathbb{R}^{d_0 \times p}$

$$W_V = \begin{bmatrix} 0_{d_0 \times (2d+1)} & I_{d_0} & 0_{d_0 \times 1} \end{bmatrix} \in \mathbb{R}^{d_0 \times (2d+d_0+2)}.$$

This yields

$$V = W_V A(X) = [\tilde{L}_1 e_{\tilde{1}}, \dots, \tilde{L}_p e_{\tilde{p}}]. \quad (\text{F.9})$$

Lastly, we use the linear transform W_O to remove the unwanted columns in (F.5)

$$W_O = \begin{bmatrix} I_n \\ 0_{(p-n) \times n} \end{bmatrix} \in \mathbb{R}^{p \times n}.$$

Until now, we finish the construction of our attention layer

$$\text{Attn} \circ A(X) = \underbrace{V}_{d_0 \times p} \underbrace{\text{Softmax}((W_K A(X))^T W_Q A(X))}_{p \times p} \underbrace{W_O}_{p \times n} \in \mathbb{R}^{d_0 \times n}.$$

Next, we derive the approximation error

$$\|\text{Attn} \circ A(X) - \underbrace{[\text{Range}_{[a,b]}(w^\top x_1 + t)e_{\tilde{k}_1}, \dots, \text{Range}_{[a,b]}(w^\top x_n + t)e_{\tilde{k}_n}]}_{d_0 \times n}\|_\infty < \epsilon.$$

Combining the column-wise results from (F.6) together with V and W_O matrices, we derive that for any $\epsilon_0 > 0$, if

$$\beta \geq \frac{\ln(n-1) - \ln(\frac{1}{2} \max\{|a|, |b|\} \epsilon_0)}{u_1 - u_2}, \quad (\text{F.10})$$

where u_1 and u_2 are the largest and second-largest entries in each column of $K^\top Q$, the following holds

$$\begin{aligned} & \|V \text{Softmax}(K^\top Q) W_O - V[e_{k_1}, e_{k_2}, \dots, e_{k_n}]\|_\infty \\ &= \|V \text{Softmax}(K^\top Q) W_O - [V e_{k_1}, V e_{k_2}, \dots, V e_{k_n}]\|_\infty \\ &= \|V \text{Softmax}(K^\top Q) W_O - [\tilde{L}_{k_1} e_{\tilde{k}_1}, \tilde{L}_{k_2} e_{\tilde{k}_2}, \dots, \tilde{L}_{k_n} e_{\tilde{k}_n}]\|_\infty \\ &< \max\{|a|, |b|\} \cdot \epsilon_0, \end{aligned} \quad (\text{F.11})$$

where the third line is by (F.9) and the fact that V multiplied by one-hot vector e_{k_i} returns its k_i -th column $V_{:,k_i}$.

The softmax error in (F.6) is at most ϵ_0 in infinity norm, but here scale by V since $|\tilde{L}_k|$ in V is at most $\max\{|a|, |b|\}$.

Note that (F.10) implies β scale with $\delta \leq u_1 - u_2$ with $O(1/\delta)$.

We now move to derive the interpolation error.

Lastly by the definition of k_i and $\tilde{L}_{k_i}$ we have:

$$|\tilde{L}_{k_i} - \text{Range}_{[a,b]}(w^\top x_i + t)| \leq \frac{b-a}{p}. \quad (\text{F.12})$$

Thus

$$\begin{aligned} & \|\text{Attn}(X)_{:,i} - \text{Range}_{[a,b]}(w^\top x_i + t) \cdot e_{\tilde{k}_i}\|_\infty \\ & \leq \|\text{Attn}(X)_{:,i} - \tilde{L}_{k_i} \cdot e_{\tilde{k}_i}\|_\infty + \|\tilde{L}_{k_i} \cdot e_{\tilde{k}_i} - \text{Range}_{[a,b]}(w^\top x_i + t) \cdot e_{\tilde{k}_i}\|_\infty \end{aligned} \quad (\text{By triangle inequality})$$

$$\leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}}, \quad \text{for } i \in [n].$$

When p goes to infinity and ϵ_0 goes to 0, the total error is arbitrary small. Thus, we set

$$\max\{|a|, |b|\} \cdot \epsilon_0 + \frac{b-a}{p} \leq \epsilon.$$

This yields

$$\|\text{Attn}(X) - [\text{Range}_{[a,b]}(w^\top x_1 + t)e_{\tilde{k}_1}, \dots, \text{Range}_{[a,b]}(w^\top x_n + t)e_{\tilde{k}_n}]\|_\infty \leq \epsilon. \quad (\text{F.13})$$

This completes the proof. $\square$

F.2.2 PROOF OF COROLLARY 3.1.1

Next, we show how to extend [Theorem 3.1](#) to approximate multiple truncated linear models $\{w_i, t_i\}_{i \in n}$.

Corollary F.1.1 (Restated of [Corollary 3.1.1](#)). For n truncated linear models $\{w_i, t_i\}_{i \in n}$, the same conditions give

$$\|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i)e_{\tilde{k}_i}\|_\infty \leq \max\{|a|, |b|\} \cdot \epsilon_0 + \frac{b-a}{p}.$$

Proof. First, we combine the bias term t into w by augmenting the input $x_i \in \mathbb{R}^d$ with 1 such that

$$x'_i := \begin{bmatrix} x_i \\ 1 \end{bmatrix} \in \mathbb{R}^{d+1}, \quad w' := \begin{bmatrix} w \\ t \end{bmatrix} \in \mathbb{R}^{d+1}.$$

Thus we have

$$\|\text{Attn}(X)_{:,i} - \text{Range}_{[a,b]}(w^\top x_i + t)e_{\tilde{k}_i}\|_\infty = \|\text{Attn}(X)_{:,i} - \text{Range}_{[a,b]}(w'^\top x'_i)e_{\tilde{k}_i}\|_\infty.$$

Then we multiply each token x'_i elementwisely by a trainable vector v'_i such that $x'_i \odot v'_i \in \mathbb{R}^{d+1}$.² Effectively,

$$w'^\top (x'_i \odot v'_i) = (w' \odot v'_i)^\top x'_i.$$

Denote $w'_i := w' \odot v'_i$,

$$w'^\top (x'_i \odot v'_i) = w'^\top_i x'_i.$$

The proof of [Theorem 3.1](#) still applies. We explain as follow. The i -th column of [\(F.5\)](#) becomes

$$S_{k,i} = -\beta^2(-2w'^\top (x'_i \odot v'_i) + \tilde{L}_1 + \tilde{L}_k)(k-1).$$

Using $w'_i = w' \odot v'_i$, this equals to

$$S_{k,i} = -\beta^2 \left(-2w'^\top_i x'_i + \tilde{L}_1 + \tilde{L}_k \right) (k-1).$$

The same algebra in [\(F.8\)](#) gives

$$\begin{aligned} k_i &:= \arg \max_k S_{k,i} = \arg \min_k |w'^\top_i x'_i - \tilde{L}_k| \\ &= \arg \min_k |w_i^\top x_i + t_i - \tilde{L}_k|, \end{aligned} \quad (\text{F.14})$$

²Such position-dependent entry-wise multiplication can be implemented by attention; see [Hu et al. \(2026b\)](#)[Lemma A.4].

where the last equality uses $w'_i = [w_i; t_i]$.

Therefore, for each column i , the softmax still selects the interpolation point closest to $w_i^\top x_i + t_i$. The finite-temperature softmax error and interpolation error are the same as in [Theorem 3.1](#). Therefore,

$$\begin{aligned} & \left\| \text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i} \right\|_\infty \\ & \leq \max\{|a|, |b|\} \cdot \epsilon_0 + \frac{b-a}{p}. \end{aligned}$$

This completes the proof. $\square$

F.2.3 PROOF OF [COROLLARY 3.1.2](#)

The proof shares the same derivation as in [Section F.2.1](#) up to [\(F.11\)](#). Hence, we start the proof after [\(F.11\)](#). By the same proof in [Section F.2.2](#), we also change [\(F.8\)](#) to the multiple truncated linear model setting as in [\(F.14\)](#).

Corollary F.1.2 (Restated of [Corollary 3.1.2](#)). Under the same conditions as [Theorem 3.1](#),

$$\left\| \text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i} \right\|_\infty \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}},$$

where k_i and $\tilde{k}_i$ are defined as [\(??\)](#), but now $G : [p] \rightarrow [d_o]$ denotes any set-to-constant function sending the interpolation index $k \in [p]$ into a constant $\tilde{k} \in [d_o]$ specifying in the desired row of the output.

Proof. Recall that [\(F.10\)](#) implies β scale with $\delta \leq u_1 - u_2$ with $O(1/\delta)$. To avoid this input dependence, this corollary turns to case (ii) of [Lemma E.1](#).

By [Lemma E.1](#) case (ii), there are two top entries in $K^\top Q$, either tied or separated by a small gap $\delta \geq 0$. In [Lemma E.1](#) we see that this case give better scaling for β since it doesn't depend on δ . However, $\tilde{k}_i$ for all $i \in [n]$ should only be a constant. This make sure when value matrix times the softmax matrix it compute the correct averaged between two interpolation points.

Let $\tilde{k}_i$ be identical for all $i \in [n]$. According to [Lemma E.1](#), the third largest $-(1/2 \cdot (x_i^\top w + t - \tilde{L}_k)^2)$ for all $k \in [n]$ is at least smaller than the largest $-(1/2 \cdot (x_i^\top w + t - \tilde{L}_{k_i})^2)$ by (without a loss of generality, assume $x_i^\top w + t - \tilde{L}_{k_i} > 0$)

$$\begin{aligned} & -\left(\frac{1}{2}(x_i^\top w + t - \tilde{L}_{k_i})^2\right) - \left[-\left(\frac{1}{2}(x_i^\top w + t - \tilde{L}_{k_i} + \Delta L)^2\right)\right] \\ & = \frac{1}{2}\Delta L[\Delta L + 2(x_i^\top w + t - \tilde{L}_{k_i})] \\ & \geq \frac{(\Delta L)^2}{2}. \end{aligned} \quad \left(\frac{(\Delta L)^2}{2} \text{ corresponds to the } \gamma \text{ in } \text{Lemma E.1}\right)$$

Let $s_i := (K^\top Q)_{:,i}$ be the i -th attention-score column. Let k_i and k'_i denote the indices of the largest and second-largest entries of s_i , respectively. Define

$$\alpha_i := \frac{1}{1 + e^{-\beta \delta_i}},$$

where δ_i is the gap between these two entries.

By case (ii) of [Lemma E.1](#), by choosing $\beta \geq \frac{\ln(n-2) - \ln(\epsilon_0/2)}{(\Delta L)^2/2}$, we have

$$\left\| \text{Softmax}(s_i) - \alpha_i e_{k_i} - (1 - \alpha_i) e_{k'_i} \right\|_1 \leq \frac{\epsilon_0}{2}.$$

Since the value matrix satisfies $V_{:,k} = \tilde{L}_k e_{\tilde{k}}$ and $|\tilde{L}_k| \leq M$, where $M := \max\{|a|, |b|\}$, multiplying by V gives

$$\begin{aligned} & \|V\text{Softmax}(s_i) - V(\alpha_i e_{k_i} + (1 - \alpha_i) e_{k'_i})\|_\infty \\ & \leq M \frac{\epsilon_0}{2}. \end{aligned}$$

Moreover, in this corollary all interpolation values are placed in the same output coordinate. Hence $e_{\tilde{k}_i} = e_{\tilde{k}'_i}$. Therefore,

$$\begin{aligned} & \|V(\alpha_i e_{k_i} + (1 - \alpha_i) e_{k'_i}) - V e_{k_i}\|_\infty \\ & = (1 - \alpha_i) \|V e_{k'_i} - V e_{k_i}\|_\infty \\ & = (1 - \alpha_i) \|\tilde{L}_{k'_i} e_{\tilde{k}'_i} - \tilde{L}_{k_i} e_{\tilde{k}_i}\|_\infty && \text{(By (F.9) and (F.11))} \\ & = (1 - \alpha_i) |\tilde{L}_{k'_i} - \tilde{L}_{k_i}| && \text{(Since } e_{\tilde{k}_i} = e_{\tilde{k}'_i}\text{)} \\ & \leq \Delta L. \end{aligned}$$

Choosing the interpolation resolution p large enough so that

$$\Delta L \leq M \frac{\epsilon_0}{2},$$

we obtain

$$\begin{aligned} \|V\text{Softmax}(s_i) - V e_{k_i}\|_\infty & \leq M \frac{\epsilon_0}{2} + \Delta L \\ & \leq M \epsilon_0. \end{aligned}$$

Applying this bound to each column $i \in [n]$, and using W_O to keep only the first n columns, yields

$$\|V\text{Softmax}(K^\top Q)W_O - V[e_{k_1}, \dots, e_{k_n}]\|_\infty \leq M \epsilon_0.$$

Equivalently, since $V e_{k_i} = \tilde{L}_{k_i} e_{\tilde{k}_i}$ as derived in (F.11), we have

$$\begin{aligned} & \left\| V\text{Softmax}(K^\top Q)W_O - \begin{bmatrix} \tilde{L}_{k_1} e_{\tilde{k}_1} & \tilde{L}_{k_2} e_{\tilde{k}_2} & \cdots & \tilde{L}_{k_n} e_{\tilde{k}_n} \end{bmatrix} \right\|_\infty \\ & \leq M \epsilon_0, \end{aligned}$$

which is exactly (F.11).

The remaining part is the same discussion of interpolation error starting from (F.8).

This completes the proof. $\square$

For later convenience, here we recast [Theorem 3.1](#) into an “arbitrary precision” version.

Corollary F.1.3 (Arbitrary Precision with Explicit Parameters). Let $a < b$ and set $M := \max\{|a|, |b|\}$. For any $\epsilon > 0$, choose

$$p \geq \max \left\{ n + 1, \left\lceil \frac{2(b-a)}{\epsilon} \right\rceil \right\} \quad \text{and} \quad \beta \geq \frac{2p^2}{(b-a)^2} (\ln(p-2) + \ln \frac{2M}{\epsilon}).$$

Then the single-layer, single-head self-attention construction in [Theorem 3.1](#) satisfies

$$\left\| \text{Attn}(X) - [\text{Range}_{[a,b]}(w^\top x_1 + t)e_{\tilde{k}_1}, \dots, \text{Range}_{[a,b]}(w^\top x_n + t)e_{\tilde{k}_n}] \right\|_\infty \leq \epsilon.$$

Proof. By [Theorem 3.1](#) we know

$$\epsilon = \underbrace{M \epsilon_0}_{\text{softmax error}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}}.$$

Choose $\epsilon_0 = \epsilon/(2M)$ so the softmax error is $\epsilon/2$. Plug it into β together with $\Delta L = (b-a)/(p-1)$, we have

$$\beta \geq \frac{2(p-1)^2}{(b-a)^2} \left(\ln(p-2) + \ln \frac{2M}{\epsilon} \right).$$

This guarantees the softmax term is $\leq \epsilon/2$. For the interpolation error, require

$$\frac{b-a}{p} \leq \frac{\epsilon}{2},$$

which is equal to

$$p \geq \frac{2(b-a)}{\epsilon}.$$

Finally take $p \geq n + 1$ to ensure $p > n$. Summing the two halves gives total error $\leq \epsilon$. □

F.3 Proof of Theorem 3.2

To approximate a truncated linear function using multi-head attention, we partition the anchors $\{\tilde{L}_k\}_{k=1}^p$ across H heads, where each head handles $n - 2$ interpolation anchors ($p = H(n - 2)$). For any scalar value z , we need to know which heads are responsible for it, that is whose interpolation range contains z . The next lemma shows that at most two adjacent heads cover the same z . This lemma enables a simplified case analysis later in the main theorem's proof.

Lemma F.2 (Cases of All Heads in Attn^H). Let p be divisible by $n - 2$ and set $H = p/(n - 2)$. For $z \in [\tilde{L}_1, \tilde{L}_p]$ and any $h \in [H]$, define three cases of the relationship between z and h :

- **Case 1:** $z \in [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)}]$,
- **Case 2:** $z \notin [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)}]$.
- **Case 3:** $z \in [\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{(h-1)(n-2)+1}] \cup [\tilde{L}_{h(n-2)}, \tilde{L}_{h(n-2)+1}]$.

These cases includes all possible situation. Then for all h , only two cases exists

- a falls in Case 1 for an h and Case 2 for all others.
- z falls in Case 3 for two adjacent h and Case 2 for all others.

Proof. Because $z \in [\tilde{L}_1, \tilde{L}_p]$ and

$$[\tilde{L}_1, \tilde{L}_p] = \bigcup_{h=1}^H [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)}],$$

there exists an $h_z \in [H]$ such that

$$z \in [\tilde{L}_{(h_z-1)(n-2)+1}, \tilde{L}_{h_z(n-2)}].$$

This leads to only two possible cases:

- Case 1*: $z \in [\tilde{L}_{(h_z-1)(n-2)+1}, \tilde{L}_{h_z(n-2)}]$.
- Case 2*: $z \in [\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}]$ for some $h_z \in [H - 1]$.

Case 1*: $z \in [\tilde{L}_{(h_z-1)(n-2)+1}, \tilde{L}_{h_z(n-2)}]$. Then z falls in Case 1 for $h = h_z$.

For any $h < h_z$, we have

$$\tilde{L}_{h(n-2)+1} \leq \tilde{L}_{(h_z-1)(n-2)} < \tilde{L}_{(h_z-1)(n-2)+1} \leq z.$$

Thus

$$z \notin [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)+1}].$$

So z falls in Case 2 for all $h < h_z$.

For any $h > h_z$, we have

$$z \leq \tilde{L}_{h_z(n-2)} < \tilde{L}_{h_z(n-2)+1} \leq \tilde{L}_{(h-1)(n-2)}.$$

Thus

$$z \notin [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)+1}].$$

So z falls in Case 2 for all $h > h_z$.

Therefore, z falls in Case 1 for one head and Case 2 for all other heads.

Case 2*: $z \in [\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}]$ for some $h_z \in [H-1]$. For head h_z , this interval is the right boundary region:

$$[\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}] \subset [\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}],$$

which is Case 3 for head h_z .

For head $h_z + 1$, the same interval is the left boundary region:

$$[\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}] = [\tilde{L}_{((h_z+1)-1)(n-2)}, \tilde{L}_{((h_z+1)-1)(n-2)+1}],$$

which is Case 3 for head $h_z + 1$.

For any other head $h \notin \{h_z, h_z + 1\}$, the interval

$$[\tilde{L}_{h_z(n-2)}, \tilde{L}_{h_z(n-2)+1}]$$

does not intersect

$$[\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{h(n-2)+1}].$$

Therefore, z falls in Case 2 for all other heads.

Thus z falls in Case 3 for two adjacent heads and Case 2 for all other heads.

This completes the proof. $\square$

Theorem F.2 (**Theorem 3.2** Restated: Multi-Head Attention Approximate Truncated Linear Models). Fix real numbers $a < b$, and let the truncation operator $\text{Range}_{[a,b]}(\cdot)$ follow **Definition 3.1**. For a precision parameter $p > n$ with $\epsilon = O(1/p)$, number of head $H = p/(n-2)$ there exists a single-layer, H -head self-attention Attn^H with a linear transformation $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(d+n) \times n}$, such that $\text{Attn}^H \circ A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d_o \times n}$ satisfies, for any $i \in [n]$,

$$\|\text{Attn}^H \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{(n-2)H-1}}_{\text{interpolation error}}.$$

Here $e_{\tilde{k}_i}$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \underset{k \in \{p\}}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k| \quad \text{where} \quad \tilde{k}_i := G(k_i) \in [d_o]. \quad (\text{F.15})$$

Here $k_i \in \{p\}$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : \{p\} \rightarrow [d_o]$ denotes any set-to-set function sending the interpolation index $k \in \{p\}$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output.

Proof. Define $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(d+n) \times n}$ for the input sequence X as

$$A(X) := \underbrace{\begin{bmatrix} I_d \\ 0_{n \times d} \end{bmatrix}}_{(d+n) \times d} X + \underbrace{\begin{bmatrix} 0_{d \times n} \\ I_n \end{bmatrix}}_{(d+n) \times n} = \begin{bmatrix} X \\ I_n \end{bmatrix} \in \mathbb{R}^{d+n}.$$

Thus, A is a token-wise linear layer augmented with positional encoding, as it applies a linear projection to each token and then adds a unique per-token bias.

Let p be a precision parameter, without loss of generality, let it be divisible by $n-2$ and denote $p/(n-2)$ as H .

Now we define the multi-head attention Attn of H heads.

For notational convenience, we extend the grid by two proof-only score locations

$$\tilde{L}_0 := \tilde{L}_1 - \Delta L, \quad \tilde{L}_{p+1} := \tilde{L}_p + \Delta L,$$

where $\Delta L = (b - a)/(p - 1)$. These two points are used only in the key-query scores for the two dummy positions of each head. The corresponding dummy values vector as in (F.19) is 0_{d_o} .

For $j \in \{0, 1, \dots, p + 1\}$, define

$$\ell_j := (j - 1)(\tilde{L}_j + \tilde{L}_1 - 2t).$$

We denote the h -th head as Attn_h , and define the weight matrices as

$$\begin{aligned} W_K^{(h)} &= -\beta \begin{bmatrix} 0_{d \times d} & -2[(h-1)(n-2) - 1]w & -2(h-1)(n-2)w & \cdots & -2h(n-2)w \\ 0_d^\top & \ell_{(h-1)(n-2)} & \ell_{(h-1)(n-2)+1} & \cdots & \ell_{h(n-2)+1} \end{bmatrix}, \\ W_Q^{(h)} &= \begin{bmatrix} I_d & 0_{d \times n} \\ 0_d^\top & 1_n^\top \end{bmatrix}, \\ W_V^{(h)} &= \begin{bmatrix} 0_{d_o \times (d+1)} & \tilde{L}_{(h-1)(n-2)+1} e_{\tilde{k}_{(h-1)(n-2)+1}} & \cdots & \tilde{L}_{h(n-2)} e_{\tilde{k}_{h(n-2)}} & 0_{d_o} \end{bmatrix}, \end{aligned}$$

for every $h \in [H]$. Here $\beta > 0$ is a coefficient we use to control the precision of our approximation. The attention reaches higher precision as β gets larger.

With the construction of weights, we are also able to calculate the K , Q , V matrices in Attn

$$K^{(h)} := W_K^{(h)} A(X) \tag{F.16}$$

$$\begin{aligned} &= -\beta \begin{bmatrix} 0_{d \times d} & -2[(h-1)(n-2) - 1]w & -2(h-1)(n-2)w & \cdots & -2h(n-2)w \\ 0_d^\top & \ell_{(h-1)(n-2)} & \ell_{(h-1)(n-2)+1} & \cdots & \ell_{h(n-2)+1} \end{bmatrix} \cdot \begin{bmatrix} X \\ I_n \end{bmatrix} \\ &= -\beta \begin{bmatrix} -2[(h-1)(n-2) - 1]w & -2(h-1)(n-2)w & \cdots & -2h(n-2)w \\ \ell_{(h-1)(n-2)} & \ell_{(h-1)(n-2)+1} & \cdots & \ell_{h(n-2)+1} \end{bmatrix} \in \mathbb{R}^{(d+1) \times n} \end{aligned} \tag{F.17}$$

where the last equality comes from multiplying X with 0, thus this is an extraction of non-zero entries in W_K .

For Q , we have

$$\begin{aligned} Q^{(h)} &:= W_Q^{(h)} A(X) \\ &= \begin{bmatrix} I_d & 0_{d \times n} \\ 0_d^\top & 1_n^\top \end{bmatrix} \cdot \begin{bmatrix} X \\ I_n \end{bmatrix} \\ &= \underbrace{\begin{bmatrix} I_d \cdot X + 0_{d \times n} \cdot I_n \\ 0_{1 \times d} \cdot X + 1_{1 \times n} \cdot I_n \end{bmatrix}}_{(d+1) \times n} \\ &= \begin{bmatrix} X \\ 1_{1 \times n} \end{bmatrix}. \end{aligned} \tag{F.18}$$

For V , we have

$$\begin{aligned} V^{(h)} &:= W_V^{(h)} A(X) \\ &= \begin{bmatrix} 0_{d_o \times (d+1)} & \tilde{L}_{(h-1)(n-2)+1} e_{\tilde{k}_{(h-1)(n-2)+1}} & \cdots & \tilde{L}_{h(n-2)} e_{\tilde{k}_{h(n-2)}} & 0_{d_o} \end{bmatrix} \cdot \begin{bmatrix} X \\ I_n \end{bmatrix} \\ &= \underbrace{0_{d_o \times d}}_{\text{dummy}} \cdot X + \underbrace{\begin{bmatrix} 0_{d_o} & \tilde{L}_{(h-1)(n-2)+1} e_{\tilde{k}_{(h-1)(n-2)+1}} & \cdots & \tilde{L}_{h(n-2)} e_{\tilde{k}_{h(n-2)}} & 0_{d_o} \end{bmatrix}}_{d_o \times n} \cdot I_n \\ &= \begin{bmatrix} 0_{d_o} & \tilde{L}_{(h-1)(n-2)+1} e_{\tilde{k}_{(h-1)(n-2)+1}} & \tilde{L}_{(h-1)(n-2)+2} e_{\tilde{k}_{(h-1)(n-2)+2}} & \cdots & \tilde{L}_{h(n-2)} e_{\tilde{k}_{h(n-2)}} & 0_{d_o} \end{bmatrix}. \end{aligned} \tag{F.19}$$

Given that all $\tilde{k}_j$, for $j \in [p]$, share the same identical number in $[d_o]$, we denote this number by k_G .

Remark F.2. This theorem have all the $\tilde{k}_j$ as the same for simplicity. This version of identical $\tilde{k}_j$ is also what subsequent theorems on universal approximations use.

Hence we rewrite $V^{(h)}$ as

$$V^{(h)} = \begin{bmatrix} 0_{d_o} & \tilde{L}_{(h-1)(n-2)+1} e_{k_G} & \tilde{L}_{(h-1)(n-2)+2} e_{k_G} & \cdots & \tilde{L}_{h(n-2)} e_{k_G} & 0_{d_o} \end{bmatrix}.$$

We define m_v as

$$m_v := \max\{|a|, |b|\}.$$

By the definition of $V^{(h)}$, we have

$$\|V\|_\infty \leq \max_{i \in [P]} \{\tilde{L}_i\} \leq m_v. \quad (\text{F.20})$$

Remark F.3 (Intuition of the Construction of $V^{(h)}$). As previously mentioned, $\tilde{L}_i$, for $i \in [p]$, are all the interpolations. In this context, $V^{(h)}$ encompasses the $(n-2)$ elements of these interpolations (i.e., $(h-1)(n-2)$ to $h(n-2)-1$). Meanwhile, the value on the two ends of V^h are both set to 0_{d_o} , because we suppress the head and let it output 0 when the input X is not close enough to the interpolations of the head.

Now we are ready to calculate the output of each Attn_h

$$\begin{aligned} & \text{Attn}_h(A(X)) \\ &= V^{(h)} \text{Softmax}((K^{(h)})^\top Q^{(h)}) \\ &= V \text{Softmax} \left(-\beta \begin{bmatrix} -2[(h-1)(n-2)-1]w & -2(h-1)(n-2)w & \cdots & -2h(n-2)w \\ \ell_{(h-1)(n-2)} & \ell_{(h-1)(n-2)+1} & \cdots & \ell_{h(n-2)+1} \end{bmatrix}^\top \begin{bmatrix} X \\ 1_{1 \times n} \end{bmatrix} \right), \end{aligned}$$

where last line is by plug in (F.16) and (F.18). Note the i -th column of the attention score matrix (the Softmax nested expression) is equivalent to the following expressions

$$\begin{aligned} & \text{Softmax}((K^{(h)})^\top Q^{(h)})_{:,i} \\ &= \text{Softmax} \left(-\beta \begin{bmatrix} -2[(h-1)(n-2)-1]w & -2(h-1)(n-2)w & \cdots & -2h(n-2)w \\ \ell_{(h-1)(n-2)} & \ell_{(h-1)(n-2)+1} & \cdots & \ell_{h(n-2)+1} \end{bmatrix}^\top \begin{bmatrix} X \\ 1_{1 \times n} \end{bmatrix} \right)_{:,i} \\ &= \text{Softmax} \left(-\beta \begin{bmatrix} -2[(h-1)(n-2)-1]w^\top x_i + \ell_{(h-1)(n-2)} \\ -2(h-1)(n-2)w^\top x_i + \ell_{(h-1)(n-2)+1} \\ \vdots \\ -2h(n-2)w^\top x_i + \ell_{h(n-2)+1} \end{bmatrix} \right) \quad (\text{pick column } i) \\ &= \text{Softmax} \left(-\beta \begin{bmatrix} [([(h-1)(n-2)-1](-2w^\top x_i + \tilde{L}_{(h-1)(n-2)} + \tilde{L}_1) - 2[(h-1)(n-2)-1]t) \\ (h-1)(n-2)(-2w^\top x_i + \tilde{L}_{(h-1)(n-2)+1} + \tilde{L}_1) - 2(h-1)(n-2)t \\ \vdots \\ h(n-2)(-2w^\top x_i + \tilde{L}_{h(n-2)+1} + \tilde{L}_1) - 2h(n-2)t \end{bmatrix} \right) \\ & \quad (\text{By } \ell_j := (j-1)(\tilde{L}_j + \tilde{L}_1 - 2t)) \end{aligned}$$

$$\begin{aligned}
 &= \text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{(h-1)(n-2)}) \cdot [(h-1)(n-2) - 1]\Delta L \\ (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{(h-1)(n-2)+1}) \cdot (h-1)(n-2)\Delta L \\ \vdots \\ (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{h(n-2)+1}) \cdot h(n-2)\Delta L \end{bmatrix} \right) \\
 &\quad \text{(By mutiplying and dividing by } \Delta L) \\
 &= \text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{(h-1)(n-2)}) \cdot (\tilde{L}_{(h-1)(n-2)} - \tilde{L}_1) \\ (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{(h-1)(n-2)+1}) \cdot (\tilde{L}_{(h-1)(n-2)+1} - \tilde{L}_1) \\ \vdots \\ (-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_{h(n-2)+1}) \cdot (\tilde{L}_{h(n-2)+1} - \tilde{L}_1) \end{bmatrix} \right) \\
 &\quad \text{(By } (j-1)\Delta L = \tilde{L}_j - \tilde{L}_1) \\
 &= \text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (-2x_i^\top w - 2t) \cdot \tilde{L}_{(h-1)(n-2)} + (\tilde{L}_{(h-1)(n-2)})^2 + (x_i^\top w + t)^2 \\ (-2x_i^\top w - 2t) \cdot \tilde{L}_{(h-1)(n-2)+1} + (\tilde{L}_{(h-1)(n-2)+1})^2 + (x_i^\top w + t)^2 \\ \vdots \\ (-2x_i^\top w - 2t) \cdot \tilde{L}_{h(n-2)+1} + (\tilde{L}_{h(n-2)+1})^2 + (x_i^\top w + t)^2 \end{bmatrix} \right) \\
 &= \text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (x_i^\top w + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (x_i^\top w + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (x_i^\top w + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right). \tag{F.21}
 \end{aligned}$$

Here, the last-second equality arises from the fact that the softmax function is shift-invariant, allowing us to subtract and add a constant across all coordinates. To be more precise, we first expand the product for the j -th coordinate of the column vector:

$$\begin{aligned}
 &(-2x_i^\top w - 2t + \tilde{L}_1 + \tilde{L}_j)(\tilde{L}_j - \tilde{L}_1) \\
 &= (-2x_i^\top w - 2t)\tilde{L}_j + \tilde{L}_1\tilde{L}_j + \tilde{L}_j^2 - (-2x_i^\top w - 2t)\tilde{L}_1 - \tilde{L}_1^2 - \tilde{L}_1\tilde{L}_j \\
 &= (-2x_i^\top w - 2t)\tilde{L}_j + \tilde{L}_j^2 - \underbrace{((-2x_i^\top w - 2t)\tilde{L}_1 + \tilde{L}_1^2)}_{\text{constant across the column vector}}.
 \end{aligned}$$

Then, dropping the constant and adding another constant $(x_i^\top w + t)^2$ across all coordinates, the above expression becomes

$$\begin{aligned}
 &(-2x_i^\top w - 2t)\tilde{L}_j + \tilde{L}_j^2 + (x_i^\top w + t)^2 \\
 &= (x_i^\top w + t - \tilde{L}_j)^2.
 \end{aligned}$$

Hence we finish the derivation of (F.21). Thus we have

$$\text{Attn}_h(A(X))_{:,i} = V\text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (x_i^\top w + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (x_i^\top w + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (x_i^\top w + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right). \tag{F.22}$$

For a specific h , we calculate the result of (F.22) column by column. Let X_i denote any column (token) of the matrix X . We partition the situation at each column into three distinct cases:

- **Case 1:** $w^\top X_i + t$ is within the active interpolation range of Attn_h :

$$w^\top X_i + t \in [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)}].$$

- **Case 2:** $w^\top X_i + t$ is outside the enlarged score range of Attn_h :

$$w^\top X_i + t \notin [\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{h(n-2)+1}].$$

- **Case 3:** $w^\top X_i + t$ lies in one of the two boundary regions of Attn_h :

$$w^\top X_i + t \in [\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{(h-1)(n-2)+1}] \cup [\tilde{L}_{h(n-2)}, \tilde{L}_{h(n-2)+1}].$$

Remark F.4 (Description of All Cases of a Single Head Attention). The H heads split the p interpolation anchors into H blocks, where each head stores $n - 2$ true interpolation anchors. The h -th head stores

$$\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{(h-1)(n-2)+2}, \dots, \tilde{L}_{h(n-2)}$$

in its value matrix. It also uses two neighboring score locations,

$$\tilde{L}_{(h-1)(n-2)} \quad \text{and} \quad \tilde{L}_{h(n-2)+1},$$

whose corresponding value entries are both 0_{d_o} .

With this understanding, the three cases have the following meaning:

- **Case 1:** $w^\top X_i + t$ falls inside the true anchor block stored by the h -th head.
- **Case 2:** $w^\top X_i + t$ falls outside even the enlarged score range of the h -th head, so the head outputs approximately 0_{d_o} .
- **Case 3:** $w^\top X_i + t$ falls in one of the two boundary regions, so two adjacent heads may contribute.

Remark F.5 (Cases of All Attention Heads). According to [Lemma F.2](#), for all heads in Attn^H , there are two possible cases:

- **Case 1*:** x falls into Case 1 for a head, and Case 2 for all other heads.
- **Case 2*:** x falls into Case 3 for two heads with adjacent interpolation ranges, and Case 2 for other heads.

This also means that when Case 1 appears in Attn^H , the situation of all head in Attn^H falls into Case 1*. And when Case 3 appears in Attn^H , the situation of all head in Attn^H falls into Case 2*. Thus, We discuss Case 2* in the discussion of Case 3.

Case 1: $w^\top X_i + t \in [\tilde{L}_{(h-1)(n-2)+1}, \tilde{L}_{h(n-2)}]$. In this case, our goal is to demonstrate this attention head outputs a value close to $\text{Range}_{[a,b]}(w^\top X_i + t)$.

Let $\tilde{L}_{(h-1)(n-2)+s}$ and $\tilde{L}_{(h-1)(n-2)+s+1}$ be the two interpolants such that

$$w^\top X_i + t \in [\tilde{L}_{(h-1)(n-2)+s}, \tilde{L}_{(h-1)(n-2)+s+1}]. \quad (\text{F.23})$$

Then, the corresponding two entries are the two largest entries in

$$-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix}.$$

Indeed,

$$\begin{aligned}
 & \operatorname{argmax}_{k \in \{(h-1)(n-2), \dots, h(n-2)+1\}} -\frac{\beta}{\Delta L} (w^\top X_i + t - \tilde{L}_k)^2 \\
 &= \operatorname{argmin}_{k \in \{(h-1)(n-2), \dots, h(n-2)+1\}} (w^\top X_i + t - \tilde{L}_k)^2 \\
 &= \operatorname{argmin}_{k \in \{(h-1)(n-2), \dots, h(n-2)+1\}} |w^\top X_i + t - \tilde{L}_k|.
 \end{aligned}$$

We also note that the distance from $w^\top X_i + t$ to interpolants beside $\tilde{L}_{(h-1)(n-2)+s}$ and $\tilde{L}_{(h-1)(n-2)+s+1}$ differs by at least

$$\tilde{L}_{r+1} - \tilde{L}_r = \Delta L.$$

This is equivalent to the occasion when $x_1 - x_3$ in [Lemma E.1](#) is larger than

$$\begin{aligned}
 & \max \left\{ \frac{\beta}{\Delta L} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+s-1})^2 - \frac{\beta}{\Delta L} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+s})^2, \right. \\
 & \quad \left. \frac{\beta}{\Delta L} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+s+2})^2 - \frac{\beta}{\Delta L} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+s+1})^2 \right\} \\
 & \geq \frac{\beta}{\Delta L} \cdot (\Delta L)^2,
 \end{aligned}$$

which is invariant to X_i .

Thus according to [Lemma E.1](#) and the fact that the two corresponding entries are the two largest entries in the i -th column of the attention score matrix, we have

$$\left\| \operatorname{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - \frac{1}{1 + e^{-\beta\delta}} \underbrace{e_s}_{n \times 1} - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} \underbrace{e_{s+1}}_{n \times 1} \right\|_\infty \leq \epsilon_2,$$

for any $\epsilon_2 > 0$.

This yields that

$$\begin{aligned}
 & \left\| V \operatorname{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - V \frac{1}{1 + e^{-\beta\delta}} e_s - V \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} e_{s+1} \right\|_\infty \\
 & \leq \left\| \operatorname{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - \frac{1}{1 + e^{-\beta\delta}} e_s - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} e_{s+1} \right\|_\infty \cdot \|V\|_\infty \\
 & \leq \|V\|_\infty \epsilon_2.
 \end{aligned}$$

This is equivalent to

$$\begin{aligned}
 & \left\| V \operatorname{Softmax}(K^\top Q)_{:,i} - \frac{1}{1 + e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s} e_{k_G} - \frac{e^{-\beta\delta}}{1 + e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s+1} e_{k_G} \right\|_\infty \\
 & \leq \|V\|_\infty \cdot \epsilon_2 \quad \quad \quad (\text{By } \|AB\| \leq \|A\| \cdot \|B\|) \\
 & \leq m_v \epsilon_2, \quad \quad \quad (\text{F.24})
 \end{aligned}$$

where the last line is by (F.20).

From (F.23), we derive that

$$\begin{aligned}
 & \left\| \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s} + \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s+1} - (w^\top X_i + t)e_{k_G} \right\|_\infty \\
 & \leq \left\| \frac{1}{1+e^{-\beta\delta}} \left(\tilde{L}_{(h-1)(n-2)+s} - (w^\top X_i + t)e_{k_G} \right) \right\|_\infty + \left\| \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \left(\tilde{L}_{(h-1)(n-2)+s+1} - (w^\top X_i + t) \right) \right\|_\infty \\
 & \quad \text{(By convex combination of } (w^\top X_i + t) \text{ and triangle inequality)} \\
 & \leq \frac{1}{1+e^{-\beta\delta}} \cdot \Delta L + \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \cdot \Delta L \quad \text{(By (F.23))} \\
 & = \Delta L. \quad \text{(F.25)}
 \end{aligned}$$

Combining (F.24) and (F.25) yields

$$\begin{aligned}
 & \left\| V\text{Softmax}(K^\top Q)_{:,i} - (w^\top X_i + t)e_{k_G} \right\|_\infty \\
 & \leq \left\| V\text{Softmax}(K^\top Q)_{:,i} - \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s} e_{k_G} - \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s+1} e_{k_G} \right\|_\infty \\
 & \quad + \left\| \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s} + \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h-1)(n-2)+s+1} - (w^\top X_i + t)e_{k_G} \right\|_\infty \quad \text{(By triangle inequality)} \\
 & \leq m_v \epsilon_2 + \Delta L, \quad \text{(F.26)}
 \end{aligned}$$

where the first inequality comes from adding and subtracting the interpolation points' convex combination and then applying triangle inequality.

Case 2: $w^\top X_i + t \notin [\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{h(n-2)+1}]$. In this case, $w^\top X_i + t$ falls outside the enlarged score range of Attn_h .

Without loss of generality, suppose $w^\top X_i + t$ lies to the left of the enlarged score range of Attn_h . Then $\tilde{L}_{(h-1)(n-2)}$ is the closest score location within Attn_h to $w^\top X_i + t$. Furthermore, the second closest score location $\tilde{L}_{(h-1)(n-2)+1}$ is at least further by ΔL , which is a constant independent of X_i .

Then by Lemma E.1, we have

$$\left\| \text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - \underbrace{e_1}_{n \times 1} \right\|_\infty \leq \epsilon_3,$$

for any $\epsilon_3 > 0$.

This yields that

$$\begin{aligned}
 & \left\| V\text{Softmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - V \underbrace{e_1}_{n \times 1} \right\|_\infty \\
 & \leq \|V\|_\infty \cdot \epsilon_3 \quad \text{(By } \|AB\| \leq \|A\| \cdot \|B\|) \\
 & \leq m_v \epsilon_3,
 \end{aligned}$$

where the last line is by (F.20).

Since the first value entry of $V^{(h)}$ is 0_{d_o} , this is equivalent to

$$\left\| \text{VSoftmax} \left(-\frac{\beta}{\Delta L} \begin{bmatrix} (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)})^2 \\ (w^\top X_i + t - \tilde{L}_{(h-1)(n-2)+1})^2 \\ \vdots \\ (w^\top X_i + t - \tilde{L}_{h(n-2)+1})^2 \end{bmatrix} \right) - 0_{d_o} \right\|_\infty \leq m_v \epsilon_3. \quad (\text{F.27})$$

The case where $w^\top X_i + t$ lies to the right of the enlarged score range is analogous, with the softmax selecting the right dummy value entry, which is also 0_{d_o} .

Case 1*. According to Lemma F.2, when Case 1 occurs for one head in the H heads of Attn^H , all other heads are in Case 2.

Combining this with the result in Case 2, we have

$$\begin{aligned} & \left\| \text{Attn}^H(A(X))_{:,i} - (w^\top X_i + t)e_{k_G} \right\|_\infty \\ & \leq \left\| \sum_{h_0 \in [H] \setminus \{h\}} \text{Attn}_{h_0} \circ A(X)_{:,i} \right\|_\infty + \left\| \text{Attn}_h \circ A(X)_{:,i} - (w^\top X_i + t)e_{k_G} \right\|_\infty \\ & \leq (H-1)m_v \epsilon_3 + m_v \epsilon_2 + \Delta L \quad (\text{By (F.26) and (F.27)}) \\ & = (H-1)m_v \epsilon_3 + m_v \epsilon_2 + \frac{b-a}{H(n-2)-1}. \end{aligned}$$

Setting ϵ_2, ϵ_3 to be

$$\begin{aligned} \epsilon_2 &= \frac{\epsilon_0}{2}, \\ \epsilon_3 &= \frac{\epsilon_0}{2(H-1)}, \end{aligned}$$

yields the final result.

Case 3 (and Case 2*): $w^\top X_i + t \in [\tilde{L}_{(h-1)(n-2)}, \tilde{L}_{(h-1)(n-2)+1}] \cup [\tilde{L}_{h(n-2)}, \tilde{L}_{h(n-2)+1}]$. Let

$$z_i := w^\top X_i + t.$$

In this case, z_i lies on the boundary region shared by two neighboring heads. Without loss of generality, suppose

$$z_i \in [\tilde{L}_{(h_0-1)(n-2)}, \tilde{L}_{(h_0-1)(n-2)+1}],$$

which is shared by Attn_{h_0-1} and Attn_{h_0} . By Lemma F.2, all other heads are in Case 2.

Combining this with Case 2, we have

$$\begin{aligned} & \left\| \text{Attn}^H(A(X))_{:,i} - \text{Attn}_{h_0} \circ A(X)_{:,i} - \text{Attn}_{h_0-1} \circ A(X)_{:,i} \right\|_\infty \\ & \leq (H-2)m_v \epsilon_3. \end{aligned}$$

By Lemma E.1, let δ denote the gap between the two largest score entries. Then

$$\left\| \text{Softmax}((K^{(h_0)})^\top Q^{(h_0)}) - \left(\frac{1}{1+e^{-\beta\delta}} e_1 + \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} e_2 \right) \right\|_\infty \leq \epsilon_4,$$

and

$$\left\| \text{Softmax}((K^{(h_0-1)})^\top Q^{(h_0-1)}) - \left(\frac{1}{1+e^{-\beta\delta}} e_{n-1} + \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} e_n \right) \right\|_\infty \leq \epsilon_5,$$

for any $\epsilon_4, \epsilon_5 > 0$.

Since

$$V^{(h_0)} e_1 = 0_{d_o}, \quad V^{(h_0)} e_2 = \tilde{L}_{(h_0-1)(n-2)+1} e_{k_G},$$

and

$$V^{(h_0-1)} e_{n-1} = \tilde{L}_{(h_0-1)(n-2)} e_{k_G}, \quad V^{(h_0-1)} e_n = 0_{d_o},$$

we obtain

$$\begin{aligned} & \left\| \text{Attn}_{h_0} \circ A(X)_{:,i} + \text{Attn}_{h_0-1} \circ A(X)_{:,i} \right. \\ & \quad \left. - \left(\frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)+1} e_{k_G} + \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)} e_{k_G} \right) \right\|_\infty \\ & \leq m_v(\epsilon_4 + \epsilon_5). \end{aligned} \tag{F.28}$$

Combining this with the contribution of all other heads gives

$$\begin{aligned} & \left\| \text{Attn}^H(A(X))_{:,i} - \left(\frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)+1} e_{k_G} + \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)} e_{k_G} \right) \right\|_\infty \\ & \leq (H-2)m_v\epsilon_3 + m_v(\epsilon_4 + \epsilon_5). \end{aligned} \tag{F.29}$$

Finally, since $z_i \in [\tilde{L}_{(h_0-1)(n-2)}, \tilde{L}_{(h_0-1)(n-2)+1}]$, the convex combination of these two adjacent anchors is within ΔL of z_i . Hence

$$\begin{aligned} & \left\| \frac{e^{-\beta\delta}}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)+1} e_{k_G} + \frac{1}{1+e^{-\beta\delta}} \tilde{L}_{(h_0-1)(n-2)} e_{k_G} - z_i e_{k_G} \right\|_\infty \\ & \leq \Delta L. \end{aligned}$$

Combining this with (F.29), we have

$$\begin{aligned} & \left\| \text{Attn}^H(A(X))_{:,i} - (w^\top X_i + t) e_{k_G} \right\|_\infty \\ & \leq \Delta L + (H-2)m_v\epsilon_3 + m_v(\epsilon_4 + \epsilon_5) \\ & = \frac{b-a}{H(n-2)-1} + (H-2)m_v\epsilon_3 + m_v(\epsilon_4 + \epsilon_5). \end{aligned}$$

Setting $\epsilon_3, \epsilon_4, \epsilon_5$ to be

$$\begin{aligned} \epsilon_3 &= \frac{\epsilon_0}{3(H-2)}, \\ \epsilon_4 &= \epsilon_5 = \frac{\epsilon_0}{3}, \end{aligned}$$

yields the final result.

This completes the proof. $\square$

F.4 Proof of Lemma 3.1

Theorem F.3 (Lemma 3.1 Restated: Explicit Construction of ReLU Neural Network as Universal Approximator). Let $f : \mathcal{X} \rightarrow \mathbb{R}$ be a continuous function defined on a compact domain $\mathcal{X} \subset \mathbb{R}^N$ for some $N \in \mathbb{N}_+$. For any $\epsilon > 0$, there exists a two-layer feed-forward neural network FFN with ReLU activation functions such that for all $x \in \mathcal{X}$

$$\|\text{FFN}(x) - f(x)\|_{L_p} \leq \epsilon. \quad (\text{F.30})$$

Proof Sketch. First, we discretize the input domain into a grid of points G_D . Around each grid point $v \in G_D$, we construct a ReLU-based bump function $R_v(x) = \sum \text{ReLU}$ that equals 1 within a small region around v and rapidly decays to 0 outside this region. Next, we define the feedforward network (FFN) as $\sum_{v \in G_D} f(v) \cdot \text{ReLU}(R_v(x) - N + 1)$, allowing us to approximate $f(x)$ as a weighted sum of function values evaluated on grid points v near x . This process yields a piecewise linear approximation of f . $\square$

Proof. We first quantize the domain into a grid, builds localized bump functions using ReLU, construct the FFN to combines the piecewise approximation in a weighted sum to approximate f , and analyze the approximation error.

Construction of Bump Function $R_v(\cdot)$. Let $x = [x_1, x_2, \dots, x_N] \in \mathbb{R}^N$. The compactness of $\mathcal{X}$ means it lies within an N -dimensional cube $[-B, B]^N$. Quantize this domain into a grid G_D with granularity g

$$G_D = \left\{ \frac{-B(g-1)}{g}, \frac{-B(g-3)}{g}, \dots, \frac{B(g-1)}{g} \right\}^N,$$

which results in g^N grid points across all dimensions.

For each point on the grid $v \in G_D$, we define a local bump function denoted as $R_v(x)$

$$\begin{aligned} R_v(x) &= \sum_{i=1}^N \phi(x_i, v_i) \\ &= \sum_{i=1}^N \left[\text{ReLU}\left(\frac{1}{\delta}\left(\frac{g(x_i - v_i)}{B} + 1\right)\right) - \text{ReLU}\left(\frac{1}{\delta}\left(\frac{g(x_i - v_i)}{B} + 1 - \delta\right)\right) \right. \\ &\quad \left. + \text{ReLU}\left(\frac{1}{\delta}\left(-\frac{g(x_i - v_i)}{B} + 1\right)\right) - \text{ReLU}\left(\frac{1}{\delta}\left(-\frac{g(x_i - v_i)}{B} + 1 - \delta\right)\right) - 1 \right]. \end{aligned}$$

The function $\phi(x_i, v_i)$ behaves as

$$\phi(x_i, v_i) = \begin{cases} 0, & |x_i - v_i| \geq \frac{B}{g}, \\ -\frac{g}{\delta B}|x_i - v_i| + \frac{1}{\delta}, & (1 - \delta)\frac{B}{g} < |x_i - v_i| < \frac{B}{g}, \\ 1, & |x_i - v_i| \leq (1 - \delta)\frac{B}{g}. \end{cases} \quad (\text{F.31})$$

for every $i \in [N]$.

Now we discuss the behavior of the bump function $R_v(x)$ for difference distance between the grid point v and x .

As shown in (F.31), the value of bump function depends on three different distance between v and x . We formally define them as follow.

First define G_v as the region centered at v with radius B/g in the ℓ_∞ norm

$$G_v := \{x \in [-B, B]^N : \|x - v\|_\infty \leq \frac{B}{g}\}.$$

Second define P_v as the core region of G_v where the bump function $R_v(x)$ is fully on (equal to N)

$$P_v := \{x \in [-B, B]^N : \|x - v\|_\infty \leq (1 - \delta) \frac{B}{g}\}. \quad (\text{F.32})$$

Third we define the shell region of G_v denoted as $G_v \setminus P_v$

$$G_v \setminus P_v := \{x \in [-B, B]^N : (1 - \delta) \frac{B}{g} \leq \|x - v\|_\infty \leq \frac{B}{g}\}.$$

Now we discuss the behavior of $R_v(x)$ under this three situations.

For $x \notin G_v$, at least one dimension of x satisfies $|x_i - v_i| \geq B/g$ for some $i \in [N]$. By examining the definition $\phi(x_i, v_i)$ in (F.31), since $R_v(x) = \sum_{i=1}^N \phi(x_i, v_i)$ and at least one $\phi(x_i, v_i) = 0$, we have

$$R_v(x) = \sum_{i=1}^N \phi(x_i, v_i) \leq N - 1 \quad \text{for } x \notin G_v. \quad (\text{F.33})$$

For $x \in P_v$, each coordinates x_i satisfies $|x_i - v_i| \leq (1 - \delta)B/g$. By (F.31), this implies $\phi(x_i, v_i) = 1$ for all i , we derive

$$R_v(x) = \sum_{i=1}^N \phi(x_i, v_i) = N \times 1 = N \quad \text{for } x \in P_v. \quad (\text{F.34})$$

For $x \in G_v \setminus P_v$, by (F.31), the corresponding $\phi(x_i, v_i) \leq 1$. Thus

$$R_v(x) \in [N - 1, N) \quad \text{for } x \in G_v \setminus P_v.$$

Until now we finish the construction of $R_v(\cdot)$ and analysis its behaviours. Next we move to the construction of FFN to approximate the target function f .

Construction of FFN. Following the above discussion, we construct the FFN to be:

$$\text{FFN}(x) = \sum_{v \in G_D} f(v) \cdot \text{ReLU}(R_v(x) - N + 1). \quad (\text{F.35})$$

The behavior of $\text{ReLU}(R_v(x) - N + 1)$ is

$$\text{ReLU}(R_v(x) - N + 1) = \begin{cases} 0, & x \notin G_v, \\ 1, & x \in P_v, \\ \text{ReLU}(R_v(x) - N + 1), & x \in G_v \setminus P_v. \end{cases}$$

By this construction, the FFN approximate $f(x)$ by weighted sum over the grid points v such that $x \in G_v$.

Now we move to analysis the approximation error of the constructed FFN.

Approximation Error Analysis. To approximate the continuous function f , we introduce the region $\mathcal{P} := \bigcup_{v \in G_D} P_v$.

We also denote μ as the Lebesgue measure in N -dimensional space for later use.

Using the uniform continuity of f and the properties of the constructed FFN, we analyze the L_p -norm error by partitioning the input domain into $\mathcal{P}$ and its complement

$$\|\text{FFN}(x) - f(x)\|_{L_p} = \left(\int_{[-B, B]^N} (\text{FFN}(x) - f(x))^p dx \right)^{\frac{1}{p}}$$

$$= \left(\int_{[-B, B]^N / \mathcal{P}} (\text{FFN}(x) - f(x))^p dx + \int_{\mathcal{P}} (\text{FFN}(x) - f(x))^p dx \right)^{\frac{1}{p}}.$$

Now we discuss the two situations in the following paragraph, and conclude our proof.

• **Case 1:** $x \in \mathcal{P}$.

For an $x \in \mathcal{P}$, let v_x denote the unique grid point such that $x \in \mathbb{P}_v$. By (F.33) and (F.34) we have

$$R_v(x) = N, \quad \text{if } v = v_x, \quad \text{and} \quad R_v(x) \leq N - 1, \quad \text{if } v \neq v_x.$$

Hence

$$\begin{aligned} \text{FFN}(x) &= \sum_{v \in G_D} f(v) \cdot \text{ReLU}(R_v(x) - N + 1) && (\text{By (F.35)}) \\ &= f(v_x) \text{ReLU}(R_{v_x}(x) - N + 1) && (\text{By } R_v(x) \leq N - 1, \text{ if } v \neq v_x) \\ &= f(v_x) \times 1 = f(v_x). \end{aligned}$$

Since the $\text{FFN}(x)$ collapse to $f(v_x)$ when $x \in \mathcal{P}$, the error $|\text{FFN}(x) - f(x)|$ becomes to approximate $f(x)$ by the function value evaluate on the closest grid point $f(v_x)$.

Because f is continuous on a closed region, it is bounded and uniformly continuous. Thus there exists a $\Delta > 0$ such that for any $x_1, x_2 \in \mathbb{R}^N$ satisfying $\|x_1 - x_2\|_\infty \leq \Delta$, the following hold

$$|f(x_1) - f(x_2)| \leq \frac{\epsilon}{2(2B)^{\frac{N}{p}} \mu(\mathcal{P})^{\frac{1}{p}}},$$

where the term $2(2B)^{\frac{N}{p}} \mu(\mathcal{P})^{\frac{1}{p}}$ is a constant to help us normalize the final error bound.

Set g to be large enough such that $2B/g < \Delta$, and since $\|v_x - v\|_\infty$ is smaller than the grid length $2B/G$, it also smaller than Δ . This yields that for $x \in \mathcal{P}$

$$|\text{FFN}(x) - f(x)| = |f(v_x) - f(x)| \leq \frac{\epsilon}{2(2B)^{\frac{N}{p}} \mu(\mathcal{P})^{\frac{1}{p}}}. \quad (\text{F.36})$$

• **Case 2:** $x \notin \mathcal{P}$.

Now we turn to analyse the approximation error outside $\mathcal{P}$.

First we know that

$$\begin{aligned} |\text{FFN}(x) - f(x)| &\leq |\text{FFN}(x)| + |f(x)| \\ &\leq \|f\|_{L_\infty} + \|f\|_{L_\infty} \\ &= 2\|f\|_{L_\infty}, \end{aligned}$$

where the second inequality coming from

$$\|f\|_{L_\infty} = \sup_{x \in [-B, B]^N} \|f(x)\|_\infty \geq f(x),$$

and by (F.35), $f(v) \leq \|f(x)\|_{L_\infty}$, and also the design of bump function make sure given x , only the one grid point closet to x contribute, hence $|\text{FFN}(x)| \leq \|f(x)\|_{L_\infty}$.

Hence the approximation error outside $\mathcal{P}$ become

$$\int_{[-B, B]^N / \mathcal{P}} (\text{FFN}(x) - f(x))^p dx \leq \int_{[-B, B]^N / \mathcal{P}} (2\|f\|_{L_\infty})^p dx \quad (\text{F.37})$$

$$= (2\|f\|_{L_\infty})^p \cdot \mu([-B, B]^N / \mathcal{P}), \quad ((2\|f\|_{L_\infty})^p \text{ is a constant.})$$

where $\mu([-B, B]^N / \mathcal{P})$ is the volume of how much of the entire domain isn't covered by $\mathcal{P}$. We calculate it as

$$\begin{aligned} & \mu([-B, B]^N / \mathcal{P}) \\ &= \mu([-B, B]^N) - \mu(\mathcal{P}) \\ &= B^N - (1 - \delta)^N B^N && (\text{By (F.32) and } \mathcal{P} := \bigcup_{v \in G_D} P_v) \\ &= (1 - (1 - \delta)^N) B^N && (\text{By associative property}) \\ &= (\delta N + \mathcal{O}(\delta^2)) B^N && (\text{By binomial expansion on } (1 - \delta)^N = 1 - N\delta + \mathcal{O}(\delta^2)) \\ &= \delta N B^N + \mathcal{O}(\delta^2). \end{aligned}$$

For any $\epsilon_1 > 0$, we select a small enough δ such that $\mu([-B, B]^N / \mathcal{P}) \leq \epsilon_1$, thus we can make the volume outside $\mathcal{P}$ as small as desired by choosing δ sufficiently small.

Thus the approximation outside $\mathcal{P}$ in (F.37) become

$$\int_{[-B, B]^N / \mathcal{P}} (\text{FFN}(x) - f(x))^p dx \leq (2\|f\|_{L_\infty})^p \cdot \mu([-B, B]^N / \mathcal{P}) \leq (2\|f\|_{L_\infty})^p \cdot \epsilon_1. \quad (\text{F.38})$$

We set ϵ_1 to be

$$\epsilon_1 = \frac{\epsilon^p}{2^{p+1} \|f\|_{L_\infty}^p}.$$

for the normalization of the final error bound.

Finally we combine (F.36) and (F.38), for any $p \in N_+$, the total approximation is

$$\begin{aligned} \|\text{FFN}(x) - f(x)\|_{L_p} &= \left(\int_{[-B, B]^N} (\text{FFN}(x) - f(x))^p dx \right)^{\frac{1}{p}} \\ &= \left(\int_{[-B, B]^N / \mathcal{P}} (\text{FFN}(x) - f(x))^p dx + \int_{\mathcal{P}} (\text{FFN}(x) - f(x))^p dx \right)^{\frac{1}{p}} \\ &\leq \left(\epsilon_1 (2\|f\|_{L_\infty})^p + \left(\frac{\epsilon}{2(2B)^{\frac{N}{p}} \mu(\mathcal{P})^{\frac{1}{p}}} \right)^p \times (2B)^N \right)^{\frac{1}{p}} && (\text{By (F.36) and (F.38)}) \\ &\leq \left(\frac{\epsilon^p}{2} + \frac{\epsilon^p}{2} \right)^{\frac{1}{p}} \\ &= \epsilon. \end{aligned}$$

This completes the proof. □

F.5 Proof of Lemma 3.2

We first outline our proof sketch.

Proof Sketch. We begin by discretizing the domain $\mathcal{X} = [-B, B]^{d \times n}$ into a finite grid G_D . For each grid point $v^{(j)} \in G_D$, we define a ‘‘bump’’ function $R_{v^{(j)}}(X)$ that is approximately 1 when X is near $v^{(j)}$ and approximately 0 otherwise. Next, using Lemma F.3 and Lemma 3.1, we construct a multi-head attention layer (plus a linear mapping) that collectively approximates these bump functions via $|G_D| \cdot d$ heads, achieving an ∞ -norm error of at most $|G_D| \cdot d \cdot \epsilon_0$. We then form a second linear map encoding the function values $[f(v^{(1)}), \dots, f(v^{(|G_D|)})]$ alongside the approximated bump functions, organizing them into a 2-row matrix. Finally, a single-head attention layer — using softmax as a near-arg max — selects the grid value $f(v^{(j)})$ associated with whichever $v^{(j)}$ is nearest to X . This yields a piecewise approximation to f within any desired error tolerance. $\square$

Now we proceed to the detailed proof.

We first present a auxiliary lemma deduced from Theorem 3.1.

Lemma F.3. Fix real numbers $a < b$, and let the truncation operator $\text{Range}_{[a,b]}(\cdot)$ follow Definition 3.1. Let $w \in \mathbb{R}^d$ and $t \in \mathbb{R}$ be such that $\|w\|_\infty \leq R_w$ and $|t| \leq R_t$. For a precision parameter $p \in \mathbb{N}_+$ satisfying $p > n$, let $\epsilon = O(\frac{1}{p})$. Then, for any $\epsilon > 0$, there exists a single-layer, single-head self-attention $\text{Attn} : \mathbb{R}^{d \times p} \rightarrow \mathbb{R}^{d \times p}$, and an layer of linear connections $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times p}$ free of activation function such that

$$\|\text{Attn} \circ A(X) - [\underbrace{0}_{d_o \times n_0}, \sum_{i=1}^n \text{Range}_{[a,b]}(w^\top x_i + t_i), \underbrace{0}_{d_o \times (p-1-n_0)}]\|_\infty < \epsilon.$$

Here $N, n_0 \in \mathbb{N}_+$ are any integer satisfying $n_0 \leq p - n$,

Proof. By Corollary F.1.3, let $\epsilon = \epsilon'/n$ with $\epsilon' > 0$, there exists a Attn^* and a A^* that satisfy

$$\|\text{Attn}^* \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_{\tilde{k}_i}\|_\infty < \frac{\epsilon'}{n}. \quad (\text{F.39})$$

By setting $t = 0$, $d_o = 1$ and $e_{\tilde{k}_i} = 1$, we have

$$\|\text{Attn}^* \circ A^*(X) - \underbrace{\text{Range}_{[a,b]}(w^\top X)}_{1 \times n}\|_\infty < \frac{\epsilon'}{n}. \quad (\text{F.40})$$

Define $A_0(Z)$

$$A_0(Z) := Z + \begin{bmatrix} \frac{t_1}{\|w\|_2^2} w & \frac{t_2}{\|w\|_2^2} w & \dots & \frac{t_n}{\|w\|_2^2} w \end{bmatrix}, \quad (\text{F.41})$$

to insert the bias terms $\{t_i\}_{i \in [n]}$ by combining with A^* and define $A := A^* \circ A_0$. The denominator $\|w\|_2^2$ is because in (F.40) every token is multiplied by w , and $\frac{\langle w, w \rangle}{\|w\|_2^2} = 1$ make sure we get t_i . Since a linear transformation followed by another linear transformation is still a linear transformation, A is a linear transformation.

Multiply the W_O in Attn^* with W_0 defined as

$$W_0 := \underbrace{\begin{bmatrix} 0_{n \times n_0} & 1_n & 0_{n \times (p-1-n_0)} \end{bmatrix}}_{n \times p}. \quad (\text{F.42})$$

And define Attn as

$$\text{Attn}(Z) = \text{Attn}^*(Z) \cdot W_0.$$

Since W_O in Attn^* multiplied with W_0 still outputs a matrix, Attn is still an attention module.

Now we calculate the difference between $\text{Attn} \circ A(X)$ with target output

$$\begin{aligned}
 & \left\| \text{Attn} \circ A(X) - \left[\underbrace{0}_{1 \times n_0}, \sum_{i=1}^n \text{Range}_{[a,b]}(w^\top x_i + t_i), \underbrace{0}_{1 \times (p-1-n_0)} \right] \right\|_\infty \\
 &= \left\| \text{Attn}^* \circ A^*(A_0(X))W_0 - \left[\underbrace{0}_{1 \times n_0}, \sum_{i=1}^n \text{Range}_{[a,b]}(w^\top x_i + t_i), \underbrace{0}_{1 \times (p-1-n_0)} \right] \right\|_\infty \quad (\text{By definition of } \text{Attn}^* \text{ and } A^*) \\
 &= \left\| \text{Attn}^* \circ A^*(A_0(X))W_0 - \left[\underbrace{0}_{1 \times n_0}, \sum_{i=1}^n \text{Range}_{[a,b]}(w^\top (x_i + \frac{t_i}{\|w\|_2^2} w)), \underbrace{0}_{1 \times (p-1-n_0)} \right] \right\|_\infty \\
 &= \left\| \text{Attn}^* \circ A^*(A_0(X))W_0 - \left[\underbrace{0}_{1 \times n_0}, \sum_{i=1}^n \text{Range}_{[a,b]}(w^\top A_0(X)_{:,i}), \underbrace{0}_{1 \times (p-1-n_0)} \right] \right\|_\infty \quad (\text{By (F.41)}) \\
 &= \left\| \text{Attn}^* \circ A^*(A_0(X))W_0 - \left[\underbrace{0}_{1 \times n_0}, \underbrace{\text{Range}_{[a,b]}(w^\top A_0(X)) \cdot 1_n}_{1 \times n}, \underbrace{0}_{1 \times (p-1-n_0)} \right] \right\|_\infty \\
 &= \left\| \text{Attn}^* \circ A^*(A_0(X))W_0 - \underbrace{\text{Range}_{[a,b]}(w^\top X) W_0}_{1 \times n} \right\|_\infty \quad (\text{By (F.42)}) \\
 &\leq \left\| \text{Attn}^* \circ A^*(A_0(X)) - \underbrace{\text{Range}_{[a,b]}(w^\top X)}_{1 \times n} \right\|_1 \cdot \|W_0\|_\infty \quad (\text{Since } \|EW_0\|_\infty \leq \|E\|_1 \|W_0\|_\infty) \\
 &\leq \left\| \text{Attn}^* \circ A^*(A_0(X)) - \underbrace{\text{Range}_{[a,b]}(w^\top X)}_{1 \times n} \right\|_\infty \cdot n \|W_0\|_\infty \quad (\text{Since } \|E\|_1 \|W_0\|_\infty \leq n \|E\|_\infty \|W_0\|_\infty) \\
 &= \frac{\epsilon'}{n} \cdot n \\
 &= \epsilon'.
 \end{aligned}$$

Because ϵ' is arbitrary, we reset the notation and denote it as ϵ for simplicity of presentation. This completes the proof. $\square$

Then we state our proof of [Lemma 3.2](#).

Lemma F.4 ([Lemma 3.2](#) Restated: Sequence-to-Scalar Universal Approximation of Attention). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when composed with linear transformations, there exists a one layer multi-head attention Attn_m stacked with one layer single-head attention Attn_s composed with linear connections A_1 and A_2 , such that

$$\|f - \text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1\|_{L_p} \leq \epsilon$$

Proof. Let $X := [X_1, X_2, \dots, X_n] \in \mathbb{R}^{d \times n}$ denotes our input. Without loss of generality, assume our inputs to come from $[-B, B]^{d \times n}$, $B \in \mathbb{R}_+$ is their bound in every dimension.

We discretize the input domain into a set of grid points G_D defined as follow.

Definition F.1 (Grid Centers). We define G_D as the set of all grid centers in $\mathbb{R}^{d \times n}$. The corresponding grids consists a quantization on $[-B, B]^{dn}$ of granularity g (meaning each dimension is equally partitioned to g intervals)

$$G_D = \left\{ \frac{-B(g-1)}{g}, \frac{-B(g-3)}{g}, \dots, \frac{B(g-1)}{g} \right\}^{d \times n},$$

and $|G_D| = g^{dn}$, since each of the dn entries can chosen from g values.

Remark F.6. For a grid point $v \in G_D$, we denote its columns as $v_i (i \in [n])$. The entry on the j -th row is denoted as $v_{i,j}$. We write it out explicitly as

$$\begin{aligned} v &:= [v_1, v_2, \dots, v_n] \in \mathbb{R}^{d \times n}, \\ v_i &= [v_{i,1} \ v_{i,2} \ \dots \ v_{i,d}]^\top, \quad i \in [n], j \in [d], \end{aligned}$$

where each $v_{i,j} \in \mathbb{R}$.

For a $v \in G_D$, define the corresponding R_v similar to that in [Lemma 3.1](#)

$$\begin{aligned} R_v(X) &:= \sum_{i=1}^d \sum_{j=1}^n [\text{ReLU}(\frac{1}{\delta}(\frac{g(X_{i,j} - v_{i,j})}{B} + 1)) - \text{ReLU}(\frac{1}{\delta}(\frac{g(X_{i,j} - v_{i,j})}{B} + 1 - \delta)) \\ &\quad + \text{ReLU}(\frac{1}{\delta}(-\frac{g(X_{i,j} - v_{i,j})}{B} + 1)) - \text{ReLU}(\frac{1}{\delta}(-\frac{g(X_{i,j} - v_{i,j})}{B} + 1 - \delta))], \end{aligned}$$

for any $v \in G_D$. We eliminate the "-1" term in the definition of $R_v(x)$ in [Lemma 3.1](#). Here δ is a coefficient we use to control the precision of our approximation in later process. We use v_i to denote its i -th column of $v \in G_D$ for $i \in [n]$, where $v_i \in [-B, B]^d$. We also label every v in G_D as $v^{(j)}$, $j \in [d^n]$ and denote this label as $l(v)$ for every v .

Next we show single-layer attention approximate $R_v(X)$.

From [Lemma F.3](#), by setting $w = g/\delta B e_i$ and $t_k = -g/\delta B v_{i,k}^{(j)}$, $k \in [n]$ there exists a single-head attention $\text{Attn}_{v^{(j)},+}^{(i)}$ attached with a linear transformation A_1 that satisfies

$$\begin{aligned} &\|\text{Attn}_{v^{(j)},+}^{(i)} \circ A_1(X) - \begin{bmatrix} 0_{j-1}^\top & \text{Range}_{[0,1]}(\frac{g(e_i^\top X_1 - v_{i,1})}{\delta B}) + \dots + \text{Range}_{[0,1]}(\frac{g(e_i^\top X_n - v_{i,n})}{\delta B}) & 0_{|G_D|-j}^\top \end{bmatrix}\|_\infty \\ &\leq \epsilon_0, \end{aligned}$$

for any $\epsilon_0 > 0$.

Also from [Theorem 3.1](#), there should also exist a single-head attention $\text{Attn}_{v^{(j)},-}^{(i)}$ attached with the same linear transformation A_1 that satisfies

$$\begin{aligned} &\|\text{Attn}_{v^{(j)},-}^{(i)} \circ A_1(X) - \begin{bmatrix} 0_{j-1}^\top & \text{Range}_{[0,1]}(\frac{g(v_{i,1} - e_i^\top X_1)}{\delta B}) + \dots + \text{Range}_{[0,1]}(\frac{g(v_{i,n} - e_i^\top X_n)}{\delta B}) & 0_{|G_D|-j}^\top \end{bmatrix}\|_\infty \\ &\leq \epsilon_0. \end{aligned}$$

Summing these two kinds of single head attention across d we approximate $R_{v^{(j)}}$ in the following remark.

Remark F.7. For every $v^{(j)} \in [G_D]$, the aggregation of all $\text{Attn}_{v^{(j)},+}^{(i)}$ and all $\text{Attn}_{v^{(j)},-}^{(i)}$ for $i \in [d]$ outputs

$$\left\| \sum_{i=1}^d (\text{Attn}_{v^{(j)},-}^{(i)}(X) + \text{Attn}_{v^{(j)},+}^{(i)}(X)) - \begin{bmatrix} 0_{1 \times (j-1)} & R_{v^{(j)}}(X) & 0_{1 \times (|G_D|-j)} \end{bmatrix} \right\|_\infty \leq d\epsilon_0. \quad (\text{F.43})$$

Since we must do this for all $v \in G_D$, we use multiple heads in parallel. We construct the multi-head attention to be

$$\text{Attn}_m(X) := \sum_{j=1}^{|G_D|} \sum_{i=1}^d (\text{Attn}_{v^{(j)},-}^{(i)}(X) + \text{Attn}_{v^{(j)},+}^{(i)}(X)). \quad (\text{F.44})$$

Then by (F.43), the output of $\text{Attn}_m \circ A(X)$ satisfies

$$\|\text{Attn}_m \circ A(X) - \sum_{j=1}^{|G_D|} \underbrace{\begin{bmatrix} 0_{1 \times (j-1)} & R_{v^{(j)}}(X) & 0_{1 \times (|G_D| - j)} \end{bmatrix}}_{1 \times |G_D|}\|_\infty \leq d\epsilon_0.$$

Thus

$$\|\text{Attn}_m \circ A(X) - [R_{v^{(1)}}(X) \ R_{v^{(2)}}(X) \ \cdots \ R_{v^{(|G_D|)}}(X)]\|_\infty \leq d\epsilon_0 \quad (\text{F.45})$$

where as previously denoted, $|G_D|$ is the total number of all grid centers.

Now we construct the second layer of attention Attn_s to pick the largest $R_{v^{(j)}}(X)$, and use a linear layer A_2 to encode the function value.

First, we construct A_2 to be

$$A_2(Z) := \begin{bmatrix} 0 & 0 & \cdots & 0 \\ f(v^{(1)}) & f(v^{(2)}) & \cdots & f(v^{(|G_D|)}) \end{bmatrix} + Z. \quad (\text{F.46})$$

By (F.45), A_2 connected after $\text{Attn} \circ A_1$ has an output satisfying

$$\|A_2 \circ \text{Attn} \circ A_1(X) - \begin{bmatrix} R_{v^{(1)}}(X) & R_{v^{(2)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \\ f(v^{(1)}) & f(v^{(2)}) & \cdots & f(v^{(|G_D|)}) \end{bmatrix}\|_\infty \leq d\epsilon_0. \quad (\text{F.47})$$

For Attn_s , we construct a single-head attention Attn_s each weight matrix to pick the desired row in Z

$$\text{Attn}_s(Z) := \underbrace{\begin{bmatrix} 0 & 1 \end{bmatrix}}_{1 \times 2} Z \text{Softmax}_\beta \left(\underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{1 \times 2} Z \right)^\top \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{1 \times 2} Z \mathbf{1}_{|G_D|}, \quad (\text{F.48})$$

where β is a parameter we use to control the precision.

Now we claim that the construct attention layer satisfies the following

$$\|\text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1(X) - [f(v^{(1)}) \ f(v^{(2)}) \ \cdots \ f(v^{(|G_D|)})] \text{Softmax}_\beta(M)\|_\infty \leq d\epsilon_0, \quad (\text{F.49})$$

which we derive by plugging (F.46), (F.47) and (F.48) into $\text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1$

$$\begin{aligned} & \underbrace{\begin{bmatrix} 0 & 1 \end{bmatrix}}_{1 \times 2} \begin{bmatrix} R_{v^{(1)}}(X) & R_{v^{(2)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \\ f(v^{(1)}) & f(v^{(2)}) & \cdots & f(v^{(|G_D|)}) \end{bmatrix} \\ & \cdot \text{Softmax}_\beta \left(\underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{1 \times 2} \begin{bmatrix} R_{v^{(1)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \\ f(v^{(1)}) & \cdots & f(v^{(|G_D|)}) \end{bmatrix} \right)^\top \underbrace{\begin{bmatrix} 1 & 0 \end{bmatrix}}_{1 \times 2} \begin{bmatrix} R_{v^{(1)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \\ f(v^{(1)}) & \cdots & f(v^{(|G_D|)}) \end{bmatrix} \\ & = [f(v^{(1)}) \ \cdots \ f(v^{(|G_D|)})] \text{Softmax}_\beta \left(\begin{bmatrix} R_{v^{(1)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \end{bmatrix}^\top \begin{bmatrix} R_{v^{(1)}}(X) & \cdots & R_{v^{(|G_D|)}}(X) \end{bmatrix} \right) \\ & = [f(v^{(1)}) \ f(v^{(2)}) \ \cdots \ f(v^{(|G_D|)})] \text{Softmax}_\beta(M), \end{aligned} \quad (\text{F.50})$$

where M is given as

$$M_{i,j} := R_{v^{(i)}}(X) R_{v^{(j)}}(X), \quad i, j \in [|G_D|].$$

Until now we complete the proof of showing two attention layers with linear transform approximate $[f(v^{(1)}) \ f(v^{(2)}) \ \cdots \ f(v^{(|G_D|)})] \text{Softmax}_\beta(M)$.

To further calculate the approximation error $\text{Softmax}_\beta(M)$, we need to review some key attributes of $R_v(X)$. Hence we

recall some results from the proof of [Lemma 3.1](#).

Here we use the following attribute of $R_v(X)$

- $R_v(X) \in \{dn, dn+1, \dots, 2dn\}$ on $[-B, B]^{d \times n}$ except for a region no larger than $1 - (1 - \delta)^{dn}$. Here $\delta > 0$ is an self-selected coefficient we defined in the construction of A_1 and Attn_m . Thus by setting δ to be sufficiently large, the region of exception can be arbitrarily small.
- Except for an arbitrarily small region, the maximal $R_{v(i)}$ equals to $2dn$ and the second largest equals to $2dn - 1$.

Since Softmax is a column-wise operation, we calculate $\text{Softmax}_\beta(M)$ by column

$$\text{Softmax}_\beta(M)_{:,i} = \text{Softmax}_\beta \left(\begin{bmatrix} R_{v(i)} X R_{v(1)} X \\ R_{v(i)} X R_{v(2)} X \\ \vdots \\ R_{v(i)} X R_{v(|G_D|)} X \end{bmatrix} \right).$$

Then by [Lemma E.1](#), when β is sufficiently large

$$\|\text{Softmax}_\beta(M)_{:,i} - e_k\|_\infty \leq \epsilon_1,$$

for any $\epsilon_1 > 0$. Here k is defined as

$$k := \underset{k \in [|G_D|]}{\text{argmax}} (R_{v(k)} X).$$

Thus

$$\| [f(v^{(1)}) \quad f(v^{(2)}) \quad \dots \quad f(v^{(|G_D|)})] \text{Softmax}_\beta(M) - f(v^{(k)}) \|_\infty \leq \epsilon_1 \cdot \|f\|_{L_\infty}.$$

This excludes an arbitrarily small region where at least two entries in $\text{Softmax}_\beta(M)_{:,i}$ are identical. We denote this region as Δ_0 .

Combine this with [\(F.49\)](#) yields

$$\|\text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1(X) - f(v^{(k)})\|_\infty \leq \epsilon_1 \cdot \|f\|_{L_\infty} + d\epsilon_0. \quad (\text{F.51})$$

Finally we calculate the approximation error including the grid point approximation. For simplicity we denote the $\text{Attn}_s \circ A_2 \circ \text{Attn}_m \circ A_1(X) := \mathcal{N}(X)$. Then, we have

$$\|f(X) - \mathcal{N}(X)\|_{L_p} = \left(\int_{X \in [-B, B]^{d \times n}} \|f(X) - \mathcal{N}(X)\|_p^p dX \right)^{1/p}.$$

We split the domain into two parts as in [Section F.4](#)

$$\begin{aligned} & \|f(X) - \mathcal{N}(X)\|_{L_p} \\ &= \left(\int_{X \in [-B, B]^{d \times n}} \|f(X) - f(v^{(k)})\|_p^p dX + \int_{X \in [-B, B]^{d \times n}} \|f(v^{(k)}) - \mathcal{N}(X)\|_p^p dX \right)^{\frac{1}{p}} \quad (\text{By triangle inequality}) \\ &\leq \left(\int_{X \in [-B, B]^{d \times n}} \|f(X) - f(v^{(k)})\|_p^p dX + \int_{X \in [-B, B]^{d \times n} \setminus \Delta_0} \|f(v^{(k)}) - \mathcal{N}(X)\|_p^p dX \right. \\ &\quad \left. + \int_{X \in \Delta_0} \|f(v^{(k)}) - \mathcal{N}(X)\|_p^p dX \right)^{\frac{1}{p}} \quad (\text{Seperate } \Delta_0 \text{ out}) \\ &\leq (\varepsilon + (2B)^{dn} (\epsilon_1 \cdot \|f\|_{L_\infty} + d\epsilon_0) + \mu(\Delta_0) \cdot (2dnM_{fN})^p)^{\frac{1}{p}} \\ &\leq \varepsilon^{\frac{1}{p}} + 2B^{\frac{dn}{p}} (\epsilon_1 \cdot \|f\|_{L_\infty} + d\epsilon_0)^{\frac{1}{p}} + 2(\mu(\Delta_0))^{\frac{1}{p}} dnM_{fN}, \end{aligned}$$

where μ denotes the Lebesgue measure on $\mathbb{R}^{d \times n}$. Here, the ε in the third row inequality can be arbitrarily small when g is large enough, according to the discussion to derive (F.36). The error ϵ_1 is the softmax approximation error, and ϵ_0 is coming from (F.43). The term M_{fN} comes from

$$\|f(X) - \mathcal{N}(X)\|_{i,j} \leq \|f(X)\|_{L_\infty} + \|\mathcal{N}(X)\|_{L_\infty} \leq 2M_{fN},$$

where M_{fN} is a mutual upper-bound of the value of f and $\mathcal{N}$. Because both f and $\mathcal{N}$ are continuous on a compact support, they are bounded in ∞ norm and hence have a mutual upper-bound.

Configure Δ_0 , ϵ_0 and set g large enough

$$\varepsilon^{\frac{1}{p}} + 2B^{\frac{dn}{p}} (\epsilon_1 \cdot \|f\|_{L_\infty} + d\epsilon_0)^{\frac{1}{p}} + 2(\mu(\Delta_0))^{\frac{1}{p}} dn \|f\|_\infty \leq \epsilon.$$

This completes the proof. $\square$

F.6 Proof of Lemma 3.3

Lemma F.5 (Lemma 3.3 Restated: Single-Layer Attention Version of Lemma 3.2). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when attached with linear transformations, there exists a one layer multi-head attention Attn_m followed by a Softmax function and attached with linear connections A_1 and A_2 , such that

$$\|f - A_2 \circ \text{Softmax} \circ \text{Attn}_m \circ A_1\|_{L_p} \leq \epsilon.$$

Proof. Starting from (F.45) we have

$$\|\text{Attn}_m \circ A(X) - [R_{v(1)} \quad R_{v(2)} \quad \cdots \quad R_{v(|G_D|)}]\|_\infty \leq d\epsilon_0. \quad (\text{F.52})$$

Applying Softmax_β to (F.52) yields

$$\|\text{Softmax}_\beta(\text{Attn}_m \circ A(X)) - \text{Softmax}_\beta([R_{v(1)} \quad R_{v(2)} \quad \cdots \quad R_{v(|G_D|)}])\|_\infty \leq d\epsilon_0. \quad (\text{F.53})$$

Define a linear map A_2 as

$$A_2(Z) := Z[f(v_1), f(v_2), \dots, f(v_{|G_D|})]^\top,$$

and apply A_2 on (F.53) we have

$$\begin{aligned} & \|A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X))) - A_2(\text{Softmax}_\beta([R_{v(1)} \quad R_{v(2)} \quad \cdots \quad R_{v(|G_D|)}]))\|_\infty \\ &= \|\text{Softmax}_\beta(\text{Attn}_m \circ A(X)) [f(v_1) \quad f(v_2) \quad \cdots \quad f(v_{|G_D|})]^\top \\ &\quad - \text{Softmax}_\beta([R_{v(1)} \quad R_{v(2)} \quad \cdots \quad R_{v(|G_D|)}]) [f(v_1) \quad f(v_2) \quad \cdots \quad f(v_{|G_D|})]^\top\|_\infty \\ &\leq B_0 d\epsilon_0, \end{aligned}$$

where B_0 denotes $\|f\|_{L_\infty}$. It is bounded in ∞ norm since f is continuous on a compact domain.

Thus

$$\begin{aligned} & \|A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X))) - \text{Softmax}_\beta([R_{v(1)} \quad \cdots \quad R_{v(|G_D|)}]) [f(v^{(1)}) \quad \cdots \quad f(v^{(|G_D|)})]^\top\|_\infty \\ &\leq B_0 d\epsilon_0. \end{aligned}$$

When β is sufficiently large, except for an arbitrarily small region Δ of measure $\mu(\Delta)$ (the region in which $2R_v^{(i)}$ are nearly identical), by Lemma E.1, the following equation

$$\text{Softmax}_\beta([R_{v(1)} \quad R_{v(2)} \quad \cdots \quad R_{v(|G_D|)}]) [f(v^{(1)}) \quad \cdots \quad f(v^{(|G_D|)})]^\top,$$

approximates

$$e_{\arg\max_{i \in [|G_D|]} R_{v^{(i)}}}^\top \cdot [f(v^{(1)}) \quad f(v^{(2)}) \quad \dots \quad f(v^{(|G_D|)})]^\top = f(v^{\arg\max_i R_{v^{(i)}}}),$$

by an arbitrarily small error, we set this to be ϵ_1 .

Since the maximal $R_{v^{(i)}}(X)$ corresponds to the $v^{(i)}$ whose corresponding grid encapsulates X .

Thus X differs from $v^{(i)}$ on each dimension by a difference no larger than the grid length B/g .

When g is sufficiently large, $\|X - v^{(i)}\|_\infty$ is sufficiently small such that by the uniform continuity of f , we have

$$\|f(X) - f(v^{(i)})\|_\infty \leq \epsilon_1.$$

Thus

$$\begin{aligned} & \|f(X) - A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X)))\|_\infty \\ & \leq \|f(X) - f(v^{(i)})\|_\infty + \|f(v^{(i)}) - A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X)))\|_\infty \quad (\text{By triangle inequality}) \\ & \leq 2\epsilon_1, \end{aligned}$$

where the second line is by the triangle inequality.

This yields that

$$\begin{aligned} & \|f - A_2 \circ \text{Softmax}_\beta \circ \text{Attn}_m \circ A\|_{L_p} \\ & \leq \left(\int_{X \in [-B, B]^{d \times n} \setminus \Delta} \|f(X) - A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X)))\|_p^p dX \right. \\ & \quad \left. + \int_{X \in \Delta} \|f(X) - A_2(\text{Softmax}_\beta(\text{Attn}_m \circ A(X)))\|_p^p dX \right)^{\frac{1}{p}} \\ & \leq ((2B)^{dn} \cdot 2\epsilon_1 + dn \cdot (2\|f\|_\infty)^p \mu(\Delta))^{\frac{1}{p}}. \end{aligned}$$

Set ϵ_1 and $\mu(\Delta)$ to satisfy that

$$((2B)^{dn} \cdot 2\epsilon_1 + dn \cdot (2\|f\|_\infty)^p \mu(\Delta))^{\frac{1}{p}} \leq \epsilon.$$

We have

$$\|f - A_2 \circ \text{Softmax}_\beta \circ \text{Attn}_m \circ A\|_{L_p} \leq \epsilon.$$

This completes the proof. □

E.7 Proof of Theorem 3.3

Theorem F.4 (Theorem 3.3 Restated: Sequence-to-Sequence Approximation of Universal Approximation of Attention). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when attached with linear transformations, there exists a two layer multi-head attention Attn_m stacked with one layer multi-head attention Attn_s , attached with linear connection A_1 and A_2 , such that

$$\|f - \text{Attn}_m^{(2)} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1\|_{L_p} \leq \epsilon.$$

We first outline our proof sketch.

Proof Sketch. We first decompose the target function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ into dn scalar subfunctions $\{f_{ij}\}$, where $f_{ij} : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ for $i \in [d], j \in [n]$. By Lemma 3.2, each f_{ij} is approximated by one-layer multi-head attention (Attn_m) combined with one-layer single-head attention (Attn_s) and linear transformations (A_1, A_2), yielding a per-subfunction error

$$\|f_{ij}(X) - \text{Attn}_s^{ij} \circ A_2 \circ \text{Attn}_m \circ A_1(X)\|_p \leq \epsilon_{\text{scaler}}.$$

The first attention layer forms bump functions $R_{v^{(k)}}(x)$ to locate the relevant region of X , which does not depend on any particular f_{ij} . We then aggregate the dn approximations into a single matrix output by defining a second multi-head attention layer as

$$\text{Attn}_m^{(2)} = \sum_{i \in [d], j \in [n]} E^{ij} \text{Attn}_s^{ij},$$

where $E^{ij} \in \mathbb{R}^{d \times n}$ is all zeros except for a single 1 in the (i, j) position. Thus, each subfunction's approximation is placed in the correct row-column entry, yielding the full sequence-to-sequence approximation of f . The same logic applies to the proof of Corollary 3.3.1. Please see Section F.7 for a detailed proof. $\square$

Now we proceed to our detailed proof.

Proof. Given $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$, we decompose f into $f_{ij} : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$, where $i \in [d], j \in [n]$ denote the entry on the i -th row and the j -th column f . Thus

$$f(X) = \begin{bmatrix} f_{11}(X) & \cdots & f_{1n}(X) \\ \vdots & \ddots & \vdots \\ f_{d1}(X) & \cdots & f_{dn}(X) \end{bmatrix}.$$

By Lemma 3.2, we approximate each f_{ij} by a multi-head attention stacked with a single-head attention in the following way

$$\|f_{ij}(X) - \text{Attn}_s^{ij} \circ A_2 \circ \text{Attn}_m \circ A_1(X)\|_p \leq \epsilon_{\text{scaler}}. \quad (\text{F.54})$$

Recall that the goal of multi-head attention Attn_m in Lemma 3.2 is to approximate the bump function $R_{v^{(j)}}$ on all the grid point $v^{(j)} \in G_D$, hence it's irrelevant to the function f_{ij} we aim to approximate. The follow-up single-head attention Attn_s^{ij} is responsible to map out the function output, hence depends on the i, j .

One thing need to modify is the definition of A_2 in (F.46), we need to append dn rows of different function value for f_{ij}

$$A_2(Z) := \begin{bmatrix} 0 & 0 & \cdots & 0 \\ f_{11}(v^{(1)}) & f_{11}(v^{(2)}) & \cdots & f_{11}(v^{(|G_D|)}) \\ f_{12}(v^{(1)}) & f_{12}(v^{(2)}) & \cdots & f_{12}(v^{(|G_D|)}) \\ \vdots & \vdots & \ddots & \vdots \\ f_{dn}(v^{(1)}) & f_{dn}(v^{(2)}) & \cdots & f_{dn}(v^{(|G_D|)}) \end{bmatrix} + Z.$$

Also the second single layer attention need slight modification to pick out function f_{ij} among dn rows

$$\text{Attn}_s^{ij}(Z) = \underbrace{\begin{bmatrix} 0 & e_k \end{bmatrix}}_{1 \times (1+dn)} Z \text{Softmax} \left(\underbrace{\left(R \begin{bmatrix} 1 & 0_{1 \times dn} \end{bmatrix} Z \right)^\top}_{1 \times (1+dn)} \underbrace{\begin{bmatrix} 1 & 0_{1 \times dn} \end{bmatrix}}_{1 \times (1+dn)} Z \right) \mathbf{1}_{|G_D|}, \quad (\text{F.55})$$

where $k = (i-1)n + j$, and one-hot vector $e_k \in \mathbb{R}^{dn}$ is used to pick out the corresponding row.

This modification doesn't change the output after (F.50), hence the approximation error remain the same.

What remain is to combine this $d \times n$ approximations into one output matrix. To combine the scalar approximations back into a $\mathbb{R}^{d \times n}$ map, we use the matrices $E^{ij} \in \mathbb{R}^{d \times n}$ whose entries is zero everywhere except for value 1 on the i -th row and the j -th column.

Combining E^{ij} with Attn_s^{ij} we construct a new one-layer multi-head attention $\text{Attn}_m^{(2)}$ defined as

$$\text{Attn}_m^{(2)} = \sum_{i \in [d], j \in [n]} E^{ij} \text{Attn}_s^{ij}, \quad (\text{F.56})$$

Then by stacking with the same Attn_m denoted as $\text{Attn}_m^{(1)}$, we construct the sequence-to-sequence approximation to be $\text{Attn}_m^{(2)} \circ \text{Attn}_m^{(1)}$.

The error of approximation $\|f(X) - \text{Attn}_m^{(2)} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1(X)\|_p$, when requiring $\epsilon_{\text{scaler}} = \epsilon / ((dn)^{1/p})$ is

$$\begin{aligned} & \|f(X) - \text{Attn}_m^{(2)} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1(X)\|_p \\ &= \left(\sum_{i \in [d], j \in [n]} |f_{ij}(X) - (\text{Attn}_m^{(2)} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1(X))_{ij}|^p \right)^{\frac{1}{p}} \\ &= \left(\sum_{i \in [d], j \in [n]} |f_{ij}(X) - \text{Attn}_s^{ij} \circ A_2 \circ \text{Attn}_m^{(1)} \circ A_1(X)|^p \right)^{\frac{1}{p}} \quad (\text{By (F.56)}) \\ &\leq (dn \epsilon_{\text{scaler}}^p)^{\frac{1}{p}} \quad (\text{By (F.54)}) \\ &= (dn)^{\frac{1}{p}} \epsilon_{\text{scaler}} \\ &= \epsilon. \quad (\text{By } \epsilon_{\text{scaler}} = \frac{\epsilon}{(dn)^{1/p}}) \end{aligned}$$

For the case of using one attention layer following by a softmax function, by Lemma 3.3 we know

$$\|f_{ij}(X) - A_2 \circ \text{Softmax} \circ \text{Attn}_m \circ A_1(X)\|_p \leq \epsilon_{\text{scaler}}.$$

Again by modifying A_2 to A_2^{ij}

$$A_2^{ij}(Z) := E^{ij} Z \underbrace{\begin{bmatrix} f_{11}(v^{(1)}) & f_{12}(v^{(1)}) & \cdots & f_{dn}(v^{(1)}) \\ f_{11}(v^{(2)}) & f_{12}(v^{(2)}) & \cdots & f_{dn}(v^{(2)}) \\ \vdots & \vdots & \ddots & \vdots \\ f_{11}(v^{(|G_D|)}) & f_{12}(v^{(|G_D|)}) & \cdots & f_{dn}(v^{(|G_D|)}) \end{bmatrix}}_{v^{|G_D|} \times dn} e_{(i-1)n+j},$$

and since the modification doesn't change the error analysis in Section F.6, we have

$$\|f_{ij}(X) - A_2^{ij} \circ \text{Softmax} \circ \text{Attn}_m \circ A_1(X)\|_p \leq \epsilon_{\text{scaler}}.$$

We have

$$\|f(X) - \sum_{i \in [d], j \in [n]} A_2^{ij} \circ \text{Softmax} \circ \text{Attn}_m^{(1)} \circ A_1(X)\|_p$$

$$\begin{aligned}
&\leq \left(\sum_{i \in [d], j \in [n]} |f_{ij}(X) - A_2^{ij} \circ \text{Softmax} \circ \text{Attn}_m \circ A_1(X)|^p \right)^{\frac{1}{p}} \\
&\leq (dn \epsilon_{\text{scaler}}^p)^{\frac{1}{p}} && (\text{By (F.54)}) \\
&= (dn)^{\frac{1}{p}} \epsilon_{\text{scaler}} \\
&= \epsilon. && (\text{By } \epsilon_{\text{scaler}} = \frac{\epsilon}{(dn)^{1/p}})
\end{aligned}$$

This completes the proof. □

F.8 Proof of Theorem E.1

In this section, we prove the sequence-to-sequence universal approximation of a two-layer attention mechanism in the ℓ_∞ norm.

We first introduce a lemma modified from (Pinkus, 1999, Theorem 3.1) and show the universal approximation theory of one layer feed-forward neural network

Lemma F.6 (Theorem 3.1 from Pinkus (1999), Universal Approximation Of One Layer Feed-Forward Neural Network). Let $\sigma : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function. The space of functions defined by single-layer neural networks

$$M(\sigma) = \left\{ g(x) = \sum_{i=1}^N \eta_i \sigma(w_i \cdot x + t_i) \mid N \in \mathbb{N}, w_i \in \mathbb{R}^d, \eta_i, t_i \in \mathbb{R} \right\},$$

is dense in $C(K)$. Here, $C(K)$ represents the space of continuous functions on any compact domain $K \subset \mathbb{R}^d$, if and only if σ is not a polynomial. In other words, for any continuous function $f \in C(K)$ and any small error tolerance $\varepsilon > 0$, there exists a function $g \in M(\sigma)$ such that the maximum difference between f and g over K is less than ε (i.e., $\|f - g\|_\infty < \varepsilon$).

The ReLU activation function, $\sigma(x) = \max(0, x)$, satisfies the conditions of above lemma because it is continuous and not a polynomial. Therefore, single-layer neural networks with ReLU activations form a dense subset of $C(K)$, meaning that they approximate any continuous function on a compact set K to arbitrary precision in the infinity norm.

Next we introduce a simplified version of Theorem 3.1, where the only difference is we force the mapping function G maps to a constant r instead of $\tilde{k}_i \in [d_o]$ no matter what input k_i is.

Theorem F.5 (Single-Head Attention Approximates Many Truncated Linear Models). Fix real $a < b$, and let $\text{Range}_{[a,b]}(\cdot)$ be the truncation operator from Definition 3.1. For a precision parameter $p > n$ with $\epsilon = O(1/p)$, there exists a single-layer, single-head self-attention Attn with a linear transformation $A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$, such that $\text{Attn} \circ A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d_o \times n}$ satisfies, for any $i \in [n]$,

$$\begin{aligned} & \|\text{Attn} \circ A(X)_{:,i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) e_r\|_\infty \\ & \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{(b-a)/p}_{\text{interpolation error}}, \quad \text{for } i \in [n], \end{aligned}$$

where e_r is a one-hot vector with a value of 1 at the r -th index and 0 elsewhere, and $r \in [d_o]$ is defined as

$$\begin{aligned} k_i &:= \underset{k \in \{0,1,2,\dots,p-1\}}{\text{argmin}} |x_i^\top w + t - \tilde{L}_k|. \\ r &:= G(k_i), \end{aligned}$$

where r is any positive integer.

Now we are ready to prove attention approximate sequence-to-sequence function with a bounded error in the infinity norm.

Theorem F.6 (Theorem E.1 Restated: Sequence-to-Sequence Approximation in Infinity Norm). For any continuous function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$ of compact support $\mathcal{X}$, and any $\epsilon > 0$, we prove that when attached with linear transformations, there exists a one layer multi-head attention Attn_m stacked with one layer multi-head attention Attn_m , such that when the precision parameter in Theorem F.5 is $p = \Omega(n^{5/2})$, for any $X \in \mathcal{X}$

$$\|f(X) - \text{Attn}_m^{(2)} \circ A \circ \text{Attn}_m^{(1)} \circ A(X)\|_\infty \leq \epsilon.$$

Proof. Given $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d \times n}$, we decompose f into $f_{ij} : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$, where $i \in [d], j \in [n]$ denote the entry on the

i -th row and the j -th column f . Thus

$$f(X) = \begin{bmatrix} f_{11}(X) & \cdots & f_{1n}(X) \\ \vdots & \ddots & \vdots \\ f_{d1}(X) & \cdots & f_{dn}(X) \end{bmatrix}.$$

We aim to construct attention layer to approximate function f_{ij} in the form of

$$\text{FFN}(\text{vec}(X)) = \sum_{i=1}^N \eta_i \text{ReLU}(w_i^\top \text{vec}(X) + t_i),$$

where $\text{vec}(X) \in \mathbb{R}^{dn}$ is the flatten operation.

We achieve this by modifying the proof of [Theorem F.5](#) and sum over the multi-head attention output to make each entry of the multi-head attention output is in the form of

$$\sum_{i=1}^N \eta_i \text{ReLU}(w_i^\top X + t_i).$$

First, we set the mapping function G to map each k_i to the same row r , that is $G(k_i) = r$. Thus the value matrix V become

$$V = \begin{bmatrix} 0_{(r-1) \times p} \\ \tilde{L}^\top \\ 0_{(d_0-r) \times p} \end{bmatrix} \in \mathbb{R}^{d_0 \times p}, \quad (\text{F.57})$$

Then we modify the W_O matrix from having an identity matrix on the upper $n \times n$ block, to having a $n \times n$ matrix with the c -th column is $\eta e_c \in \mathbb{R}^n$ entry and other entry is 0

$$W_O = \begin{bmatrix} \eta e_c^\top \\ 0_{(p-n) \times n} \end{bmatrix} \in \mathbb{R}^{p \times n}, \quad (\text{F.58})$$

where $e_c \in \mathbb{R}^n$, then

$$V \text{Softmax}(K^\top Q) W_O = \underbrace{\begin{bmatrix} 0_{(r-1) \times p} \\ \tilde{L}^\top \\ 0_{(d_0-r) \times p} \end{bmatrix}}_{d_0 \times p} \underbrace{\begin{bmatrix} \alpha_1 & \alpha_2 & \cdots & \alpha_n & 0_{p \times (p-n)} \end{bmatrix}}_{p \times p} \underbrace{\begin{bmatrix} \eta e_c^\top \\ 0_{(p-n) \times n} \end{bmatrix}}_{p \times n},$$

then the following approximation error

$$\|V \text{Softmax}(K^\top Q) W_O - V[e_{k_1}, e_{k_2}, \dots, e_{k_n}]\|_\infty < |b| \cdot \epsilon_0,$$

should become

$$\|V \text{Softmax}(K^\top Q) W_O - \sum_{i=1}^n \eta_i \tilde{L}_{k_i} e_r e_c^\top\|_\infty < \|\eta\|_\infty \cdot n \cdot |b| \cdot \epsilon_0,$$

where the outer product $e_r e_c^\top$ create a matrix with 1 at (r, c) and 0 elsewhere. We denote the attention with modifications in [\(F.57\)](#) and [\(F.58\)](#) as $\text{Attn}_{r,c}$.

Lastly, the error of the interpolation point is

$$\left| \sum_{i=1}^n \eta_i \tilde{L}_{k_i} - \sum_{i=1}^n \eta_i \text{Range}_{[a,b]}(w_i^\top x_i + t_i) \right| \leq \sum_{i=1}^n \eta_i \left| \tilde{L}_{k_i} - \text{Range}_{[a,b]}(w_i^\top x_i + t_i) \right| \quad (\text{By triangle inequality})$$

$$= \|\eta\|_\infty \cdot n \cdot \frac{b-a}{p}.$$

Thus we have

$$\|\text{Attn}_{r,c}(X) - \cdot (\sum_{i=1}^n \eta_i \text{Range}_{[a,b]}(w_i^\top x_i + t_i)) e_r e_c^\top\|_\infty \leq \|\eta\|_\infty \cdot n \cdot (|b| \cdot \epsilon_0 + \frac{b-a}{p}). \quad (\text{F.59})$$

In fact, if we assume for every i we have $a \leq (w_i^\top x_i + t_i) \leq b$ and $\eta_i = 1$ for $i \in [n]$, the term $\sum_{i=1}^n \text{Range}_{[a,b]}(w_i^\top x_i + t_i)$ become

$$\sum_{i=1}^n \text{Range}_{[a,b]}(w_i^\top x_i + t_i) = \sum_{i=1}^n (w_i^\top x_i + t_i) = \tilde{w}^\top \tilde{x} + \tilde{t}, \quad (\text{F.60})$$

where $\tilde{x} \in \mathbb{R}^{dn}$ is the flatten vector of input sequence X , with $\tilde{w} = [w_1^\top, \dots, w_n^\top] \in \mathbb{R}^{dn}$ and $\tilde{t} = [t_1^\top, \dots, t_n^\top] \in \mathbb{R}^{dn}$.

Hence (F.59) become

$$\|\text{Attn}_{r,c}(X) - \cdot (\tilde{w}^\top \tilde{x} + \tilde{t}) e_r e_c^\top\|_\infty \leq n \cdot (|b| \cdot \epsilon_0 + \frac{b-a}{p}). \quad (\text{F.61})$$

Recall that we aim to approximate $f_{rc}(\cdot) : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}$ by showing attention mechanism approximate FFN with N neurons $\sum_{i=1}^N \eta_i \text{ReLU}(\tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i)$ in the (r, c) -th entry of the attention output. Until now we success to construct one-layer single-head attention layer $\text{Attn}_{r,c}(\cdot)$ whose output (r, c) entry is a linear model on the whole sequence $\eta(\tilde{w}^\top \tilde{x} + \tilde{t})$ by (F.61).

What left is to use a second attention layer in Theorem F.5 to create the ReLU function and sum them up. We know by Theorem F.5, the attention layer take one input token into truncated linear model.

We construct the first layer multi-head attention with $d_o n$ head, each head is $\text{Attn}_{r,c}^{(1)}(\cdot)$ for $r \in [d_o], c \in [n]$.

Pass it to the second layer of another $\text{Attn}_{r',c'}^{(2)}(\cdot)$ where $r' \in [d_o], c' \in [n]$, and $k \in [d_o]$ for later use, we have

$$\begin{aligned} \text{Attn}_{r',c'}^{(2)}\left(\sum_{r \in [d_o], c \in [n]} \text{Attn}_{r,c}^{(1)}(X)\right) &\approx \text{Attn}_{r',c'}^{(2)}\left(\underbrace{\begin{bmatrix} \tilde{w}_{1,1}^\top \tilde{x} + \tilde{t}_{1,1} & \cdots & \tilde{w}_{1,n}^\top \tilde{x} + \tilde{t}_{1,n} \\ \vdots & \vdots & \vdots \\ \tilde{w}_{d_o,1}^\top \tilde{x} + \tilde{t}_{d_o,1} & \cdots & \tilde{w}_{d_o,n}^\top \tilde{x} + \tilde{t}_{d_o,n} \end{bmatrix}}_{d_o \times n}\right) \quad (\text{By (F.61)}) \\ &\approx \sum_{i=1}^n \eta_i \text{Range}_{[a,b]}(w_k^\top \begin{bmatrix} \tilde{w}_{1,i}^\top \tilde{x} + \tilde{t}_{1,i} \\ \tilde{w}_{2,i}^\top \tilde{x} + \tilde{t}_{2,i} \\ \vdots \\ \tilde{w}_{d_o,i}^\top \tilde{x} + \tilde{t}_{d_o,i} \end{bmatrix} + t_k) e_{r'} e_{c'}^\top \quad (\text{By (F.59)}) \\ &= \sum_{i=1}^n \eta_i \text{ReLU}(\tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_{k,i}) e_{r'} e_{c'}^\top. \\ &\quad (\text{By letting } w_k = e_k, t_k = 0, a = 0, \text{ and } \tilde{w}_{k,i}^\top \tilde{x} + \tilde{t}_{k,i} \leq b) \end{aligned}$$

Denote $M := \sum_{r \in [d_o], c \in [n]} \text{Attn}_{r,c}^{(1)}(X)$ and $Y := \begin{bmatrix} \tilde{w}_{1,1}^\top \tilde{x} + \tilde{t}_{1,1} & \cdots & \tilde{w}_{1,n}^\top \tilde{x} + \tilde{t}_{1,n} \\ \vdots & \vdots & \vdots \\ \tilde{w}_{d_o,1}^\top \tilde{x} + \tilde{t}_{d_o,1} & \cdots & \tilde{w}_{d_o,n}^\top \tilde{x} + \tilde{t}_{d_o,n} \end{bmatrix}.$

The approximation error is

$$\begin{aligned}
 & \|\text{Attn}_{r',c'}^{(2)}(M) - \sum_{i=1}^n \eta_i \text{ReLU}(\tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i) e_r e_c^\top\|_\infty \\
 & \leq \|\text{Attn}_{r',c'}^{(2)}(M) - \text{Attn}_{r',c'}^{(2)}(Y)\|_\infty + \|\text{Attn}_{r',c'}^{(2)}(Y) - \sum_{i=1}^n \eta_i \text{ReLU}(\tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i) e_r e_c^\top\|_\infty \\
 & \leq \|\text{Attn}_{r',c'}^{(2)}(M) - \text{Attn}_{r',c'}^{(2)}(Y)\|_{2,\infty} + \|\text{Attn}_{r',c'}^{(2)}(Y) - \sum_{i=1}^n \eta_i \text{ReLU}(\tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i) e_r e_c^\top\|_\infty \\
 & \quad \quad \quad (\text{By } \|A\|_{\infty,\infty} \leq \|A\|_{2,\infty}) \\
 & \leq \|\text{Attn}_{r',c'}^{(2)}(M) - \text{Attn}_{r',c'}^{(2)}(Y)\|_{2,\infty} + n(|b| \cdot \epsilon_0 + \frac{b}{p}) \quad (\text{By (F.59)}) \\
 & \leq \|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \|A(M) - A(Y)\|_{2,\infty} + n(|b| \cdot \epsilon_0 + \frac{b}{p}) \\
 & \quad \quad \quad (\text{By lipschitzness of attention modifying from (Edelman et al., 2022, Lemma A.14)}) \\
 & = \|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \|M - Y\|_{2,\infty} + n(|b| \cdot \epsilon_0 + \frac{b}{p}) \quad (A \text{ preserves column norm}) \\
 & \leq \|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \sqrt{n} \|M - Y\|_\infty + n(|b| \cdot \epsilon_0 + \frac{b}{p}) \quad (\text{By } \|A\|_{2,\infty} \leq \|A\|_\infty) \\
 & \leq \|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \cdot d_o n^{\frac{5}{2}} (|b| \cdot \epsilon_0 + \frac{b}{p}) + n(|b| \cdot \epsilon_0 + \frac{b}{p}) \\
 & = (|b| \cdot \epsilon_0 + \frac{b}{p}) \cdot (\|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \cdot d_o n^{\frac{5}{2}} + n), \quad (\text{F.62})
 \end{aligned}$$

where we show $A(\cdot)$ does not change the maximum column norm and the term $\|M - Y\|_\infty$ is bounded as follow. To see why $A(\cdot)$ preserves the maximum column norm, from (F.4) we know

$$\begin{aligned}
 A(X) &= \underbrace{\begin{bmatrix} I_d \\ 0_{(d+d_o+2) \times d} \end{bmatrix}}_L X \underbrace{\begin{bmatrix} I_n, 0_{n \times (p-n)} \end{bmatrix}}_R + \text{Constant} \\
 &= LXR + \text{Constant},
 \end{aligned}$$

and we have

$$A(X_1) - A(X_2) = L(X_1 - X_2)R.$$

Let $X = X_1 - X_2$, we aim to show

$$\|LXR\|_{2,\infty} = \|X\|_{2,\infty}.$$

By definition we have

$$\|LXR\|_{2,\infty} = \max_{j \in [p]} \|(LXR)_{:,j}\|_2,$$

that is, the maximum Euclidean norm of column vector of LXR . However, we know the effect of XR is to create 0 column vector on the right of X , and $L(XR)$ just create zero row vector to X . Hence, the maximum Euclidean norm of column vector of X is the same as that of LXR , that is $\|LXR\|_{2,\infty} = \|X\|_{2,\infty}$.

To bound the term $\|M - Y\|_\infty$, first denote $E_{r,c} = \text{Attn}_{r,c}^{(1)}(X) - (\tilde{w}_{r,c}^\top \tilde{x} + \tilde{t}_{r,c}) e_r e_c^\top$, then $M - Y = \sum_{r,c} E_{r,c}$. We have

$$\begin{aligned}
 \|M - Y\|_\infty &= \left\| \sum_{r=1}^{d_o} \sum_{c=1}^n E_{r,c} \right\|_\infty \\
 &\leq \sum_{r=1}^{d_o} \sum_{c=1}^n \|E_{r,c}\|_\infty && \text{(By triangle inequality)} \\
 &\leq \sum_{r=1}^{d_o} \sum_{c=1}^n n(|b| \cdot \epsilon_0 + \frac{b}{p}) && \text{(By (F.61))} \\
 &= d_o n^2 (|b| \cdot \epsilon_0 + \frac{b}{p}).
 \end{aligned}$$

For the bound on (F.62), ϵ_0 is arbitrarily small when β in softmax function is sufficiently large. If we further set $p = \Omega(n^{5/2})$, (F.62) is bounded or be arbitrary small when n increase.

Hence for now we construct a multihead attention whose output (r', c') entry is an approximation of an FFN that is a universal approximator of every continuous function defined on compact domain $f_{r',c'} : \mathbb{R}^{dn} \rightarrow \mathbb{R}$.

Thus combine the error of attention approximate ReLU neural network and the error of ReLU network approximate target function we have

$$\begin{aligned}
 &\left| \left(\text{Attn}_{r',c'}^{(2)} \left(\sum_{r \in [d_o], c \in [n]} \text{Attn}_{r,c}^{(1)}(X) \right) \right)_{r',c'} - f_{r',c'}(X) \right| \\
 &\leq \left| \left(\text{Attn}_{r',c'}^{(2)} \left(\sum_{r,c} \text{Attn}_{r,c}^{(1)}(X) \right) \right)_{r',c'} - \sum_{i=1}^n \tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i \right| + \left| \sum_{i=1}^n \tilde{w}_{n(r'-1)+c',i}^\top \tilde{x} + \tilde{t}_i - f_{r',c'}(X) \right| \\
 &\leq (|b| \cdot \epsilon_0 + \frac{b}{p}) \cdot (\|W_O\|_\infty \|W_V^\top\|_2 (1 + 4\|W_K^\top W_Q\|_2) \cdot d_o n^{\frac{5}{2}} + n) + \varepsilon \leq \epsilon, \quad \text{for } p = \Omega(n^{5/2}). \quad (\text{F.63})
 \end{aligned}$$

where ε is the approximation error of the ReLU network, and the second inequality comes from triangle inequality. The universal approximation theory of ReLU neural network in infinity norm is shown in (Pinkus, 1999, Theorem 3.1).

Note that we remove the restriction that the neurons of FFN we aim to approximate is restricted by n by increasing the output sequence length of the first layer $\text{Attn}_{r,c}$. We achieve simply by increasing n in the matrix W_O of our attention to arbitrary positive integer N .

Finally, by constructing dn head of this second layer attention $\text{Attn}_{r',c'}^{(2)}$, and set $d_o = dn$ in the first layer attention $\sum_{r \in [d_o], c \in [N]} \text{Attn}_{r,c}^{(1)}$ we get

$$\begin{aligned}
 &\|f(X) - \sum_{r' \in [d], c' \in [n]} \text{Attn}_{r',c'}^{(2)} \left(\sum_{r \in [d_o], c \in [N]} \text{Attn}_{r,c}^{(1)}(X) \right)\|_\infty \\
 &= \max_{r' \in [d], c' \in [n]} \left| f_{r',c'}(X) - \left(\text{Attn}_{r',c'}^{(2)} \left(\sum_{r \in [d_o], c \in [N]} \text{Attn}_{r,c}^{(1)}(X) \right) \right)_{r',c'} \right| \\
 &\leq \epsilon. && \text{(By (F.63) each } (r, c)\text{-th difference is at most } \epsilon, \text{ the } \max_{r,c} \text{ is also most } \epsilon)
 \end{aligned}$$

This completes the proof. $\square$

F.9 Proofs of Theorem C.1

Remark F.8 (Key Technique).

$$\begin{aligned}
 & \operatorname{argmin}_{k \in [p]} (w^\top x_i + t - \tilde{L}_k)^2 \\
 &= \operatorname{argmin}_{k \in [p]} (-2w^\top x_i - 2t) \cdot \tilde{L}_k + \tilde{L}_k^2 + (w^\top x_i + t)^2 \\
 &= \operatorname{argmin}_{k \in [p]} (-2w^\top x_i - 2t) \cdot (\tilde{L}_k - \tilde{L}_1) - \tilde{L}_1^2 + \tilde{L}_k^2 && ((w^\top x_i + t) \text{ and } \tilde{L}_1 \text{ are constant w.r.t. } k) \\
 &= \operatorname{argmin}_{k \in [p]} (-2w^\top x_i - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (\tilde{L}_k - \tilde{L}_1) \\
 &= \operatorname{argmin}_{k \in [p]} (-2w^\top x_i - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (k - 1) \Delta L \\
 &= \operatorname{argmin}_{k \in [p]} (-2w^\top x_i - 2t + \tilde{L}_1 + \tilde{L}_k) \cdot (k - 1). && (\text{Multiply a positive constant doesn't change } \operatorname{argmin}(\cdot))
 \end{aligned}$$

We first prove the in-context version of our main theorem.

Theorem F.7 (Theorem C.1 Restated). Fix real numbers $a < b$, and let the truncation operator $\operatorname{Range}_{[a,b]}(\cdot)$ follow Definition 3.1. Let the input be

$$X = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ w & w & \cdots & w \\ t & t & \cdots & t \end{bmatrix} \in \mathbb{R}^{(2d+1) \times n},$$

where $\{w, x_i\}_{i \in [n]}$ are bounded. Let $\epsilon_0 \geq 0$. For a precision parameter $p > n$, there exists a single-layer, single-head self-attention Attn with a linear transformation $A : \mathbb{R}^{(2d+1) \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$, such that $\operatorname{Attn} \circ A : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{d_o \times n}$ satisfies, for any $i \in [n]$,

$$\|\operatorname{Attn} \circ A(X)_{:,i} - \operatorname{Range}_{[a,b]}(w^\top x_i + t) e_{\tilde{k}_i}\|_\infty \leq \underbrace{\max\{|a|, |b|\} \cdot \epsilon_0}_{\text{finite-}\beta \text{ softmax error}} + \underbrace{\frac{b-a}{p}}_{\text{interpolation error}}.$$

Here $e_{\tilde{k}_i}$ is a one-hot vector with a value of 1 at the $\tilde{k}_i$ -th index and 0 elsewhere, and

$$k_i := \operatorname{argmin}_{k \in [p]} |x_i^\top w + t - \tilde{L}_k| \quad \text{where} \quad \tilde{k}_i := G(k_i) \in [d_o].$$

Here $k_i \in [p]$ is the index of the interpolation point closest to the i -th token (i -th truncated linear model). For all $i \in [n]$, $G : [p] \rightarrow [d_o]$ denotes any set-to-set function sending the interpolation index $k \in [p]$ into a position index $\tilde{k} \in [d_o]$ specifying in the desired row index of the output. By setting $\beta \geq (\ln(n-1) - \ln \epsilon_0)/\delta$, we make ϵ_0 arbitrarily small, though the theorem fails on a arbitrarily small volume in $\mathbb{R}^{d \times n}$. When $\tilde{k}_i$ a constant for all $i \in [n]$, by setting $\beta \geq (\ln(n-2) - \ln \epsilon_0)/((\Delta L)^2/2)$, we achieve arbitrary small ϵ_0 without any failure region.

Proof. Before we plug the input token to the self-attention, we preprocess it with linear transformations $A : \mathbb{R}^{(2d+1) \times n} \rightarrow \mathbb{R}^{(2d+d_o+2) \times p}$. Without loss of generality, we set the precision parameter $p \in \mathbb{N}$ defined in Definition 3.3 to be larger than input sequence length n .

Following Definition 3.3, define

$$\ell_k := (k-1)(\tilde{L}_k + \tilde{L}_1 - 2t), \quad k \in [p].$$

Define a linear transform A such that

$$\begin{aligned}
 A(X) &= \underbrace{\begin{bmatrix} I_d & 0_{d \times d+1} \\ 0_{(d+d_o+2) \times d} & 0_{(d+d_o+2) \times (d+1)} \end{bmatrix}}_{(2d+d_o+2) \times (2d+1)} X \underbrace{\begin{bmatrix} I_n, 0_{n \times (p-n)} \end{bmatrix}}_{n \times p} \\
 &+ \underbrace{\begin{bmatrix} 0_{d \times d} & 0_{d \times d} & 0_d \\ 0_{d \times d} & I_d & 0_d \\ 0_{1 \times d} & 0_{1 \times d} & -2 \\ 0_{(d_o+1) \times d} & 0_{(d_o+1) \times d} & 0_{d_o+1} \end{bmatrix}}_{(2d+d_o+2) \times (2d+1)} X \underbrace{\begin{bmatrix} 0 & 1 & \cdots & (p-1) \\ 0_{n-1} & 0_{n-1} & \cdots & 0_{n-1} \end{bmatrix}}_{n \times p} \\
 &+ \underbrace{\begin{bmatrix} 0_d & 0_d & \cdots & 0_d & 0_d & \cdots & 0_d \\ 0_d & 0_d & \cdots & 0_d & 0_d & \cdots & 0_d \\ 0 & \tilde{L}_2 + \tilde{L}_1 & \cdots & (n-1)(\tilde{L}_n + \tilde{L}_1) & n(\tilde{L}_{n+1} + \tilde{L}_1) & \cdots & (p-1)(\tilde{L}_p + \tilde{L}_1) \\ & & & \tilde{L}_{d_o \times p} & & & \\ 1 & 1 & \cdots & 1 & 0 & \cdots & 0 \end{bmatrix}}_{(2d+d_o+2) \times p} \\
 &= \underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n & 0 & \cdots & 0 \\ 0_d & w & \cdots & (n-1)w & nw & \cdots & (p-1)w \\ 0 & \ell_2 & \cdots & \ell_n & \ell_{n+1} & \cdots & \ell_p \\ & & & \tilde{L}_{d_o \times p} & & & \\ 1 & 1 & \cdots & 1 & 0 & \cdots & 0 \end{bmatrix}}_{(2d+d_o+2) \times p},
 \end{aligned}$$

where $\tilde{L} = [\tilde{L}_1 e_{\tilde{1}}, \dots, \tilde{L}_p e_{\tilde{p}}] \in \mathbb{R}^{d_o \times p}$.

The output of this linear mapping is the same as the linear mapping output (F.4) in Section F.2.1, hence the remaining proof is the same.

This completes the proof. $\square$

F.10 Proofs of Theorem C.2

Theorem F.8 (Restate of Theorem C.2). Let $l : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ be any C^1 convex loss function defined on $(w^\top x_i, y_i)$. With input X in the form of Definition 4.1, when X is bounded, there exists a multi-head self-attention Attn_m whose parameters are irrelevant X , with skip connections and each attached with a linear layer, such that for any $\epsilon > 0$, we have

$$\left\| \text{Attn}_m \circ A(X) - \begin{bmatrix} x_1 & \cdots & x_n \\ y_1 & \cdots & y_n \\ w - \eta \nabla L(w) & \cdots & w - \eta \nabla L(w) \\ 1 & \cdots & 1 \end{bmatrix} \right\|_\infty \leq \epsilon,$$

where η denotes the learning rate and $L(w) := (1/n) \sum_{i=1}^n l(w^\top x_i, y_i)$ is an empirical loss upon the given input-output pairs.

Proof. The main goal of the proof is to show multihead attention approximate $\nabla L(w) = (1/n) \sum_{i=1}^n \frac{\partial}{\partial w} l(w^\top x_i, y_i)$.

Our proof consists of the following steps:

1. Approximate the derivative of loss function $l_w(w^\top x_i, y_i) := \frac{\partial}{\partial w} l(w^\top x_i, y_i)$ by classical ReLU neural network.
2. Use single-head attention to approximate a ReLU nested linear function by Theorem C.1.
3. Use multihead attention to aggregate the ReLU function to approximate ReLU neural network (NN).
4. Combine the error of multihead attention approximate ReLU NN and the error of ReLU NN approximate $\frac{\partial}{\partial w} l(w^\top x_i, y_i)$.
5. Design W_O^* matrix to sum over $\frac{\partial}{\partial w} l(w^\top x_i, y_i)$ on different in-context example (x_i, y_i) to get $\nabla L(w)$.

We now begin our proof.

Since l is C^1 , the derivative of l to w is continuous. By standard universal approximation results (Pinkus, 1999), there exists a set of ReLU neural network with parameter $a_h^{(r)}, b_h^{(r)}, c_h^{(r)} \in \mathbb{R}$ bounded by B_R , for all $h \in [H]$. The subscript $r \in [d]$ indicates the r -th coordinate of partial derivative $l_w(w^\top x_i, y_i)$, such that for any $\epsilon_0 > 0$

$$\| (l_w(w^\top x_i, y_i))_{r,\cdot} - \sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \|_\infty \leq \epsilon_0. \quad (\text{F.64})$$

We begin to construct multihead attention with linear mapping to approximate $\sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)})$.

Construct a linear transform $L_{h,r} \in \mathbb{R}^{(2d+4) \times (2d+2)}$ to be

$$L_{h,r} := \begin{bmatrix} \overbrace{\text{diag}(a_h^{(r)} 1_{1 \times d}, b_h^{(r)})}^{(d+1) \times (d+1)} & 0_{d \times d} & 0_{(d+1) \times 1} \\ 0_{1 \times (d+1)} & 0_{1 \times d} & 1 \\ 0_{(d+1) \times (d+1)} & I_{d \times d} & 0_{d \times 1} \\ 0_{1 \times (d+1)} & 0_{1 \times d} & c_h^{(r)} \end{bmatrix}.$$

$L_{h,r}(X)$ outputs

$$L_{h,r}(X) = \begin{bmatrix} \overbrace{\text{diag}(a_h^{(r)} 1_{1 \times d}, b_h^{(r)})}^{(d+1) \times (d+1)} & 0_{d \times d} & 0_{(d+1) \times 1} \\ 0_{1 \times (d+1)} & 0_{1 \times d} & 1 \\ 0_{(d+1) \times (d+1)} & I_{d \times d} & 0_{d \times 1} \\ 0_{1 \times (d+1)} & 0_{1 \times d} & c_h^{(r)} \end{bmatrix} \underbrace{\begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n \\ w & w & \cdots & w \\ 1 & 1 & \cdots & 1 \end{bmatrix}}_{(2d+2) \times n}$$

$$= \underbrace{\begin{bmatrix} a_h^{(r)} x_1 & a_h^{(r)} x_2 & \cdots & a_h^{(r)} x_n \\ b_h^{(r)} y_1 & b_h^{(r)} y_2 & \cdots & b_h^{(r)} y_n \\ 1 & 1 & \cdots & 1 \\ w & w & \cdots & w \\ 1 & 1 & \cdots & 1 \\ c_h^{(r)} & c_h^{(r)} & \cdots & c_h^{(r)} \end{bmatrix}}_{(2d+4) \times n}.$$

View $[x_i^\top, y_i, 1]$ as a whole input vector corresponding to the x_i in [Theorem C.1](#), view $[w^\top, 1, c_h^{(r)}]$ as the w in [Theorem C.1](#), and set $t = 0$. Let B_1 denote the bound of $\|X\|_1 = \sum_{i,j} |X_{ij}|$, then according to [Theorem C.1](#), there exists a Attn_h^* and A_h^* such that the i -th column of output satisfy

$$\begin{aligned} & \|\text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \text{Range}_{[0, B_R B_1^2]}((w, 1, c_h^{(r)})^\top \underbrace{\begin{bmatrix} a_h^{(r)} x_i \\ b_h^{(r)} y_i \\ 1 \end{bmatrix}}_{(2d+2) \times 1})\|_\infty \\ & \quad \text{(By selecting the output dimension in [Theorem C.1](#) to be 1)} \\ & \leq \epsilon_1, \quad \text{for } i \in [n], \end{aligned} \tag{F.65}$$

for any $\epsilon_1 > 0$.

Notice that

$$\begin{aligned} \left| (a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \right| & \leq |a_h^{(r)}| B_1^2 + |b_h^{(r)}| B_1 + |c_h^{(r)}| \quad (\text{By } \|x_i\|_1 \leq B_1, |y_i| \leq B_1 \text{ and } \|w\|_1 \leq B_1.) \\ & \leq B_R B_1^2, \end{aligned}$$

the truncated linear model $\text{Range}_{[0, B_R B_1^2]} \cdot \text{ReLU}(\cdot)$

$$\text{Range}_{[0, B_R B_1^2]}((w, 1, c_h^{(r)})^\top \begin{bmatrix} a_h^{(r)} x_i \\ b_h^{(r)} y_i \\ 1 \end{bmatrix}) = \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}).$$

Hence (F.65) become

$$\|\text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1}\|_\infty \leq \epsilon_1, \quad i \in [d]$$

Thus, summing the H head output we get

$$\left\| \sum_{h=1}^H \text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} \right\|_\infty \leq H \epsilon_1, \quad r \in [d].$$

Until now we success to construct multihead attention $\sum_{h=1}^H \text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))$ to approximate $\sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)})$ on the $(d+1+r, i)$ entry.

Combine the above expression for all $r \in [d]$ we have

$$\left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \sum_{h=1}^H \sum_{r=1}^d \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} \right\|_\infty \leq d \cdot H \epsilon_1 \tag{F.66}$$

We first bound the error for ReLU neural network to approximate $l_w(w^\top x_i, y_i)$.

By (F.64) we derive

$$\begin{aligned}
 & \left\| \sum_{h=1}^H \sum_{r=1}^d \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} - \begin{bmatrix} \overbrace{l_w(w^\top x_i, y_i)}^{(d+1) \times 1} \\ d \times 1 \\ 0 \end{bmatrix} \right\|_\infty \\
 & \leq \sum_{r=1}^d \left\| \sum_{h=1}^H \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} - \begin{bmatrix} \overbrace{l_w(w^\top x_i, y_i)}^{(d+1) \times 1} \\ d \times 1 \\ 0 \end{bmatrix} \right\|_\infty \quad (\text{By triangle inequality}) \\
 & \leq d\epsilon_0,
 \end{aligned}$$

where the last line is from (F.64).

Combine this with (F.66), we have

$$\begin{aligned}
 & \left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \begin{bmatrix} \overbrace{l_w(w^\top x_i, y_i)}^{(d+1) \times 1} \\ d \times 1 \\ 0 \end{bmatrix} \right\|_\infty \\
 & \leq \left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r}^* \circ A_{h,r}^*(L_{h,r}(X))_{:,i} - \sum_{h=1}^H \sum_{r=1}^d \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} \right\|_\infty \\
 & \quad + \left\| \sum_{h=1}^H \sum_{r=1}^d \text{ReLU}(a_h^{(r)} w^\top x_i + b_h^{(r)} y_i + c_h^{(r)}) \underbrace{e_{d+1+r}}_{(2d+2) \times 1} - \begin{bmatrix} \overbrace{l_w(w^\top x_i, y_i)}^{(d+1) \times 1} \\ d \times 1 \\ 0 \end{bmatrix} \right\|_\infty \\
 & \leq dH\epsilon_1 + d\epsilon_0.
 \end{aligned}$$

What left is to sum up the derivative of loss function $l_w(w^\top x_i, y_i)$ on n in-context example x_i, y_i for $i \in [n]$ to form $\nabla L(w) = (1/n) \sum_{i=1}^n l_w(w^\top x_i, y_i)$.

We construct W_O^* and integrate it into the original W_O^* of Attn to turn the loss gradient into a step of gradient descent

$$W_O^* := \begin{bmatrix} -\frac{\eta}{n} 1_n & -\frac{\eta}{n} 1_n & \cdots & -\frac{\eta}{n} 1_n \end{bmatrix}.$$

Now we define the final form of our network

$$\begin{aligned}
 \text{Attn}_{h,r}^* &= \text{Attn}_{h,r}^*(Z) W_O^*, \\
 A_h(Z) &= A_{h,r}^* \circ L_{h,r}(Z)
 \end{aligned}$$

Thus we have

$$\begin{aligned}
 & \left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r} \circ A_{h,r}(X) - \left(-\eta \begin{bmatrix} \underbrace{0}_{(d+1) \times 1} \\ \underbrace{\nabla L(w)}_{d \times 1} \\ 0 \end{bmatrix} \right) \right\|_{\infty} \\
 &= \left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r} \circ A_{h,r}(X) - \left(-\frac{\eta}{n} \sum_{i=1}^n \begin{bmatrix} \underbrace{0}_{(d+1) \times 1} \\ \underbrace{l_w(w^{\top} x_i, y_i)}_{d \times 1} \\ 0 \end{bmatrix} \right) \right\|_{\infty} \quad (\text{By } W_O^*) \\
 &\leq dH\epsilon_1 + d\epsilon_0.
 \end{aligned}$$

With skip connections, we have

$$\left\| \sum_{h=1}^H \sum_{r=1}^d \text{Attn}_{h,r} \circ A_{h,r}(X) + X - \left(X - \frac{\eta}{n} \sum_{i=1}^n \begin{bmatrix} \underbrace{0}_{(d+1) \times 1} \\ \underbrace{l_w(w^{\top} x_i, y_i)}_{d \times 1} \\ 0 \end{bmatrix} \right) \right\|_{\infty} \leq dH\epsilon_1 + d\epsilon_0,$$

where

$$X - \frac{\eta}{n} \sum_{i=1}^n \begin{bmatrix} \underbrace{0}_{(d+1) \times 1} \\ \underbrace{l_w(w^{\top} x_i, y_i)}_{d \times 1} \\ 0 \end{bmatrix} = \begin{bmatrix} x_1 & x_2 & \cdots & x_n \\ y_1 & y_2 & \cdots & y_n \\ w - \eta \nabla L(w) & w - \eta \nabla L(w) & \cdots & w - \eta \nabla L(w) \\ 1 & 1 & \cdots & 1 \end{bmatrix}.$$

Setting $dH\epsilon_1 + d\epsilon_0 \leq \epsilon$ yields the final result.

This completes the proof. $\square$

G ReLU, Hard Tanh and Clipped ReLU Activation Functions

Example 1 (Truncated Linear Model Subsumes ReLU). When $a = 0$ and $b \rightarrow \infty$, $\text{Range}_{[a,b]}(w^{\top} x + t)$ reduces to the standard ReLU (ramp function). Conversely, choosing finite values for a and b saturates the function on both ends, effectively making $\text{Range}_{[a,b]}(w^{\top} x + t)$ a double-sided ReLU. It retains the piecewise linearity essential for universal approximation while bounding the output values.

Example 2 (Truncated Linear Model Subsumes Hard Tanh). Consider $a = -1$ and $b = +1$. Then $\text{Range}_{[a,b]}(x)$ becomes the hard tanh activation:

$$\text{HardTanh}(x) = \begin{cases} -1, & x \leq -1, \\ x, & -1 < x < +1, \\ +1, & x \geq +1. \end{cases}$$

Thus, truncated linear functions recover this bounded, piecewise-linear activation.

Example 3 (Truncated Linear Model Subsumes Clipped ReLU). When $a = 0$ and $b > 0$ is finite, $\text{Range}_{[0,b]}(x)$ matches a

clipped ReLU. That is,

$$\text{ClippedReLU}_{[0,b]}(x) = \max\{0, \min\{x, b\}\},$$

which maintains linearity in the interval $[0, b]$ and saturates at both ends.

H Transformer-Native Sequence-to-Sequence Universal Approximation with Softmax Attention

This section extends the softmax attention sequence-to-sequence approximation result of [Theorem 3.3](#) to a more Transformer-native setting.

Lemma H.1 (Attention Simulates Column-Wise Linear Transformations). Let $X \in \mathbb{R}^{d \times n}$ and let

$$\ell(X) := AXB \in \mathbb{R}^{d_{\text{out}} \times n}, \quad A \in \mathbb{R}^{d_{\text{out}} \times d}, \quad B \in \mathbb{R}^{n \times n}$$

be a linear map that is token-wise in A and sequence-wise in B . Assume that all entries of B are strictly positive.^a Consider the augmented input

$$Z := \begin{bmatrix} X & 0_d \\ I_n & 0_n \\ 0_{1 \times n} & 1 \end{bmatrix} \in \mathbb{R}^{(d+n+1) \times (n+1)},$$

where $0_d \in \mathbb{R}^{d \times 1}$ be the all-zeros vector^b. Then for any $\epsilon > 0$, there exists a single-head attention

$$\text{Attn}(Z) = W_V Z \cdot \text{Softmax}((W_K Z)^\top (W_Q Z))$$

such that

$$\left\| \text{Attn}(Z) - \begin{bmatrix} \ell(X) & 0_{d_{\text{out}}} \end{bmatrix} \right\|_\infty \leq \epsilon.$$

^aAny matrix B admits an decomposition $B^+ - B^-$ with $B^+, B^- \geq 0$. The attention construction for positive matrices applies separately to B^+ and B^- , and combine through multi-head architecture yields the general case.

^bThe all-zeros vector is common in real-world transformer. The attention learn and attend to zero-value tokens (such as [SEP]) when current input have sufficient information and model want to do nothing (Xiao et al., 2024; Bondarenko et al., 2023; Hu et al., 2024; Ansari et al., 2025).

Proof Sketch. The goal is to realize the linear map $X \mapsto AXB$ in the first n output columns and to keep the last (padding) column close to 0. The construction proceeds in four steps:

1. Choose W_V so that the values store $3MAX$ for real tokens and 0 for the padding token.
2. Choose W_K and W_Q so that the first n attention columns implement mixing by $B/(3M)$.
3. Use a large parameter T in W_Q so that the last attention column concentrates on the padding token, which yields an output close to 0 there.

□

Proof. For each column $i \in [n]$ of B , set

$$s_i := \sum_{r=1}^n B_{ri}, \quad S := (s_1, \dots, s_n) \in \mathbb{R}^{1 \times n},$$

and let

$$M := \max_{i \in [n]} s_i.$$

Strict positivity of the entries implies $0 < B_{ri} \leq M$ and

$$3M - s_i \geq 2M > 0$$

for all r, i . The constant M will serve as a common denominator for the softmax normalization.

Step 1: Values V store AX and Ignore the Padding Token.

Define

$$W_V := 3M \begin{bmatrix} A & 0_{d_{\text{out}} \times (n+1)} \end{bmatrix} \in \mathbb{R}^{d_{\text{out}} \times (d+n+1)}.$$

Then

$$V := W_V Z = 3MA \begin{bmatrix} X & 0_d \end{bmatrix} \in \mathbb{R}^{d_{\text{out}} \times (n+1)}.$$

Step 2: Keys and Queries Implement Mixing by B .

Let $1_n \in \mathbb{R}^{n \times 1}$ be the all-ones vector and let $T > 0$ be a scalar parameter (chosen later). Set

$$W_Q := \begin{bmatrix} 0_{n \times d} & I_n & T1_n \end{bmatrix} \in \mathbb{R}^{n \times (d+n+1)}.$$

Writing $Z = [z_1, \dots, z_{n+1}]$, one obtains

$$Q_{:,j} := W_Q z_j = \begin{cases} e_j, & j \leq n, \\ T1_n, & j = n+1, \end{cases}$$

so

$$Q = W_Q Z = [e_1, \dots, e_n, T1_n] \in \mathbb{R}^{n \times (n+1)}.$$

For the keys, define

$$W_K := \begin{bmatrix} 0_{n \times d} & \ln(B^\top) & \ln(3M1_n - S^\top) \end{bmatrix} \in \mathbb{R}^{n \times (d+n+1)},$$

where the logarithm appears entrywise and $\ln(3M1_n - S^\top) \in \mathbb{R}^{n \times 1}$ has k -th entry $\ln(3M - s_k)$. Then $K := W_K Z \in \mathbb{R}^{n \times (n+1)}$ satisfies, for $i \leq n$,

$$K_{:,i} = \ln(B^\top) e_i,$$

whose r -th entry equals $(K_{:,i})_r = \ln B_{ir}$, and

$$K_{:,n+1} = \ln(3M1_n - S^\top),$$

whose r -th entry equals $(K_{:,n+1})_r = \ln(3M - s_r)$.

Now consider the score matrix

$$Y := K^\top Q \in \mathbb{R}^{(n+1) \times (n+1)},$$

with entries $Y_{ij} = \langle K_{:,i}, Q_{:,j} \rangle$.

For $i \leq n$ and $j \leq n$,

$$Y_{ij} = \langle \ln(B^\top) e_i, e_j \rangle = \ln B_{ij}.$$

For $i = n+1$ and $j \leq n$,

$$Y_{n+1,j} = \langle \ln(3M1_n - S^\top), e_j \rangle = \ln(3M - s_j).$$

Thus, for each $j \leq n$,

$$Y_{:,j} = \begin{bmatrix} \ln B_{1j} \\ \vdots \\ \ln B_{nj} \\ \ln(3M - s_j) \end{bmatrix}.$$

Apply column-wise softmax and write

$$W_{ij} := (\text{Softmax}(Y))_{ij} = \frac{\exp(Y_{ij})}{\sum_{r=1}^{n+1} \exp(Y_{rj})}.$$

Then

$$\exp(Y_{ij}) = \begin{cases} B_{ij}, & i \leq n, \\ 3M - s_j, & i = n + 1, \end{cases}$$

and

$$\sum_{r=1}^{n+1} \exp(Y_{rj}) = s_j + (3M - s_j) = 3M.$$

Hence, for $j \leq n$,

$$W_{ij} = \begin{cases} B_{ij}/(3M), & i \leq n, \\ (3M - s_j)/(3M), & i = n + 1, \end{cases}$$

or in block form,

$$W_{:,1:n} = \begin{bmatrix} B/(3M) \\ 1_{1 \times n} - S/(3M) \end{bmatrix}.$$

Combining this with $V = 3MA[X \ 0_d]$ yields for the first n columns

$$\text{Attn}(Z)_{:,1:n} = VW_{:,1:n} = 3MAX \cdot \frac{B}{3M} = AXB = \ell(X). \quad (\text{H.1})$$

So the first n tokens already match the desired output exactly.

Step 3: The Padding Column Stays Close to Zero. The remaining task is to control the last column $j = n + 1$. Here $q_{n+1} = T1_n$ enters, and one obtains

$$Y_{i,n+1} = \begin{cases} TH_1(i), & i \leq n, \\ TH_2, & i = n + 1, \end{cases}$$

where

$$H_1(i) := \sum_{r=1}^n \ln B_{ir}, \quad H_2 := \sum_{r=1}^n \ln(3M - s_r).$$

Since $0 < B_{ir} \leq M$ and $3M - s_r \geq 2M$,

$$\begin{aligned} H_1(i) &\leq n \ln M, \\ H_2 &\geq n \ln(2M), \end{aligned}$$

so $H_2 - H_1(i) \geq n \ln 2 > 0$ for all $i \leq n$. Thus the last column of Y has a strictly larger entry at index $n + 1$ than at any index $i \leq n$. For any $\delta > 0$, a sufficiently large choice of T yields

$$\max_{i \leq n} W_{i,n+1} \leq \delta, \quad |W_{n+1,n+1} - 1| \leq \delta.$$

In words, the last attention column concentrates on the padding token.

Recall that $V_{:,n+1} = 0_{d_{\text{out}}}$. Writing the last column of W as

$$W_{:,n+1} = \begin{bmatrix} a_0(T) \\ a_1(T) \end{bmatrix},$$

with $a_0(T) \in \mathbb{R}^{n \times 1}$ and $a_1(T) \in \mathbb{R}$, gives

$$\text{Attn}(Z)_{:,n+1} = VW_{:,n+1} = 3MAXa_0(T),$$

since the contribution from the padding token vanishes. The bounds on $W_{i,n+1}$ imply $\|a_0(T)\|_\infty \leq \delta$ and $\|a_0(T)\|_1 \leq n\delta$, so

$$\|\text{Attn}(Z)_{:,n+1}\|_\infty \leq 3M\|AX\|_\infty \|a_0(T)\|_1 \leq 3Mn\|AX\|_\infty \delta.$$

For a given $\epsilon > 0$, choose δ and then T so that $3Mn\|AX\|_\infty \delta \leq \epsilon$.

Step 4: Final Error Bound. The first n output columns equal $\ell(X)$ exactly according to (H.1), and the last column has infinity norm at most ϵ by the choice of T . Therefore

$$\left\| \text{Attn}(Z) - \begin{bmatrix} \ell(X) & 0_{d_{\text{out}}} \end{bmatrix} \right\|_\infty \leq \epsilon,$$

which completes the proof. $\square$

Theorem H.1 (Sequence-to-Sequence Universal Approximation of Multi-Head Softmax Attention). Let $1 \leq p < \infty$. Let $\mathcal{X} \subset \mathbb{R}^{d \times n}$ be a compact domain of input sequences. Let $f : \mathcal{X} \rightarrow \mathbb{R}^{d \times n}$ be a continuous sequence-to-sequence function. For any $\epsilon > 0$, there exists a network Φ composed of three multi-head attention layers such that

$$\|\Phi(X) - f(X)\|_{L_p} < \epsilon.$$

Proof Sketch. We divide the proof into three stage.

1. We first show that there exist a multi-head attention approximating the pre-activation of ReLU neural network.
2. We then construct the second attention layer to reorganize and share information across tokens through Lemma H.1.
3. Finally, we construct the third attention layer to approximate ReLU activation and the final linear combination to approximate ReLU neural network.

$\square$

Proof. Let $X = [x_1, x_2, \dots, x_n] \in \mathbb{R}^{d \times n}$ denote the input sequence. We first establish the result for a function $f : \mathbb{R}^{d \times n} \rightarrow \mathbb{R}^{1 \times n}$ acting on a sequence with a single output dimension. Generalization to multiple output dimensions follows by stacking such constructions. By standard universal approximation theorems for Feed-Forward Networks (FFNs) (Pinkus, 1999), for any $\epsilon_{FFN} > 0$, there exists an approximation of f taking the form of a sum of ReLUs (flatten input $\mathbb{R}^{dn}$). Let $\sum_{k=1}^N a_{i,k} \text{ReLU}(\sum_{j=1}^n w_{i,k,j}^\top x_j)$, $a_{i,k} \in \{-1, 1\}$ ³, $w_{i,k,j} \in \mathbb{R}^d$ denote the FFN approximation of the i -th token of the target function f_i .

Preprocessing.

³Because ReLU is positively homogeneous, we absorb $|a_{i,k}|$ into the weights inside the ReLU and keep only $\text{sign}(a_{i,k}) \in \{-1, 1\}$ outside.

Before feeding the input to the network, we pad the input with a zero token and append a positional encoding at the bottom. We denote this augmented input as

$$X_p := \begin{bmatrix} X & 0_d \\ I_n & 0_n \\ 0_{1 \times n} & 1 \end{bmatrix} \in \mathbb{R}^{(d+n+1) \times (n+1)}.$$

First Layer: Token-Wise Linear Pre-Activations.

In this layer, we first show that there exist multi-head attention Attn_1 approximating the pre-activation linear model of ReLU neural network.

According to [Theorem 3.2](#), for every $k \in [N]$, there exists a multi-head attention $\text{Attn}_{1,k}^*$ that for any $\epsilon_0 > 0$ approximates

$$\|\text{Attn}_{1,k}^*(X_p) - e_{(i-1)N+k}^{(nN+n+1)} [w_{i,k,1}^\top x_1 \quad w_{i,k,2}^\top x_2 \quad \cdots \quad w_{i,k,n}^\top x_n \quad 0]\|_\infty \leq \epsilon_0.$$

Next we construct an attention head Attn_I to preserve the identity matrix at the bottom

$$\text{Attn}_I(X_p) := \begin{bmatrix} 0_{nN \times (2n+1)} \\ S_I \end{bmatrix} X_p \text{Softmax}(\beta(S_I X_p)^\top S_I X_p),$$

where

$$S_I := \begin{bmatrix} 0_{(n+1) \times n} & I_{n+1} \end{bmatrix}.$$

Multiplication by S_I simply discards the top n rows of X_p and keeps the bottom $(n+1)$ rows, hence

$$S_I X_p = I_{n+1}.$$

The value, key, and query projections used in Attn_I are therefore

$$\begin{aligned} V &= \begin{bmatrix} 0_{nN \times (2n+1)} \\ S_I \end{bmatrix} X_p = \begin{bmatrix} 0_{nN \times (n+1)} \\ I_{n+1} \end{bmatrix}, \\ K &= S_I X_p = I_{n+1}, \quad Q = S_I X_p = I_{n+1}. \end{aligned}$$

The score matrix is

$$\beta K^\top Q = \beta I_{n+1}.$$

Hence $\text{Softmax}(\beta K^\top Q)$ becomes arbitrarily close to the identity when β is large. As a result, the attention output $V \text{Softmax}(\beta K^\top Q)$ nearly copies the bottom identity block unchanged while keeping the upper rows zero.

now we know Attn_I satisfies

$$\|\text{Attn}_I(X_p) - \begin{bmatrix} 0_{nN \times (n+1)} \\ I_{n+1} \end{bmatrix}\|_\infty \leq \epsilon_0.$$

Summing these heads ($\text{Attn}_{1,k}^*$ for $k \in [N]$ and Attn_I) defines Attn_1

$$\text{Attn}_1 := \text{Attn}_1^* + \text{Attn}_I.$$

It satisfies

$$\|\text{Attn}_1 - \begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix}\| = \|\text{Attn}_1^*(X_p) + \text{Attn}_I(X_p) - \begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix}\|_\infty \leq \epsilon_0,$$

where W_0 stacks the pre-activation blocks

$$W_0 := \begin{bmatrix} W_1 \\ W_2 \\ \vdots \\ W_n \end{bmatrix},$$

with each $W_i \in \mathbb{R}^{N \times (n+1)}$ defined as

$$W_i := \begin{bmatrix} w_{i,1,1}^\top x_1 & \dots & w_{i,1,n}^\top x_n & 0 \\ w_{i,2,1}^\top x_1 & \dots & w_{i,2,n}^\top x_n & 0 \\ \vdots & \ddots & \vdots & \vdots \\ w_{i,N,1}^\top x_1 & \dots & w_{i,N,n}^\top x_n & 0 \end{bmatrix}.$$

Second Layer: Reorganization and Mix Information across Tokens.

Now we construct the second layer.

Define $A_i, i \in [N]$ as

$$A_i := \begin{bmatrix} 0_{N \times (i-1)N} & I_N & 0_{N \times (n-i)N} & 0_{1 \times (n+1)} \\ 0_{N \times (i-1)N} & 0_{N \times N} & 0_{N \times (n-i)N} & I_{n+1} \end{bmatrix}.$$

This matrix will select the rows in the output of the first layer which are needed for the i -th head in the second layer.

According to [Lemma H.1](#), for any $\epsilon_2 > 0$ there exists an attention $\text{Attn}_{2,i}$ that satisfies

$$\|\text{Attn}_{2,i}(\begin{bmatrix} W_i \\ I_{n+1} \end{bmatrix}) - (\sum_{s=0}^{N-1} \underbrace{e_{sn+i}^{(nN+n+1)} (e_{s+1}^{(N)})^\top}_{E_{sn+i,s+1}}) W_i \begin{bmatrix} 0_{(n+1) \times (i-1)} & 1_{n+1} & 0_{(n+1) \times (n-i)} \end{bmatrix}\|_\infty \leq \epsilon_2.$$

Here $W_i := (W_0)_{k,:}$ is the i -th row of W_0 .

Since

$$A_i \cdot \begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix} = \begin{bmatrix} W_i \\ I_{n+1} \end{bmatrix},$$

we have

$$\|\text{Attn}_{2,i} \circ A_i(\begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix}) - (\sum_{s=0}^{N-1} \underbrace{e_{sn+i}^{(nN+n+1)} (e_{s+1}^{(N)})^\top}_{E_{sn+i,s+1}}) W_i \begin{bmatrix} 0_{(n+1) \times (i-1)} & 1_{n+1} & 0_{(n+1) \times (n-i)} \end{bmatrix}\|_\infty \leq \epsilon_2.$$

We note that $\text{Attn}_{2,i} \circ A_i$ is also an attention.

Summing the above constructed heads yields

$$\left\| \sum_{i=1}^n \text{Attn}_{2,i} \circ A_i \left(\begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix} \right) - \begin{bmatrix} \text{diag}(w_{1,1}^\top \bar{X}, w_{2,1}^\top \bar{X}, \dots, w_{n,1}^\top \bar{X}) & 0 \\ \text{diag}(w_{1,2}^\top \bar{X}, w_{2,2}^\top \bar{X}, \dots, w_{n,2}^\top \bar{X}) & 0 \\ \vdots & \vdots \\ \text{diag}(w_{1,N}^\top \bar{X}, w_{2,N}^\top \bar{X}, \dots, w_{n,N}^\top \bar{X}) & 0 \\ 0_{(n+1) \times n} & 0_{n+1} \end{bmatrix} \right\| \leq \epsilon_2,$$

in which

$$\bar{X} := \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

and

$$w_{i,k} := \begin{bmatrix} w_{i,k,1} \\ w_{i,k,2} \\ \vdots \\ w_{i,k,n} \end{bmatrix},$$

where

$$w_{i,k}^\top \bar{X} = \sum_{j=1}^n w_{i,k,j} x_j.$$

We also use an attention head to preserve the identity matrix, constructed like the previous head we used to preserve the identity matrix

$$\text{Attn}'_I(X_p) := \begin{bmatrix} 0_{N \times (nN+n+1)} \\ S'_I \end{bmatrix} X_p \text{Softmax}(\beta(S'_I X_p)^\top S'_I X_p),$$

in which S'_I is

$$\begin{bmatrix} 0_{(n+1) \times (nN)} & I_{n+1} \end{bmatrix}.$$

adding this head yields the final output of the second layer to satisfy

$$\left\| \sum_{i=1}^n \text{Attn}_{2,i} \circ A_i \left(\begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix} \right) + \text{Attn}'_I \left(\begin{bmatrix} W_0 \\ I_{n+1} \end{bmatrix} \right) - \begin{bmatrix} \text{diag}(w_{1,1}^\top \bar{X}, w_{2,1}^\top \bar{X}, \dots, w_{n,1}^\top \bar{X}) & 0_n \\ \text{diag}(w_{1,2}^\top \bar{X}, w_{2,2}^\top \bar{X}, \dots, w_{n,2}^\top \bar{X}) & 0_n \\ \vdots & \vdots \\ \text{diag}(w_{1,N}^\top \bar{X}, w_{2,N}^\top \bar{X}, \dots, w_{n,N}^\top \bar{X}) & 0_n \\ I_n & 0_n \\ 0_{1 \times n} & 1 \end{bmatrix} \right\|_\infty \leq \epsilon_2.$$

This defines the second layer Attn_2

$$\text{Attn}_2 := \sum_{i=1}^n \text{Attn}_{2,i} \circ A_i + \text{Attn}'_I.$$

Last Layer: ReLU and Aggregation.

We now show that the third attention layer implements the nonlinearity and the final signed aggregation of the FFN. For notation simplicity later, we redefine the output after Attn_2 we as

$$Z^{(2)} = \begin{bmatrix} \text{diag}(s_{1,1}, \dots, s_{n,1}) & 0_n \\ \text{diag}(s_{1,2}, \dots, s_{n,2}) & 0_n \\ \vdots & \vdots \\ \text{diag}(s_{1,N}, \dots, s_{n,N}) & 0_n \\ I_n & 0_n \\ 0_{1 \times n} & 1 \end{bmatrix} \in \mathbb{R}^{(Nn+n+1) \times (n+1)},$$

where

$$s_{i,k} := w_{i,k}^\top \bar{X}, \quad \bar{X} = \begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}.$$

Let's get some intuition of our final goal. Stacking these pre-activations as rows yields an $N \times n$ matrix

$$S := \begin{bmatrix} s_{1,1} & \dots & s_{n,1} \\ \vdots & \ddots & \vdots \\ s_{1,N} & \dots & s_{n,N} \end{bmatrix},$$

so that the k -th block on top of $Z^{(2)}$ is exactly $\text{diag}(S_{k,:})$.

The FFN approximation of the function with a single output dimension form

$$f(X) = [f_1(X) \quad \dots \quad f_n(X)] \approx \sum_{k=1}^N a^{(k)} \odot \text{ReLU}(S_{k,:}),$$

where $a^{(k)} := (a_{1,k}, \dots, a_{n,k}) \in \{\pm 1\}^n$ and $\odot$ denotes elementwise multiplication. Thus the third layer Attn_3 must map the block-diagonal structure in $Z^{(2)}$ to this signed ReLU combination.

Now let's start to construct the attention weight matrices to achieve this goal.

Values, keys, and queries. For a fixed $k \in [N]$, define the value projection

$$W_V^{(k)} := \begin{bmatrix} 0_{1 \times (k-1)n} & 1_{1 \times n} & 0_{1 \times ((N-k)n+n+1)} \end{bmatrix} \in \mathbb{R}^{1 \times (Nn+n+1)}.$$

By construction, $W_V^{(k)}$ picks out the k -th $n \times (n+1)$ block at the top of $Z^{(2)}$ and sums its rows. Since that block is diagonal, we obtain

$$V_k := S_k Z^{(2)} = \begin{bmatrix} S_{k,1} & \dots & S_{k,n} & 0 \end{bmatrix} \in \mathbb{R}^{1 \times (n+1)},$$

that is, the k -th row of S appears as the first n entries of V_k , and the value at the padding token is 0. In the notation of the attention operator above, this means

The common key projection is chosen as

$$W_K := \begin{bmatrix} 0_{n \times nN} & I_n & 0_{n \times 1} \end{bmatrix} \in \mathbb{R}^{n \times (Nn+n+1)}.$$

Multiplying by $Z^{(2)}$ selects the identity block in the bottom $(n+1) \times (n+1)$ portion:

$$K := W_K Z^{(2)} = \begin{bmatrix} I_n & 0_{n \times 1} \end{bmatrix} \in \mathbb{R}^{n \times (n+1)}.$$

Query projections for the positive and negative parts. For a fixed $k \in [N]$, define

$$\begin{aligned} C_1 &:= \text{diag}(1_{a_{1,k}=1}, \dots, 1_{a_{n,k}=1}), \\ C_2 &:= \text{diag}(1_{a_{1,k}=-1}, \dots, 1_{a_{n,k}=-1}), \\ D &:= -1_{n \times (n+1)} + \begin{bmatrix} I_n & 1_n \end{bmatrix} \in \mathbb{R}^{n \times (n+1)}. \end{aligned}$$

The matrices C_1 and C_2 select positions with positive and negative coefficients $a_{i,k}$, respectively, while D compares with a reference (the padding token) as we describe below.

The query projections for the two heads corresponding to hidden unit k are

$$\begin{aligned} W_Q^{(1,k)} &:= \begin{bmatrix} 0_{n \times (k-1)n} & C_1 & 0_{n \times (N-k)n} & D \end{bmatrix}, \\ W_Q^{(2,k)} &:= \begin{bmatrix} 0_{n \times (k-1)n} & C_2 & 0_{n \times (N-k)n} & D \end{bmatrix}. \end{aligned}$$

Writing $Z^{(2)}$ in block form as

$$Z^{(2)} = \begin{bmatrix} \text{diag}(S_{1,:}) & 0 \\ \vdots & \vdots \\ \text{diag}(S_{N,:}) & 0 \\ I_n & 0 \\ 0_{1 \times n} & 1 \end{bmatrix} = \begin{bmatrix} Z^{\text{top}} \\ Z^{\text{bottom}} \end{bmatrix},$$

with $Z^{\text{top}} \in \mathbb{R}^{Nn \times (n+1)}$ and $Z^{\text{bottom}} \in \mathbb{R}^{(n+1) \times (n+1)}$, we obtain

$$\begin{aligned} W_Q^{(1,k)} Z^{(2)} &= C_1 \text{diag}(S_{k,:}) + D Z^{\text{bottom}}, \\ W_Q^{(2,k)} Z^{(2)} &= C_2 \text{diag}(S_{k,:}) + D Z^{\text{bottom}}. \end{aligned}$$

Since Z^{bottom} is the $(n+1) \times (n+1)$ identity,

$$D Z^{\text{bottom}} = D,$$

and hence

$$\begin{aligned} Q_k^{(1)} &:= W_Q^{(1,k)} Z^{(2)} = C_1 \text{diag}(S_{k,:}) + D, \\ Q_k^{(2)} &:= W_Q^{(2,k)} Z^{(2)} = C_2 \text{diag}(S_{k,:}) + D. \end{aligned}$$

Score matrices and their structure. The score matrices for the first heads associated with hidden unit k are

$$Y_k^{(1)} := K^\top Q_k^{(1)} = K^\top C_1 \text{diag}(S_{k,:}) + K^\top D.$$

Since

$$K^\top = \begin{bmatrix} I_n \\ 0_{1 \times n} \end{bmatrix},$$

we can compute the product with D explicitly:

$$K^\top D = \begin{bmatrix} I_n \\ 0_{1 \times n} \end{bmatrix} (-1_{n \times (n+1)} + \begin{bmatrix} I_n & 1_n \end{bmatrix}) = E,$$

where

$$E = \begin{bmatrix} 0 & -1 & \dots & -1 & 0 \\ -1 & 0 & \dots & -1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & \dots & 0 & 0 \\ 0 & 0 & \dots & 0 & 0 \end{bmatrix} \in \mathbb{R}^{(n+1) \times (n+1)}.$$

Thus

$$Y_k^{(1)} = \text{diag}(1_{a_{1,k}=1}S_{k,1}, \dots, 1_{a_{n,k}=1}S_{k,n}) + E,$$

Equivalently, in full matrix form,

$$Y_k^{(1)} = \begin{bmatrix} 1_{a_{1,k}=1}S_{k,1} & -1 & \dots & -1 & 0 \\ -1 & 1_{a_{2,k}=1}S_{k,2} & \dots & -1 & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ -1 & -1 & \dots & 1_{a_{n,k}=1}S_{k,n} & 0 \\ 0 & 0 & \dots & 0 & 0 \end{bmatrix},$$

and $Y_k^{(2)}$ is obtained by replacing $1_{a_{i,k}=1}$ with $1_{a_{i,k}=-1}$ on the diagonal.

Softmax and signed ReLU. For the first head associated with hidden unit k , define the output row vector

$$H_k^{(1)} := V_k \text{Softmax}(\beta Y_k^{(1)}) \in \mathbb{R}^{1 \times (n+1)},$$

where $V_k = [S_{k,1} \dots S_{k,n} 0]$. Thus the first n coordinates of $H_k^{(1)}$ can be written as

$$H_{k,1:n}^{(1)} = [S_{k,1} \dots S_{k,n} 0] \text{Softmax}(\beta Y_k^{(1)})_{:,1:n}.$$

From the explicit form of $Y_k^{(1)}$ above, each column $j \leq n$ has the structure

$$(Y_k^{(1)})_{:,j} = \begin{bmatrix} -1 \\ \vdots \\ -1 \\ 1_{a_{j,k}=1}S_{k,j} \\ -1 \\ \vdots \\ -1 \\ 0 \end{bmatrix},$$

that is, a diagonal entry $1_{a_{j,k}=1}S_{k,j}$ at position $i = j$, a baseline value -1 in all other rows $i \leq n$, and 0 at the padding index $i = n + 1$. For large β , the column-wise Softmax therefore concentrates on either the diagonal entry $i = j$ (when $a_{j,k} = 1$ and $S_{k,j} > 0$) or on the padding index $i = n + 1$ (when $a_{j,k} = 1$ and $S_{k,j} \leq 0$ or when $a_{j,k} = -1$)

Formally, let $\widetilde{W}_k^{(1)} \in \mathbb{R}^{(n+1) \times (n+1)}$ denote the *ideal hardmax weight matrix* whose j -th column is

$$(\widetilde{W}_k^{(1)})_{:,j} := \begin{cases} e_j, & a_{j,k} = 1 \text{ and } S_{k,j} > 0, \\ e_{n+1}, & \text{otherwise,} \end{cases}$$

so that $\text{Softmax}(\beta Y_k^{(1)})_{:,j} \rightarrow (\widetilde{W}_k^{(1)})_{:,j}$ as $\beta \rightarrow \infty$. Define the corresponding “hard” output

$$\widetilde{H}_k^{(1)} := V_k \widetilde{W}_k^{(1)} \in \mathbb{R}^{1 \times (n+1)}.$$

By the definition of $V_k = [S_{k,1} \ \dots \ S_{k,n} \ 0]$ and the columns of $\widetilde{W}_k^{(1)}$, the j -th coordinate of $\widetilde{H}_k^{(1)}$ satisfies

$$\widetilde{H}_{k,j}^{(1)} = \begin{cases} S_{k,j}, & a_{j,k} = 1 \text{ and } S_{k,j} > 0, \\ 0, & \text{otherwise,} \end{cases} = 1_{a_{j,k}=1} \text{ReLU}(S_{k,j}).$$

Hence, in vector form,

$$\widetilde{H}_{k,1:n}^{(1)} = (1_{a_{1,k}=1} \text{ReLU}(S_{k,1}), \dots, 1_{a_{n,k}=1} \text{ReLU}(S_{k,n})) \in \mathbb{R}^{1 \times n}.$$

Since $\text{Softmax}(\beta Y_k^{(1)})$ converges column-wise to $\widetilde{W}_k^{(1)}$ as $\beta \rightarrow \infty$, we have

$$H_k^{(1)} = V_k \text{Softmax}(\beta Y_k^{(1)}) \approx \widetilde{H}_k^{(1)},$$

and in particular

$$H_{k,1:n}^{(1)} \approx (1_{a_{1,k}=1} \text{ReLU}(S_{k,1}), \dots, 1_{a_{n,k}=1} \text{ReLU}(S_{k,n})).$$

The second head is treated analogously. Writing

$$H_k^{(2)} := V_k \text{Softmax}(\beta Y_k^{(2)}),$$

the same reasoning applied to $Y_k^{(2)}$ yields

$$H_{k,1:n}^{(2)} \approx (1_{a_{1,k}=-1} \text{ReLU}(S_{k,1}), \dots, 1_{a_{n,k}=-1} \text{ReLU}(S_{k,n})).$$

The module $\text{Attn}_{3,k}$ is defined as the difference of these two single-head attentions,

$$\text{Attn}_{3,k}(Z^{(2)}) := V_k \text{Softmax}(\beta Y_k^{(1)}) - V_k \text{Softmax}(\beta Y_k^{(2)}) = H_k^{(1)} - H_k^{(2)},$$

so that its first n coordinates satisfy

$$(\text{Attn}_{3,k}(Z^{(2)}))_{1:n} \approx (a_{1,k} \text{ReLU}(S_{k,1}), \dots, a_{n,k} \text{ReLU}(S_{k,n})) = a^{(k)} \odot \text{ReLU}(S_{k,:}).$$

Summing over k yields

$$(\text{Attn}_3(Z^{(2)}))_{1:n} = \sum_{k=1}^N (\text{Attn}_{3,k}(Z^{(2)}))_{1:n} \approx \sum_{k=1}^N a^{(k)} \odot \text{ReLU}(S_{k,:}),$$

which exactly matches the length- n output of the FFN approximation. Therefore there exists $\beta > 0$ such that

$$\left\| \text{Attn}_3 \left(\begin{bmatrix} \text{diag}(s_{1,1}, \dots, s_{n,1}) & 0_n \\ \vdots & \vdots \\ \text{diag}(s_{1,N}, \dots, s_{n,N}) & 0_n \\ I_n & 0_n \\ 0_{1 \times n} & 1 \end{bmatrix} \right) - \left[\sum_{k=1}^N a_{1,k} \text{ReLU}(S_{k,1}) \quad \dots \quad \sum_{k=1}^N a_{n,k} \text{ReLU}(S_{k,n}) \right] \right\|_{\infty} \leq \epsilon_2.$$

Finally we truncate the padded token and define Φ as

$$T \circ \text{Attn}_3 \circ \text{Attn}_2 \circ \text{Attn}_1 \circ P$$

where P is the preprocessing step that pads 1 zero token and T is the truncation step that delete the last token.

Error Analysis and Convergence in L_p .

We now demonstrate that the accumulated error through the three layers can be bounded arbitrarily in the L_p norm. Unlike analyses relying on Lipschitz constants, which may explode with large attention weights, we utilize the uniform continuity of the target operators on compact domains.

Let L_1, L_2, L_3 denote the ideal mathematical operators approximated by the three layers (component extraction, column summation, and ReLU aggregation, respectively). Let $H_0 = X_p$ be the input. We define the sequence of ideal feature maps as $H_1 = L_1(H_0)$, $H_2 = L_2(H_1)$, and $H_3 = L_3(H_2)$, where H_3 corresponds to the target FFN output. Conversely, let $\tilde{H}_1 = \text{Attn}_1(H_0)$, $\tilde{H}_2 = \text{Attn}_2(\tilde{H}_1)$, and $\tilde{H}_3 = \text{Attn}_3(\tilde{H}_2)$ denote the actual outputs of the constructed layers.

Let C_d be a constant such that $\|M\|_p \leq C_d \|M\|_\infty$ for matrices of the relevant dimensions. We seek to show that for any $\epsilon > 0$, the parameters of the attention layers can be chosen such that $\|\tilde{H}_3 - H_3\|_p < \epsilon$.

We proceed via a backward induction argument. Consider the final layer. By the triangle inequality,

$$\begin{aligned} \|\tilde{H}_3 - H_3\|_p &= \|\text{Attn}_3(\tilde{H}_2) - L_3(H_2)\|_p \\ &\leq \|\text{Attn}_3(\tilde{H}_2) - L_3(\tilde{H}_2)\|_p + \|L_3(\tilde{H}_2) - L_3(H_2)\|_p. \end{aligned}$$

The operator L_3 involves ReLU functions and linear sums, which are continuous. Since the domain of valid feature maps is compact, L_3 is uniformly continuous. Therefore, there exists a $\delta_2 > 0$ such that for any inputs Y, Y' satisfying $\|Y - Y'\|_p < \delta_2$, we have $\|L_3(Y) - L_3(Y')\|_p < \epsilon/2$. Furthermore, by the construction of the third layer, specifically by increasing the Softmax scaling parameter β , we can limit the approximation error such that $\|\text{Attn}_3(Z) - L_3(Z)\|_p < \epsilon/2$ for all Z in the compact range. Thus, the total error is bounded by ϵ provided that $\|\tilde{H}_2 - H_2\|_p < \delta_2$.

We apply the same logic to the second layer. We require $\|\tilde{H}_2 - H_2\|_p < \delta_2$. Decomposing the error yields

$$\|\tilde{H}_2 - H_2\|_p \leq \|\text{Attn}_2(\tilde{H}_1) - L_2(\tilde{H}_1)\|_p + \|L_2(\tilde{H}_1) - L_2(H_1)\|_p.$$

The operator L_2 is linear and therefore uniformly continuous. Thus, there exists a $\delta_1 > 0$ such that $\|\tilde{H}_1 - H_1\|_p < \delta_1$ implies $\|L_2(\tilde{H}_1) - L_2(H_1)\|_p < \delta_2/2$. By [Lemma H.1](#), we can choose the parameters of Attn_2 such that the approximation error $\|\text{Attn}_2(\tilde{H}_1) - L_2(\tilde{H}_1)\|_p$ is less than $\delta_2/2$. This condition holds if $\|\tilde{H}_1 - H_1\|_p < \delta_1$.

Finally, for the first layer, we ensure $\|\tilde{H}_1 - H_1\|_p < \delta_1$. The input H_0 is exact, so there is no propagated error. By [Theorem 3.2](#), we can construct Attn_1 with precision $\epsilon_0 = \delta_1/C_d$ such that

$$\|\tilde{H}_1 - H_1\|_p \leq C_d \|\text{Attn}_1(H_0) - L_1(H_0)\|_\infty \leq C_d \epsilon_0 = \delta_1.$$

By choosing the construction parameters corresponding to ϵ_0, ϵ_2 and β derived from the moduli of continuity δ_1 and δ_2 , we guarantee that the final output satisfies

$$\|\tilde{H}_3 - H_3\|_p < \epsilon$$

for all $X \in \mathcal{X}$. Equivalently,

$$\sup_{X \in \mathcal{X}} \|\tilde{H}_3 - H_3\|_p \leq \epsilon.$$

Since the target FFN H_3 can approximate the function f to arbitrary precision, the pure attention network in the one-row case is a pointwise universal approximator on $\mathcal{X}$.

Extension to d Output Rows.

The discussion above treats $f : \mathcal{X} \rightarrow \mathbb{R}^{1 \times n}$. For a general sequence-to-sequence map $f : \mathcal{X} \rightarrow \mathbb{R}^{d \times n}$, write

$$f^{(r)}(X) := f(X)_{r,:} \in \mathbb{R}^{1 \times n}, \quad r \in [d],$$

and construct, for each r , a three-layer attention network $\Phi^{(r)}$ that approximates $f^{(r)}$ with

$$\|\Phi^{(r)}(X) - f^{(r)}(X)\|_p \leq \varepsilon_{\text{row}} \quad \text{for all } X \in \mathcal{X}.$$

Define the combined network

$$\Phi(X) := \begin{bmatrix} \Phi^{(1)}(X) \\ \vdots \\ \Phi^{(d)}(X) \end{bmatrix} \in \mathbb{R}^{d \times n},$$

which corresponds to placing the heads for all $\Phi^{(r)}$ inside the same three multi-head attention layers and concatenate their outputs in the feature dimension as usual. For the full error $E(X) := \Phi(X) - f(X)$, we have

$$\|E(X)\|_p^p = \sum_{r=1}^d \|\Phi^{(r)}(X) - f^{(r)}(X)\|_p^p \leq d \varepsilon_{\text{row}}^p,$$

so

$$\|E(X)\|_p \leq d^{1/p} \varepsilon_{\text{row}}.$$

Choosing $\varepsilon_{\text{row}} := \varepsilon / d^{1/p}$ yields

$$\|\Phi(X) - f(X)\|_p \leq \varepsilon \quad \text{for all } X \in \mathcal{X}.$$

By the definition of the L_p norm in (2.1) (applied with $\Omega = \mathcal{X}$ and $f(X) = \|\Phi(X) - f(X)\|_p$), we have

$$\begin{aligned} \|\Phi - f\|_{L_p}^p &= \int_{\mathcal{X}} \|\Phi(X) - f(X)\|_p^p dX \\ &\leq \int_{\mathcal{X}} \varepsilon^p dX. \end{aligned}$$

Since $\mathcal{X}$ is compact, the integral $\int_{\mathcal{X}} 1 dX$ is finite. Given any target $\epsilon > 0$, we can run the above construction with a pointwise tolerance ε small enough so that the right-hand side is at most ϵ^p , which yields

$$\|\Phi - f\|_{L_p} < \epsilon.$$

This complete the proof. □